Integrating Wildfire Risk in Forest Road Network Planning

A Linear Optimization Approach to Select a Cost-Minimizing and Fire Response Enhancing Road Network in Timber Forests

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Declaration

I declare that this document is an original work of my own authorship and that it fulfills all the requirements of the Code of Conduct and Good Practices of the Universidade de Lisboa.

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Abstract

Wildfires are an increasingly severe threat to Mediterranean forests, where climate, vegetation, and topography amplify fire risk. Forest road networks play a dual role in timber forests: they are indispensable for harvesting operations and can provide access for firefighting crews. Yet, current forest road planning approaches focus almost exclusively on minimizing road construction and maintenance costs for timber extraction, without accounting for wildfire resilience.

This master's dissertation develops a linear optimization model that integrates wildfire risk considerations into multiperiod forest road network planning. The model minimizes total road construction and maintenance costs across five ten-year periods, while ensuring both mechanized harvesting access and ground-based fire vehicle access to stands with low fire resistance. Constraints are based on stand-level wildfire resistance indicators, enabling managers to adjust thresholds that determine the extent of protection. Two road width options are incorporated: narrow roads sufficient for timber extraction and wider roads allowing fire response operations.

The model is applied to the Paiva forest intervention zone in northern Portugal, a fire-prone landscape managed collectively by smallholders. Results demonstrate how the framework generates cost-minimal road networks that also guarantee firefighting access to the least resilient areas. While fire-blind networks would be cheaper in the short term, the integrated approach highlights the long-term value of strategic investment in infrastructure that protects both timber resources and broader landscape resilience. This work contributes decision support for forest managers and can be generalized to other fire-prone regions. Aligning infrastructure planning with climate adaptation needs, the work supports integrated forest management.

Keywords

Forest Road Network Planning; Wildfire Resilience; Operational Research; Decision Support; Linear Programming

Resumo

Os incêndios florestais constituem uma ameaça crescente para as florestas mediterrânicas, onde o clima, a vegetação e a topografia amplificam o risco de ignição e propagação. As redes de estradas florestais desempenham um papel duplo nestas paisagens: são indispensáveis para a exploração madeireira e fornecem acesso essencial às equipas de combate. Contudo, as abordagens atuais de planeamento concentram-se quase exclusivamente na minimização dos custos de construção e manutenção, sem integrar a resiliência ao fogo.

Esta dissertação de mestrado apresenta um modelo de otimização linear que incorpora o risco de incêndio no planeamento multiperíodo da rede viária florestal. O modelo minimiza os custos totais de construção e manutenção ao longo de cinco períodos de dez anos, assegurando simultaneamente o acesso mecanizado à exploração florestal e o acesso de veículos de combate a incêndios a áreas de baixa resistência ao fogo. As restrições baseiam-se em indicadores de resistência ao incêndio ao nível do povoamento, permitindo ajustar limiares de proteção. Foram consideradas duas tipologias de estrada: vias estreitas para extração de madeira e vias largas aptas a suportar operações de combate.

A aplicação do modelo à Zona de Intervenção Florestal do Paiva, no norte de Portugal — uma paisagem propensa a incêndios e gerida coletivamente por pequenos proprietários — demonstra como a abordagem proposta gera redes de custo mínimo que garantem também o acesso dos meios de combate às áreas mais vulneráveis. Embora as redes concebidas sem considerar o risco sejam mais económicas a curto prazo, a integração da resiliência evidencia o valor estratégico de investir em infraestruturas que protegem os recursos madeireiros e reforçam a resiliência da paisagem. O modelo constitui uma ferramenta prática de apoio à decisão e pode ser generalizado a outras regiões suscetíveis a incêndios.

Palavras-chave

Planeamento de Redes Viárias Florestais; Resiliência face a Incêndios Florestais; Investigação Operacional; Apoio à Decisão; Programação Linear

Resumo Alargado

Os incêndios florestais constituem uma ameaça crescente e cada vez mais complexa para as florestas mediterrânicas, em particular no contexto português. As condições climáticas, a composição da vegetação e a topografia acentuada criam um ambiente propício à ignição e à propagação do fogo, agravado pelas alterações climáticas e pelo abandono progressivo das práticas tradicionais de gestão. Este cenário traduz-se num risco elevado de grandes incêndios, com impactos profundos nos ecossistemas, na economia rural e na segurança das populações. Neste contexto, as infraestruturas florestais — e em especial as redes de estradas — assumem um papel central, pois servem simultaneamente a exploração madeireira e o acesso dos meios de prevenção e combate. No entanto, as abordagens de planeamento atualmente utilizadas continuam a privilegiar quase exclusivamente a minimização dos custos de construção e manutenção das vias associadas à exploração de madeira, raramente considerando a importância da resiliência ao fogo e da acessibilidade operacional em situações de emergência.

A presente dissertação de mestrado teve como objetivo desenvolver um modelo de otimização linear que integre o risco de incêndio no planeamento multiperíodo de redes viárias florestais. Pretendeu-se, assim, encontrar um equilíbrio entre a eficiência económica das operações de exploração e a necessidade de garantir o acesso rápido e seguro dos meios terrestres de combate a incêndios, sobretudo nas áreas mais vulneráveis. O trabalho propõe um enquadramento metodológico inovador, capaz de identificar configurações de rede que minimizam o custo total de investimento e manutenção, assegurando simultaneamente a acessibilidade necessária à exploração madeireira e às intervenções de combate.

O modelo foi formulado como um problema de otimização linear inteira mista (MILP) e aplicado a um horizonte temporal de cinquenta anos, dividido em cinco períodos de dez anos. O objetivo consiste em minimizar o custo total de construção, manutenção e eventual alargamento das estradas, garantindo que, em cada período, todos os povoamentos acessíveis sejam servidos por vias adequadas. Foram consideradas duas tipologias de estrada: vias estreitas, destinadas à extração de madeira, e vias largas, capazes de suportar viaturas pesadas de combate a incêndios. Esta distinção permite representar de forma realista as diferentes exigências operacionais e os custos associados, assegurando que as decisões de investimento refletem as funções complementares da rede.

As restrições do modelo garantem que todos os povoamentos programados para exploração são acessíveis por vias adequadas e que as zonas com menor resistência ao fogo dispõem de acessos compatíveis com os veículos de combate. O conceito de resistência foi incorporado através de indicadores ao nível do povoamento, obtidos a partir de variáveis biofísicas como combustível, declive, composição florestal e exposição solar. Estes indicadores permitem ajustar limiares de acessibilidade conforme o grau de vulnerabilidade de cada área, dando ao gestor florestal flexibilidade para calibrar o nível de proteção desejado. Para assegurar a conectividade da rede, introduziram-se variáveis auxiliares de fluxo que garantem a ligação de cada povoamento à rede pública, evitando trajetos que atravessassem zonas não acessíveis ou de gestão restrita.

A aplicação prática do modelo realizou-se na Zona de Intervenção Florestal (ZIF) do Paiva, localizada no norte de Portugal, uma região com elevada propensão a incêndios e marcada pela fragmentação fundiária. A área de estudo, com cerca de 6 700 hectares, apresenta povoamentos heterogéneos e uma rede viária incompleta. Para a representação espacial, foi criada uma base de dados geográfica integrada, com informação sobre povoamentos, rede hidrográfica, curvas de nível, rede viária existente e histórico de áreas ardidas. A rede foi convertida num grafo dirigido e simplificada através de procedimentos de pré-processamento que incluíram a fusão de segmentos redundantes e a remoção de nós desnecessários, reduzindo substancialmente o número de elementos sem perda de conectividade ou de custo acumulado.

O modelo foi implementado em Python, recorrendo à API do solver CPLEX, e executado em notebooks Jupyter, que integraram as fases de preparação de dados, formulação do modelo, resolução e análise dos resultados. Para reduzir a complexidade computacional, aplicaram-se técnicas de decomposição geográfica e reforço do modelo, limitando a criação de variáveis apenas a arcos viáveis e introduzindo restrições de dominância que eliminaram redundâncias. O problema resultante foi resolvido com algoritmos de branch-and-cut, assegurando soluções ótimas ou quase ótimas dentro de um limite de tolerância definido, com tempos de execução compatíveis com a dimensão da rede.

Para testar o modelo, foram analisados três cenários de planeamento: (i) MaxWood, centrado na minimização dos custos de exploração; (ii) MaxRes, que privilegia a acessibilidade das zonas menos resistentes ao fogo; e (iii) Stakeholder, que reflete as prioridades definidas em colaboração com os gestores da ZIF. Os resultados demonstraram que o cenário MaxWood, embora mais económico a curto prazo, conduz a redes que não asseguram cobertura suficiente das áreas vulneráveis. Já os cenários MaxRes e Stakeholder revelaram-se mais equilibrados, garantindo acessos a mais de 90

A análise de sensibilidade mostrou que o modelo é robusto face a variações moderadas nos custos de construção e manutenção, mantendo soluções estáveis e economicamente racionais. No entanto, a acessibilidade global depende fortemente da continuidade fundiária e da disponibilidade de terrenos, uma vez que a ausência de adesão de alguns proprietários à ZIF impede a ligação de determinados povoamentos. Apesar destas limitações, o desempenho computacional foi satisfatório e o solver apresentou comportamento estável, o que confirma a aplicabilidade prática da abordagem proposta.

Os contributos principais deste trabalho podem ser organizados em três níveis: metodológico, prático e estratégico. Em termos metodológicos, destaca-se o desenvolvimento de um modelo de otimização multiperíodo que conjuga, de forma inédita, os objetivos de exploração madeireira e de acessibilidade ao combate a incêndios. No plano prático, a aplicação ao caso real demonstra o potencial do modelo como ferramenta de apoio à decisão, capaz de identificar soluções de investimento que conciliam eficiência económica e reforço da resiliência. Finalmente, do ponto de vista estratégico, o enquadramento apresentado promove uma abordagem integrada de gestão florestal, alinhada com as necessidades de adaptação às alterações climáticas e com as políticas de ordenamento e defesa da floresta contra incêndios.

Ainda assim, é importante reconhecer as limitações do estudo. A resolução dos dados geográficos utilizados não permite captar todas as variações locais relevantes; a ausência de restrições hidrológicas detalhadas pode gerar trajetos pouco realistas em zonas de linhas de água; e as estimativas de custo, por dependerem de médias regionais, podem não refletir condições locais específicas. Além disso, fatores ambientais e sociais, como impactos ecológicos ou acessos comunitários, não foram integrados nesta primeira formulação. Estas lacunas abrem caminho a futuros desenvolvimentos, nomeadamente a incorporação de modelos de comportamento do fogo, a inclusão de custos ambientais e o uso de abordagens híbridas que combinem otimização exata com técnicas heurísticas ou de inteligência artificial, capazes de lidar com incerteza e escalabilidade.

Em síntese, esta dissertação propõe uma abordagem inovadora para o planeamento de redes viárias florestais, que alia a racionalidade económica da otimização matemática à urgência de reforçar a resiliência dos territórios face ao fogo. Ao integrar o risco de incêndio como um elemento estrutural do processo de decisão, o modelo demonstra que é possível conceber redes de estradas simultaneamente eficientes, seguras e sustentáveis. O trabalho oferece, assim, uma ferramenta prática e flexível para gestores florestais, entidades públicas e associações de produtores, contribuindo para decisões de investimento mais informadas e alinhadas com os desafios de um contexto mediterrânico cada vez mais vulnerável ao fogo. Através desta integração entre ciência de dados, planeamento espacial e gestão florestal, o estudo evidencia o papel que a modelação pode desempenhar na construção de uma silvicultura mais adaptativa, colaborativa e resiliente.

Contents

List of Tables

List of Symbols

List of Figures

Acronyms

AFVS Associação Florestal de Vale do Sousa

AI Artificial Intelligence

bwd backward

CAOF Comissão de Acompanhamento para as Operações Florestais

CPLEX IBM® ILOG® CPLEX® Optimization Studio

CRS Coordinate Reference System

EWE Extreme Wildfire Event

fwd forward

FFPR Forest Fire Prevention Road

GIS Geographic Information System

IP Integer Programming

LB Lower Bound

MILP Mixed Integer Linear Programming

UB Upper Bound

ZIF Zona de Intervenção Florestal

Glossary

digraph

A digraph is a Graph in which edges have orientations. It is written as a pair $G = (V, A)$, where V is a set of elements called *vertices*, *nodes*, or *points*, and A is a set consisting of ordered pairs of vertices (v_i, v_j) called *arcs*, *directed edges*, *arrows*, or *directed lines*. 49

FIRE-RES

4-year project (2021-2025) dedicated to developing an integrated forest management strategy to address the challenges posed by Extreme Wildfire Events (EWEs) in Europe. It is led by the Forest Science and Technology Centre of Catalonia, Spain, and funded under the European Union's H2020 research and innovation program. See also appendix A . . . . xix, 9, 90

Graph

Mathematical structure used to model pairwise relationships between objects. It consists of a set of vertices (also called nodes) and a set of edges that connect pairs of vertices. Graphs can be directed (edges have a direction) or undirected (edges have no direction), and they are fundamental in fields such as computer science, network analysis, and combinatorics. xviii, 35, 47

IBM® ILOG® CPLEX® Optimization Studio

IBM® ILOG® CPLEX® Optimization Studio is a commercial decision optimization software developed by IBM for building and solving complex optimization models. Analytical decision support toolkit for rapid development and deployment of optimization models using mathematical and constraint programming. It combines an integrated development environment with the powerful Optimization Programming Language and high-performance CPLEX and CP Optimizer solvers. . . . . xvi, 64

Integer Programming

In Integer Programming (IP), a subset of Linear Programming, the decision variables are restricted to integer values. . . . . xvi, xviii, 12

Linear Programming

Mathematical optimization method used to achieve the best outcome in a mathematical model, where the objective and constraints are expressed by linear relationships. By systematically selecting values for the decision variables using techniques such as branch-and-bound or the simplex method, the goal is to find the values that optimize the objective function while satisfying all the constraints. xviii

Living Lab

name for case study regions within the FIRE-RES project 9, 90

nonindustrial private forest

Privately owned forests that are not managed for industrial purposes but for other objectives such as recreation, conservation, or personal use. 9

QGIS

Quantum Geographic Information System, an open-source GIS software for spatial data visualization and analysis . . . . vii, 18, 19

1

Introduction

Contents

1.1 Background and Motivation

FIRE occurrences play a significant role in the dynamics of forest ecosystems, particularly in the Mediterranean region. Although they can have beneficial ecological effects (Keeley et al. 2011; Pereira et al. 2021; Davim et al. 2022), each year, some extreme wildfires occur globally that exceed suppression capabilities, cause extensive damage, and frequently result in fatalities (Tedim et al. 2018; Schwörer et al. 2024). Large wildfires release large quantities of greenhouse gases and air pollutants, thereby deteriorating air quality and compromising ecosystems, their provisioning services, and public health (Wang et al. 2023; Bolan et al. 2025; Carreras-Sospedra et al. 2024). They lead to widespread vegetation loss, which accelerates soil erosion, disrupts water and nutrient cycles, and can increase the risks of floods and landslides (Shakesby and Doerr 2006; Francos et al. 2024; Roces-Díaz et al. 2022; Sutanto et al. 2024; Culler et al. 2023). Hence, preventing wildfires from escalating into such extreme events is a major environmental and societal concern.

However, both the frequency and intensity of the most extreme wildfire events worldwide have increased in recent decades (Cunningham et al. 2024). A growing consensus among scientists, fire managers, and agencies recognizes that wildland areas are increasingly exposed to challenging fire weather, longer fire seasons, and more extensive fires — all exacerbated by climate change (San-Miguel-Ayanz et al. 2017; Jolly et al. 2015; Bowman et al. 2017; Robinne et al. 2018; Ellis et al. 2022). In 2017, Portugal experienced an unprecedented wildfire season, with approximately 500,000 hectares burned and more than 120 fatalities reported (Turco, Jerez, et al. 2019). Climate change is expected to further intensify fire-conducive conditions, thereby increasing the risk of large wildfires (Dupuy et al. 2020; Turco, Rosa-Cánovas, et al. 2018). Projections for Europe suggest that, by 2090, the annual area burned could rise by 120–270% compared to the 2000–2010 average (Robinne et al. 2018; FAO 2020). Specifically, in the Mediterranean Basin, the frequency of heat-induced fire-weather types — closely associated with large wildfire activity — is projected to increase by 14% under the RCP4.5 scenario and by 30% under the RCP8.5 scenario by the end of the century, significantly raising the likelihood of extreme wildfire events (Ruffault et al. 2020).

As wildfires are projected to become more frequent and intense, particularly in fire-prone regions such as the Mediterranean basin, it is increasingly important to adopt a holistic approach to wildfire management. This includes moving away from zero-burn policies[1] and prioritizing

[1]The suppression of low-intensity fires can lead to fuel accumulation, increasing the severity of subsequent wildfires (Mateus and Fernandes 2014; Collins et al. 2013; Kreider et al. 2024).

preventive measures and fuel management, such as prescribed burning and agro-silvo-pastoral practices (Pais et al. 2023; Oliveras Menor et al. 2025; Mauri et al. 2023). Within the broader context of integrated forest management, it is both reasonable and necessary to incorporate wildfire response considerations into forest road network planning. As Stefanović et al. (2016) noted, forest roads are the backbone of ground-based firefighting access:

> *"The importance of forest roads in fire protection is great, because they represent the basic infrastructure necessary for effective fire prevention and suppression."*

Developed forest road networks facilitate faster fire response and are associated with reduced fire size (burned area) – but they also increase the risk of anthropogenic wildfire ignitions due to improved public access (Viegas and Ribeiro 2022; Thompson, Gannon, et al. 2021; Narayanaraj and Wimberly 2012; Pinto et al. 2020). In a study on the influence of forest roads on the spatial patterns of human- and lightning-caused wildfire ignitions, Narayanaraj and Wimberly (2012) found that although most fires occurred in roaded areas, they accounted for only a small proportion of the total burned area, whereas a few large fires, typically lightning-induced in roadless areas, accounted for most of the burned area. Similarly, in an analysis on the impact of several landscape features on the occurrence and size of forest fires, Pinto et al. (2020) observed that while higher road density and more firebreaks were associated with increased fire occurrence, these features also correlated with smaller fire sizes. The authors suggested a link to the improved fire suppression: Shorter travel distances between ignition points and fire brigade locations can reduce response time and enable earlier control, whereas sparse road networks lead to a longer period of uncontrolled spread, resulting in larger burned areas.

In areas without road access and beyond the operational reach of fire vehicle pumps, fire suppression at the front line requires either aerial deployment, ground crews advancing on foot[2], or a strategic "awaiting"[3] of the fireline until it reaches accessible terrain -— at which point roads can act as natural control lines and support corridors. Where such infrastructure is lacking or distant, this strategy allows fires to spread unhindered until ground-based firefighting becomes feasible — significantly increasing the risk of large-scale ecological and economic loss. While aerial suppression can effectively support containment efforts — and enjoys broad public and media appeal — it has notable limitations, including high costs, limited availability, and operational constraints (Peña et al. 2022; Thompson, Calkin, et al. 2013) such as dependence on

[2]This reliance on manual methods limits the amount and type of equipment deployable and increases physical risks for personnel.

[3]This does not mean inaction by fire responders; it refers to indirect tactics such as preparing containment zones, conducting backburns, creating defensible space, staging resources, and evacuating at-risk areas.

weather conditions and is not a sufficient substitute for accessible, strategically located roads. Yocom et al. (2019) identified distance to roads as the most influential variable in predicting fire perimeters, noting that approximately 26% of observed perimeters aligned with existing roads. This suggests that responders routinely utilize roads as containment boundaries in areas with adequate road infrastructure.

> *"The first thought as forest engineers is to use constructions that already exist (road network, water springs, etc) so as to become useful for forest protection and fire fighting."* (Stergiadou 2014)

Building on this idea, the importance of incorporating wildfire protection and response considerations into the early stages of forest infrastructure planning becomes evident. In fire-prone landscapes, road infrastructure is not only essential for access, but also serves as a strategic asset for wildfire control. When road networks are planned and developed with dual purposes in mind, they can significantly enhance the effectiveness of suppression efforts. Roads designed to support wildfire response enable fire crews to intervene earlier and more safely, reducing the risk of fires escalating beyond containment capacity and causing extensive ecological or economic damage. Given the increasing frequency and severity of wildfires, the question is no longer whether roads can serve multiple purposes — but why they are not routinely designed to do so.

1.2 Integrating Wildfire Risk in Forest Road Network Planning – An Overview of Related Work

Integrating wildfire risk into forest road network planning represents a small, yet existing field of research. The crucial role of forest roads in wildfire management is well documented: there is broad agreement that early suppression efforts are essential for containing wildfire spread and that forest roads provide the critical access needed for both prevention and emergency response (Viegas and Ribeiro 2022; Thompson, Gannon, et al. 2021; Majlingova 2012; Stefanović et al. 2016; Stergiadou 2014; Laschi et al. 2019). To the best of the author's knowledge, no existing method directly addresses the selection of road segments for construction from a predefined set of alternatives while explicitly accounting for wildfire risk. Nonetheless, a variety of related research efforts — ranging from Geographic Information System (GIS)-based spatial analyses to multi-criteria decision-making tools and spatial decision support systems — incorporates wildfire considerations in forest road network planning or offers valuable conceptual foundations. These approaches, while not directly solving the constrained selection problem, inform how spatial wildfire vulnerability and operational objectives can be jointly considered. Some of these contributions are presented below.

Liampas et al. (2019) observed that while forest opening up was initially driven by the need for timber transportation and connectivity between forest settlements, it is increasingly acknowledged as a vital preventive measure for forest protection, particularly within the context of multipurpose forestry in the Greek Mediterranean mountains. In response to this shift in perspective, they proposed a method for forest road planning that employs least cost path analysis and multi-criteria analysis within a GIS environment. Although wildfire risk was not explicitly integrated into their model, the criteria and indicators used in the multi-criteria analysis were informed by a previous work developing a forest fire risk zone map based on satellite imagery and GIS data (Dong et al. 2005).

Psilovikos et al. (2009) examined the role of forest roads in enhancing wildfire suppression effectiveness in the suburban forests of Thessaloniki, Greece. Focusing on two major forest roads that had been the subject of environmental and functional debates, the authors assessed their contribution to fire protection in terms of spatial coverage and critical response time. Specifically, they evaluated whether fire-fighting vehicles traveling along these roads could cover the entire forest area and reach ignition points within the crucial 15-minute window considered essential for successful early fire suppression. Their findings emphasized the significance of strategically placed roads in achieving rapid response and effective protection while also addressing the need to balance technical efficiency with environmental compatibility, and offers practical and theoretical guidance for improving forest road networks as an integral component of wildfire defense planning.

Kravanja and Potočnik (2007) examined the role of Forest Fire Prevention Roads (FFPRs) as a proactive fire protection measure in the Kras Forest Management Region in Slovenia, an area characterized by high wildfire risk due to climatic, geological, and anthropogenic factors. They found that although FFPRs are critical for enabling access for emergency interventions, the existing and even the planned FFPR network in the Kras I Forest Management Unit falls short of providing sufficient openness to support modern fire suppression techniques. Their work highlights the importance of aligning road density and layout with fire-prone zones to ensure effective protection, underscoring the spatial mismatch that can exist between planned infrastructure and operational firefighting needs.

Akay et al. (2017) delineated “fire-access zones” in Turkey by combining terrain slope, road position, and firetruck hose-reach capacities. Their GIS analysis highlighted areas beyond current access that warrant new road construction. Eastaugh and Molina (2012) introduced a GIS metric comparing reference fire intensities at roads and fuelbreaks to forest averages. They found that roads generally coincide with average-risk zones, whereas fuelbreaks often intersect higher-intensity areas, guiding targeted siting and maintenance. Zhang et al. (2020) proposed a new firefighting distance criterion to evaluate the actual firefighting coverage of the road network and an approach that aims at mapping the optimal forest road network based on the firefighting distance criterion.

Drosos et al. (2014) developed an optimization model for the W. Nestos forest (Greece) that uses uphill/downhill buffer zones (150 m/250 m, 200 m/400 m, 300 m/500 m) around fire-truck positions to quantify protected area given hose-length limits. Their GIS analysis showed that, despite effective coverage by 4×4 patrols with 500 m hoses and early warning systems, expanding road density is economically warranted to ensure comprehensive fire protection and efficient timber operations.

Stefanović et al. (2016) evaluated four forest road network variants in Serbia using a multi-criteria decision-making framework that incorporated fire frequency data, the behavior of a recent large wildfire, and seven criteria weighted through the entropy weight coefficient method. These criteria were selected to reflect trade-offs between construction costs, forest accessibility, and the effectiveness of firefighting operations. Their findings revealed that although one road layout was more expensive from a construction cost perspective, its superior capacity to reduce fire-related tree losses justified the investment as the avoided damage costs would outweigh the initial expenditure.

These contributions are highly relevant; yet there remains a need to further support decision-making in forest road network planning in a more practical manner, like already done in other areas: For instance, Sakellariou et al. (2020) presented a spatial decision support system that allocates fire vehicles based on critical response times and dynamically reroutes them around evolving hazards, which provides a practical decision-making tool for emergency services. By contrast, no comparable tool exists that integrates wildfire resistance considerations into forest road network planning. What is still lacking is a decision support tool targeted at timber forest managers and addressing an earlier, fundamental step: guiding construction and maintenance decisions for dual-use forest roads that serve both timber harvesting and fire vehicle access. Such a tool would assist in prioritizing which roads should be built or upgraded to a wider standard, thereby integrating fire vehicle access to critical low-resilience areas directly into the road planning process.

1.3 Problem Statement, Research Objective and Contributions

Although integrated forest management practices include fuel reduction, ecosystem restoration, and other preventive measures, forest road network design does not typically account for access requirements of ground-based fire suppression resources. For timber forests, no comprehensive framework currently integrates multiperiod economic cost minimization with timber and fire vehicle accessibility requirements, nor offers a decision support tool for dual-purpose road planning. Existing optimization models for forest road network planning focus primarily on minimizing construction and maintenance costs or maximizing timber-harvesting efficiency (e.g., Mesquita et al. 2022). In contrast, this master's dissertation develops a decision support tool for forest managers that, over multiple harvesting periods, identifies road segments for construction and maintenance, thereby recommending a cost-efficient road network that supports timber harvesting when and where needed, while also ensuring fire vehicle access to the least fire-resistant areas.

Problem Statement

Mediterranean-type timber forests, where climate and topography amplify fire risk, require road infrastructure that simultaneously supports timber extraction and wildfire containment. This work addresses a case study in the fire-prone Vale do Sousa[4] region (described in detail in section 1.4), where private forest owners engage in collaborative planning of timber harvests over ten-year periods. In each period, some forest stands (also called management units) are scheduled for harvesting and therefore require access roads wide enough to accommodate timber extraction operations. These roads may either be newly constructed in that period or maintained if built in earlier periods. At the same time, certain stands are less wildfire-resistant than others, and the least fire-resistant zones require wide access roads to enable effective ground-based firefighting in the event of a nearby fire.

With the goal of ensuring equitable cost distribution among landowners, road construction is limited to stand boundaries. The selected road network should be optimized to minimize total construction and maintenance costs.

[4] https://www.valsousa.pt/valsousa/o-vale-do-sousa

Research Objective

The aim of this Master's dissertation is to develop a linear optimization model for selecting road segments along stand boundaries for construction and maintenance across five ten-year periods, which:

- minimizes total economic costs of road construction and maintenance over all periods,
- ensures mechanized harvesting access according to the harvesting schedule,
- provides ground-based fire vehicle access to stands with low fire resistance,
- while taking into account two different road width options (small ones suitable for timber harvest and transport, and wider ones suitable for fire-fighting operations).

Contributions

To the best of the author's knowledge, this work introduces a novel linear integer programming model that integrates fire vehicle accessibility into cost-optimized forest road network planning, bridging timber logistics and wildfire risk management. Building upon traditional models focused solely on timber harvest and transport cost minimization, this approach explicitly incorporates constraints ensuring fire vehicle access to stands with low fire resistance[5].

The model supports multi-period planning, capturing the dynamic interplay between forest growth and fire risk. Its application to the study region Paiva demonstrates how stand-boundary constraints and collaborative management can be aligned with wildfire response goals. In the context of increasing wildfire frequency and intensity in Mediterranean-type ecosystems, this framework offers a practical tool for climate-resilient forest infrastructure planning.

In light of prior research, key contributions include:

- **Integrated Modeling Framework**: Combining cost efficiency, multi-period timber harvest accessibility and firefighting accessibility in a single optimization model.
- **Fire-Adapted Constraints**: Novel inclusion of fire vehicle access requirements for vulnerable stands, based on stand-level fire resistance indicators.

[5] The designation of these high-risk zones is based on the stand-level fire wildfire resistance indicator developed by Ferreira et al. (2015) and implemented in form of a quantile threshold. The level of this threshold is to be adjusted by forest managers to determine the degree of emphasis placed on wildfire protection (i.e., how many areas are classified as high risk and therefore prioritized for wider road access).

- **Real-World Application**: Case study in Castelo da Paiva illustrating the integration of timber road planning with wildfire resilience considerations.

1.4 Description of the Study Area

Study area for this research is the Zona de Intervenção Florestal (ZIF)[6] Paiva[7], located in Portugal's northwestern Aveiro district near the Douro River. The ZIF Paiva encompasses approximately 7,620 hectares and is representative of the forest structure in Northern Portugal, characterized by small and scattered stands and a high incidence of wildfires. Its administrative scope includes the entirety of the municipality Castelo de Paiva with all its parishes. The Associação Florestal de Vale do Sousa (AFVS)[8] serves as the managing entity, overseeing the participation of 185 registered members. Paiva lies within the broader region of Vale do Sousa (see fig. 1.1), which serves as a Living Lab for the Horizon 2020 project FIRE-RES (2021–2025), aimed at increasing wildfire resilience across European landscapes. (Ministério da Agricultura, do Desenvolvimento Rural e das Pescas 2008; AFVS – Associação Florestal do Vale do Sousa 2024)

Vale do Sousa extends over 14,316 hectares and comprises small-scale properties. The current dominant tree species include *Eucalyptus globulus* (blue gum) and *Pinus pinaster* (maritime pine) – both of which are fire-prone and managed primarily for timber production. Apart from forestland, the landscape composition features agricultural plots, urban regions, as well as shrubland and abandoned agricultural areas. Stakeholders include nonindustrial private forest owners, forest industry actors, investment funds, and municipal entities - all coordinated under the umbrella of the AFVS and collaboratively managed by over 360 landowners.

The region is highly vulnerable to wildfires: Approximately 46% of the ZIF has burned between 2014-2023, highlighting the urgent need for more strategic and integrated risk-reduction strategies. In response, stakeholders are implementing a landscape-scale joint management plan that seeks to balance wildfire mitigation with sustained forest productivity and ecosystem services. (FIRE-RES 2023)

Despite its wildfire exposure, Paiva is a touristically attractive region known for its scenic landscapes and cultural heritage.[9]

[6]For details on the concept of forest intervention zones in Portugal, see `https://www.icnf.pt/florestas/zif`
[7]For details on the forest intervention zone 075/07 — Paiva, see Portaria n.º 1515/2008
[8]English: Association of Forest Producers of Vale do Sousa
[9]To access a tourist map of Paiva, see this post on the Castelo de Paiva Turismo facebook page.

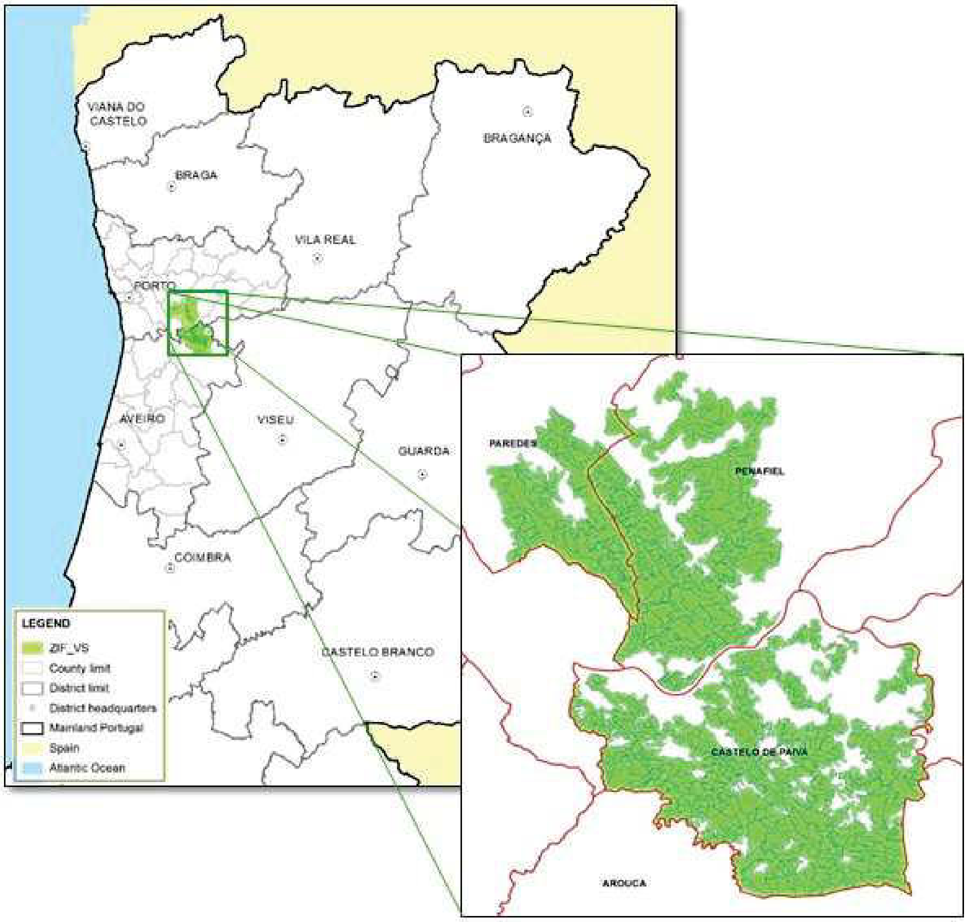

Figure 1.1: Location of ZIF Vale do Sousa. Source: Marques, Marto, et al. (2017)

1.5 About the Structure of this Document

This document comes in five chapters: Following the introductory chapter which introduces the reader to the research context, problem statement and proposed solution, chapter 2 provides a detailed overview of the data used, including characteristics, preprocessing steps and challenges and outlines the implementation details of the preprocessing. In chapter 3 a single-objective integer programming model is developed and the solution approach is described, while chapter 4 presents and discusses the resulting solution, i.e. the forest road network design considered optimal under given constraints for the case study area, as well as potential limitations. Concluding with a summary of the main findings and contributions of the work and some indications on potential further research, chapter 5 constitutes the final part of this work.

2

Data

Contents

THIS chapter presents the data used in this work and describes the preprocessing steps necessary to transform it into a suitable input for the Integer Programming (IP) model. Each section within this chapter covers a different dataset, all of which are essential to addressing the problem. These sections are further organized into subsections, beginning with an introduction that presents the data source and a general data description, followed by subsections on preprocessing performed. Each section ends with a brief summary of the most relevant preprocessing results.

As a foundation for this study, we begin by analyzing the geospatial data relevant to the problem. This includes an examination of the existing public road infrastructure (section 2.1) and the forest management units (section 2.2) within the study area. A key aspect of this analysis is assessing the spatial relationships between existing roads and forest stands to determine which stands are already accessible and can therefore be excluded from the optimization model. Additionally, we review the harvesting schedule data (section 2.3), identifying the relevant information for this study and ensuring its compatibility with the subsequent modeling steps. Finally, we estimate the costs associated with constructing, maintaining, and upgrading forest roads on a per-kilometer basis (section 2.4). This structured approach transforms the data provided into the necessary foundation for this project. Then, the problem was decomposed into smaller, more manageable subproblems through spatial segmentation (section 2.5). Finally, the graph structure representing the network potential forest road segments was created (section 2.6).

Before delving into the data and the preprocessing steps undertaken to make it usable, some preliminary remarks regarding the scope and relevance of the data discussed might help the reader's understanding of the presented extracts: In all datasets used for this study, a wide range of attributes is available, as the data were originally collected for various purposes beyond the scope of this work. Many of these attributes pertain to other domain-specific aspects which, while valuable in their original context, are not directly relevant to the problem addressed here. Given the breadth of available information, it is neither practical nor necessary to present and discuss every attribute in detail. Instead, the following sections focus solely on the attributes that are essential for this study — those that contribute directly or indirectly to the analysis and modeling performed.

Regarding the preprocessing presented, only the key steps are outlined here for the sake of brevity and readability. Standard cleaning operations, validation checks, and other routine procedures have been omitted but can be found in the accompanying Jupyter notebooks, along with the full list of libraries used. Data manipulation and geospatial analysis were primarily conducted using the `pandas` and `GeoPandas` libraries, with mostly generated through Matplotlib. Supplementary libraries were incorporated as required throughout the process.

2.1 Public Road Infrastructure (Geospatial Data)

The study area is intersected by public roads which serve as a critical component of the transportation infrastructure for timber and ground-based firefighting forces within the region. This data is particularly relevant to our problem for two reasons: First, because some forest stands are already accessible via existing roads, which reduces the complexity of the problem by limiting the need for new road construction; and second, because any future road network must connect to this existing infrastructure to ensure continuity and accessibility.

This section provides a brief description of the road dataset, followed by an overview of the preprocessing steps applied. Key steps performed on this data included the removal of roads marked as ”inoperable” in the dataset (see 2.1.2), filling relevant data gaps (see 2.1.3), and distinguishing between two different road networks (see 2.1.4): one consisting of standard timber roads, and the other consisting exclusively of wider roads deemed suitable for fire vehicle access.

2.1.1 Introduction to the Road Dataset

The data on the roads considered in this study originates[1] from a geospatial dataset provided as a shapefile `RVF_CPv_clip.shp`, projected in the Lisboa Hayford Gauss IGeoE (ESRI:102164) coordinate system.[2] The dataset comprises 311 road segments in the municipality of Castelo de Paiva, each represented as a vector linestring. In total, the road network spans approximately 390 kilometers. Figure 2.1 visualizes the geometries of these segments.

Apart from the spatial geometries, the following attributes were used in this work:

Attribute	Description
`Id`	Unique identifier for the road
`COMPRIM`	Length of the road
`OPERAC`	Operational status of the road
`LARGURA`	Width of the road
`DESIGNACAO`	Designation or name of the road

[1] According to its provider (Susete Marques, PhD), this dataset was — prior to its inclusion in this work — clipped from a bigger dataset originally obtained from the Municipal Forest Office.

[2] A projected coordinate system typically used for mapping in mainland Portugal; see https://epsg.io/102164.

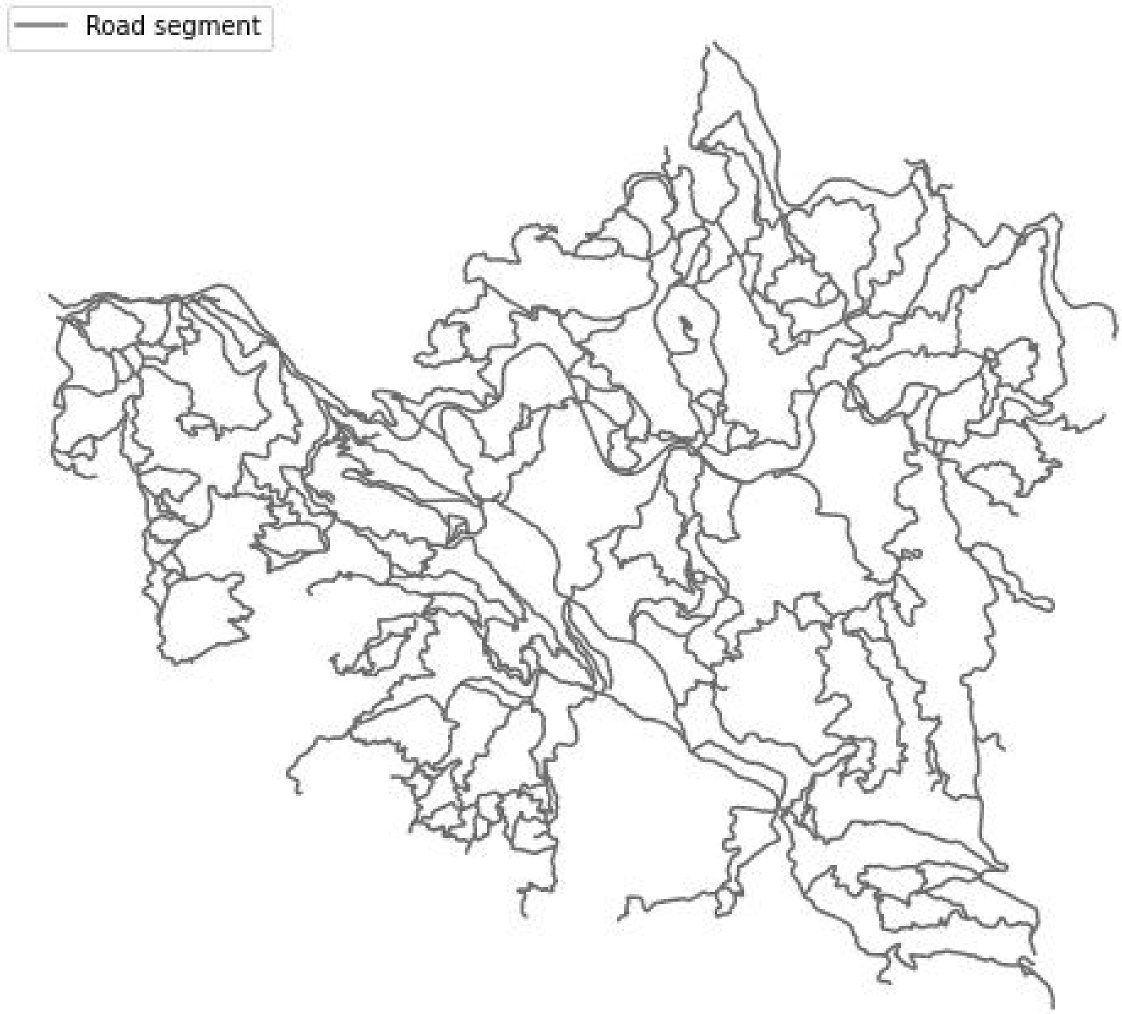

Figure 2.1: Road network in the study area of Paiva, prior to preprocessing

2.1.2 Removal of Inoperable Roads

Some roads in the dataset are marked as inoperable and can not be used. Only operable roads are relevant to this study. An assessment of road operability based on the `OPERAC` column revealed that 13 road segments are inoperable (visualized as red dotted lines in fig. D.1). As these roads do not contribute to the problem under investigation, they were removed from the dataset.

→ After this adjustment, 298 operable road segments remain with a total length of approximately 376 km.

2.1.3 Addressing Data Deficiencies: Road Width Values Reported as Zero

Upon examining the frequency distribution of the road width values (column `LARGURA`, see Table 2.1 and a corresponding plot in fig. D.2), it becomes evident that there are some ”holes” in the data. Out of 298 operable road segments, 70 segments have a reported width of 0 meters, which is clearly implausible. Expected road widths typically fall between 3 meters and 10 meters, and they should always be greater than zero. After further data exploration, these zero values were deemed erroneous and attributed to data quality deficiencies requiring correction.

Table 2.1: Frequency table for `LARGURA` with implausible values highlighted in orange

LARGURA [m]	0	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6.5	7	7.5
Count	70	1	54	38	45	22	54	5	5	1	1	1	1

Given the high proportion of missing values (70 out of 298 segments), the corresponding features could not be removed from the dataset. Instead, a new column `ROADWIDTH` was created, and new road width values were imputed for segments with missing data. An estimate for the missing values can be derived from the `REDE_DFCI` attribute, following the classification system defined in Categorias de classificação da rede viária florestal (Ministério da Agricultura 2014). According to these guidelines, **first-order roads** (*1.ª ordem*) should have a usable lane width of at least 6 m; **second-order roads** (*2.ª ordem*) should range between 4 m and 6 m; and **complementary roads** (*complementar*) can be narrower. Based on this classification:

- 67 segments (`REDE_DFCI` = 1) were assigned a width of 6 m;
- 1 segment (`REDE_DFCI` = 2) was assigned a width of 4 m;
- 2 segments (`REDE_DFCI` = 3) were assigned a width of 3 m.

This approach ensures a consistent and interpretable road width variable across the entire dataset, enabling more reliable analysis in subsequent modeling steps. The resulting dataset includes a complete `ROADWIDTH` column, as illustrated in fig. D.3.

2.1.4 Distinguishing Fire Roads from the Full Set of Roads by Width

To enable timber transport, roads must have a minimum width of 3 m. To accommodate two-way traffic for firefighting vehicles, a wider minimum width of 6 m is required, which calls for the creation of a separate subset of wide roads. While fig. D.4 shows the entire network of *"timber roads"*, fig. D.5 displays only the wider *"fire roads"*.

Note that forest stands adjacent to roads wider than 6 meters are fully accessible for both timber extraction and firefighting activities, and can therefore be excluded from the scope of this study. This will be addressed in the following section.

2.1.5 Summary of Road Data for Further Usage

Summarizing the relevant results from this section, in the area under study, we have:

- a **timber road network**, consisting of 205 roads wider than at least 3 meters suitable for timber transport (see fig. D.4),
- a **wider road network**, subset of the above timber roads, consisting of 70 roads wider than at least 6 meters suitable for both timber transport and firefighting traffic (see fig. D.5).

2.2 Forest Management Units (Geospatial Data)

Following the preparation of the existing road network, we now turn our attention to the forest management units (stands) within the study area — equally essential elements in the problem under investigation. A forest management unit is a defined area of forest land that is managed as a unit. These units are central to the planning of forest road infrastructure.

The significance of forest stands in the modeling framework for this problem stems from both their access requirements and spatial boundaries. First, the characteristics of each stand determine its specific needs for road access. These needs are incorporated as constraints in the linear programming model (see chapter 3, section 3.2.4.C), ensuring that any proposed road construction accounts for the stand-level factors of timber availability and fire resilience. Second, due to the fragmented ownership of stands across different entities, the design of future road infrastructure must respect stand boundaries to uphold property divisions and support coordinated management. In particular, this approach allows for a fair distribution of the area losses associated with road placement among different landowners, ensuring that no single entity bears a disproportionate impact on timber production or economic returns.

This section begins with a description of the dataset used (section 2.2.1) and the corrections necessary to make it usable (section 2.2.2), followed by an analysis of road accessibility of the forest stands (section 2.2.3) - because only inaccessible stands are part of the decision problem under study.[3]

2.2.1 Introduction to the Forest Management Units Dataset

The data on forest management units used in this study originates from a geospatial dataset provided in the form of a shapefile `CasteloPaiva.shp`, made available by Susete Marques, PhD. According to its provider, the dataset was derived from Pléiades satellite imagery with a spatial resolution of 0.5 meters. Each forest management unit is represented as a polygon (vector data). The dataset is projected in ETRS89 / Portugal TM06 (EPSG:3763)[4]. It comprises **687 forest management units** that vary in shape and size (see fig. 2.2). Each unit (stand) is homogeneous with respect to species composition, slope, age and site index (soil quality). For the purposes of this study, the only attributes used — aside from the spatial geometry — are

[3]Nevertheless, all stands are relevant for the creation of the road network.

[4]Standard projected Coordinate Reference System (CRS) for mainland Portugal, for details see `https://epsg.io/3763`.

ID_UG, the unique identifier for each management unit, and Declive, representing the stand's average slope.[5]

Figure 2.2: Castelo da Paiva: Forest stands

2.2.2 Spatial Correction and Topology Cleaning of Stands Data in QGIS

During verification in QGIS using multiple baselayers (e.g., Google Maps, OpenStreetMap), a noticeable spatial misalignment in the dataset became evident. The geometries of the forest stands appeared systematically shifted relative to recognizable geographic features such as rivers and roads. In the absence of accompanying metadata and after unsuccessful reprojection attempts, the issue was ultimately resolved by **manually shifting** the entire layer. This

[5] According to the data provider, the polygons were designed to have relatively homogeneous slopes.

correction was particularly important, as subsequent analyses involve spatial relationships between forest stands and the road network, for which accurate alignment is essential.

Following this spatial realignment, additional topological inconsistencies were identified. Visual inspection and the QGIS Topology Checker tool revealed numerous small gaps and overlaps between adjacent polygons - errors that could compromise spatial operations such as intersection and buffering. Specifically, the tool flagged **165 overlaps** and **709 gaps**, a quantity impractical to address manually. To resolve these issues, the following steps were undertaken:

1. **Snapping:** The dataset was snapped to itself using a 1-meter tolerance to improve the alignment of adjacent polygon boundaries.
2. **Geometry Validation:** Invalid geometries were identified and repaired using QGIS's Fix Geometry.
3. **Second Topology Check:** A second run of the Topology Checker reported **84 gaps** and **32 overlaps**.
4. **Manual Inspection and Cleaning:** Upon review, **65 gaps** were retained as legitimate features (e.g., clearings or administrative boundaries), while **19 gaps** and **all 32 overlaps** were manually corrected using the Vertex Tool to ensure topological consistency.
5. **Multipolygon Correction:** During the previous steps, **two multipolygon features** were inadvertently created. These were manually inspected and compared with the original geometries, then corrected back into simple polygons using the Vertex Tool to ensure consistency across the dataset and simplify further data processing.

These corrections were essential for preparing the data to be suitable for the spatial operations and optimization modeling conducted in later stages of this work - particularly for the creation of the network of potential roads along the stand boundaries (see section 2.6).

2.2.3 Classification of Forest Stands by Road Accessibility

In this work, a road is considered suitable for timber logging and transport operations if it has a minimum width of 3 meters, hereafter referred to as a "*timber road*".[6] A stand that is adjacent to

[6]This definition does not account for legal ownership constraints, nor for other physical or environmental limitations that may affect actual usability. All roads in the provided dataset that are marked as operable and have a width of at least 3 meters are considered suitable for timber logging.

a timber road (i.e., one with a minimum width of 3 meters) is considered "*timber-accessible*".[7] A stand that is adjacent to a wide road (i.e., one with a minimum width of 6 meters) is considered "*fully accessible*". The decision problem addressed in this study includes only those stands that are currently not timber-accessible, hereafter referred to as *"inaccessible"*.[8]

A corresponding spatial analysis of the relationships between forest stands and the surrounding road network was performed using the `geometry.intersects` method from GeoPandas, which determines whether two geometric objects share any points or areas. To ensure spatial accuracy and address initial data imprecision, preprocessing steps included aligning the CRSs of the datasets and applying a 6-meter spatial buffer around the stands. In a first classification step, a set of **203 fully accessible stands** (see fig. D.6 in appendix D.2, shown in green) was identified.[9]

In a second step, the remaining 484 stands not accessible via wide roads (orange-hatched in fig. D.6) were further classified. Figure D.7 shows the resulting sets: There are **169 timber-accessible stands** (light green) and **315 inaccessible stands** (orange cross-hatched).

Only the 315 inaccessible stands (orange cross-hatched in fig. D.7 and solid orange in fig. D.8) are considered in the optimization. However, all stands remain relevant for the creation of the road network, as roads should follow stand boundaries.

2.2.4 Summary of Stand Accessibility Classification

The accessibility analysis yields the following classification:

- **315 inaccessible stands** (see fig. D.8)
  ➔ These are included in the optimization model; their access needs are to be extracted from the harvesting schedule data (see section 2.3).
- **372 stands that are already accessible**
  ➔ Excluded from the optimization, but relevant for geometry since stand boundaries constrain road placement.

[7]Since wider fire roads are a subset of timber roads, stands adjacent to them are also timber-accessible.

[8]Determining which narrow roads should be widened to improve firefighting access constitutes a separate planning problem and falls outside the scope of this study (discussed in chapter 4).

[9]Since both timber extraction and firefighting access to these stands are already supported by the existing road network, their access needs are considered fulfilled and must be excluded from the decision problem. This was done by filtering out the corresponding prescription data (see section 2.3). This exclusion reduces the required road construction, thereby lowering overall costs and simplifying the optimization by reducing its size and complexity.

2.3 Harvesting Schedules (Tabular Data)

This section outlines how the provided harvesting schedules - detailing when, where, and how much timber is to be removed - inform the road access requirements used in the optimization model. Starting from three pre-defined harvesting schedules (referred to as “solutions”),[10] we determine, for each 10-year-period, which stands require road access and what type (timber road vs. wider fire road). Although this information is not explicitly included in the original datasets, it can be systematically inferred through a sequence of processing steps. After reformatting the stakeholder solution to match the structure of the other two (section 2.3.1), each solution was then merged with the full prescription dataset to incorporate the key attributes `Vremovido`, `rait0/rait5/rait10`, for each `period` (see section 2.3.2). Fire resistance values were assigned based on shrub cleaning periodicity (`ShPer`), which may occur every 5, or 10 years, or not at all (section 2.3.3). Special cases and anomalies in the data were identified and analyzed (section 2.3.4). Finally, the dataset was filtered to include only stands that are part of the decision problem, based on the results of the accessibility classification from section 2.2. Ultimately, the road access requirements for each stand and period were derived (section 2.3.5).

2.3.1 Introduction to the Harvesting Schedules Dataset

The harvesting schedules used in this study were developed by Susete Marques, PhD, Olha Nahorna, and Felipe Silva, based on data collected in the fall of 2019 and processed in 2020. These schedules represent the outcome of simulations of multiple forest management alternatives, using growth anulled simulators/yield-tables (e.g., SIMFLOR; for more details, see appendix B) developed in collaboration with stakeholders from the AFVS. Each management alternative is referred to as a ”solution,” with every solution assigning a specific prescription to each forest stand.

The dataset includes **three alternative harvesting schedules**, each representing a distinct management strategy:

- **MaxWood solution** — optimized for maximum wood production,
- **MaxRes solution** — optimized for maximum fire resistance,

[10] These were generated through a prior optimization process conducted independently of this study.

Stakeholder solution — selected by stakeholders[11].

All three schedules cover the same set of forest management units and share a 50-year planning horizon, divided into five 10-year periods. The dataset was provided in the form of an Excel workbook Presc_dataset21_missingstakeholders_sol.xlsx containing three worksheets, along with a separate log file:

- Presc — includes the full set of possible prescriptions (section 2.3.1.A);
- MaxWood — details the MaxWood solution (section 2.3.1.B);
- MaxFireRES — details the MaxRes solution (section 2.3.1.C);
- Stakeholders_Solution.log — logfile containing the stakeholder-selected solution (section 2.3.1.D).

Please note that the harvesting schedule data includes stands not only within Paiva (the study area for this work, introduced in section 1.4 and with corresponding stands detailed in section 2.2) but also from the neighboring ZIF Entre Douro e Sousa. Since this Master's dissertation focuses exclusively on Paiva, stands outside the study area must be excluded. However, to ensure a comprehensive understanding of the dataset and maintain consistency during preprocessing, the harvesting schedule data were initially analyzed in full. Most preprocessing steps were applied to the complete dataset, with filtering of out-of-scope stands performed only at the final stage.

2.3.1.A Prescription data

The prescription data Presc is extensive, containing 41 attributes (columns) and 317,883 entries (rows), where each row corresponds to a possible prescription for a forest stand in a given period. Most of the columns include information irrelevant to the current problem, and a large portion of the rows represent prescriptions not selected in any solution. Therefore, only a subset of this data was used in this project. The attributes relevant to the problem under study are detailed in the grey box below and exemplified by the excerpt in Table 2.2 in the appendix.

[11] According to the data provider, this selection was based on a Pareto front involving four optimization criteria (carbon, biodiversity, fire resistance and erosion), selected by the stakeholders in a 2021 meeting in Oporto

Remark. The reader may notice that some attribute names contain spaces, such as `Na sol MaxRES`. While this formatting is not ideal for programmatic handling and such spaces should ideally be removed, the names are shown here as-is to reflect the original dataset and to improve readability in the accompanying exemplary table extracts.

Attribute	**Description**
`UG`	Unique identifier for the forest stand
`presc`	Unique identifier for the prescription
`species`	Species prescribed for the stand
`area`	Area in hectares occupied by the species
`ug_Sp`	Control column for data consistency (`UG_species`)
`Control`	Control column for data consistency (`UG.presc`)
`Na sol MaxRES`	Indicator whether the prescription is included in MaxRES solution
`Na sol MaxWood`	Indicator whether the prescription is included in MaxWood solution
`period`	Harvesting period
`vthin (ton)`	Volume of the stand thinned in tons
`vharv (ton)`	Volume of the stand harvested in tons
`Vremovido (ton)`	Total biomass removed (in tons)
`rait0`	Fire resistance indicator[1] for no occurring shrub cleaning
`rait5`	Fire resistance indicator[1] for shrub cleaning interval of 5 years
`rait10`	Fire resistance indicator[1] for shrub cleaning interval of 10 years

Table 2.2: Illustration of relevant columns from the prescription data sheet

UG	ug_Sp	area	presc	Na sol MaxRES	Na sol MaxWood	period	species	vthin (ton)	vharv (ton)	Vremovido (ton)	rait0	rait5	rait10
1	1_Ec	42.6018	1	#N/A	#N/A	1	Ec	0	2999.2720	2999.2720	132.0655	166.147	136.3257
1	1_Ec	42.6018	1	#N/A	#N/A	2	Ec	0	5575.8278	5575.8278	140.5859	166.147	140.5859
...	...	...	...	...	...	...	...	...	...	...	...	...	...

Full table size: 317,883 rows across 41 columns.

2.3.1.B MaxWood Solution

The `MaxWood` solution was provided in tabular format, comprising 1,512 entries across the seven columns described in the grey box below. This dataset, illustrated in Table 2.3, specifies the prescriptions selected under the MaxWood scenario — that is, the set of management actions aimed at maximizing wood production.

[1] Deeper explanation of the `rait` values: They originate from a fire resistance indicator calculated in prior work (methods of Ferreira et al. 2015, adjusted and applied to Vale do Sousa by Marques, Marto, et al. (2017)). Ranging from 0 to 1, the indicator was multiplied by the stand's area, producing the present numerical `rait` values in value range $\mathbb{R}^+$.

Attribute	**Description**
`Decision_Variable_Name`	Identifier of the decision variable, formed by concatenating `Presc`, `UG`, `Sp`, and `ShPer`
`Control`	Control attribute for data consistency (`UG.Presc`)
`Presc`	Unique identifier of the selected prescription
`UG`	Unique identifier of the forest stand
`Sp`	Prescribed species
`ShPer`	The prescribed periodicity of shrub cleaning (0, 5 or 10)
`Sol`	Indicator if prescription is included in the solution (`Yes`/`No`)

Table 2.3: Excerpt from MaxWood Solution sheet

Decision Variable Name	Control	Presc	UG	Sp	ShPer	Sol
Presc3502_Pa1_Ec_0	1.3502	3502	1	Ec	0	Yes
Presc19_Pa2_Ec_0	2.19	19	2	Ec	0	Yes
Presc21_Pa3_Ec_10	3.21	21	3	Ec	10	Yes
...	...	...	...	...	...	Yes

Full table length: 1512 rows

Remark. The length of the MaxWood solution table (1512 rows) does not correspond with the number of stands contained in it (1406 unique values in column `UG`), indicating the presence of stands that have two different prescriptions within one and the same harvesting plan. This will be further addressed in section 2.3.4.

2.3.1.C MaxRES solution

Similarly, the sheet ⊞ `MaxFireRES` identifies the selected prescriptions for the MaxRes solution — that is, the prescriptions recommended to maximize the fire resistance of the forest. It was also provided in tabular format (1,512×7), as illustrated in Table 2.4. The columns are identical to those in the MaxWood Solution sheet; for a description, please refer to the grey box above.

Table 2.4: Excerpt from MaxRES Solution sheet

Decision Variable Name	Control	Presc	UG	Sp	ShPer	Sol
Presc1601_Pa1_Ec_5	1.1601	1601	1	Ec	5	Yes
...	...	...	...	...	...	Yes

Full table length: 1512 rows

Remark. As with the previous solution, the number of stands does not match the row count: while there are 1,512 rows, the solution contains prescriptions for 1,406 stands, of which 106 have two prescriptions instead of one. The reason for this discrepancy is explained in section 2.3.4.

2.3.1.D Stakeholder Solution

The stakeholder solution was provided in a different format — essentially as raw output in the form of a logfile generated directly by the optimizer. To make the data usable, the decision variable names and their corresponding solution values were extracted from the file and converted into a tabular format (see Table 2.5 below at the left). At this stage, the dataset contained only the two columns described in the grey box below. The stakeholder solution data contains 1,514 rows.

Attribute	**Description**
`Variable Name`	name of the decision variable, concatenating `Presc, UG, Sp, ShPer`
`Solution Value`	fraction of area of stand `UG` which is assigned to prescription `presc`

The data in the `Variable Name` column are not immediately usable, as they contain concatenated identifiers — specifically the prescription ID, forest stand ID, species, and shrub cleaning periodicity (`Presc`, `UG`, `Sp`, and `ShPer`). To enable further processing, these components had to be parsed and restructured into separate attributes, matching the format used in the MaxWood and MaxRes solutions. This was accomplished using regular expressions to extract the relevant elements: the numeric values following the prefixes “Presc” and “Pa” were assigned to the `Presc` and `UG` columns, respectively, and the shrub cleaning periodicity (`ShPer`) was also extracted. A species column was not created at this stage, as that information already exists in the complete prescription dataset, which would later be merged with each solution. An excerpt of the resulting structured data is shown in Table 2.6 at the right as an illustrative example.

During the transformation of the stakeholder solution, two management units were identified that contained multiple prescriptions in the same period, with fractional solution values between 0 and 1 (see Table 2.7). Consultation with the subject-matter expert indicated that the solution space was formulated using continuous decision variables bounded within $[0, 1]$, where each variable denotes the proportion of a stand’s area allocated to a specific prescription. Consequently, the fractional values for each management unit (`UG`) collectively sum to 1, representing full area coverage.

Table 2.5: Stakeholder solution (from logfile)

Variable Name	Solution Value
Presc503_Pa101_Pb_10	1.000000
Presc503_Pa1015_Pb_5	1.000000
Presc500602_Pa1016_Pb_0	1.000000
...	...

Full table length: 1514 rows

Table 2.6: Stakeholder solution (processed)

Presc	UG	ShPer	Solution Value
503	101	10	1.000000
503	1015	5	1.000000
500602	1016	0	1.000000
...	...	...	...

Full table length: 1514 rows

Table 2.7: Fractional solution values in stakeholder solution

UG	Presc	ShPer	Solution Value
1253	36503	10	0.494295
1253	36	5	0.505705
1339	19700	0	0.567668
1339	19318	0	0.432332

Before delving deeper into the further processing, let us define the intended outcome:

2.3.1.E Objective of Preprocessing the Harvesting Schedule Data

The goal is to identify, for each period, the **set of stands that require road access** and determine the **appropriate road width**. This relies on two key stand-level attributes:

- **Timber removal volume (`Vremovido`)** – Any stand scheduled for timber harvesting or thinning necessitates a path connected to the road network.
- **Fire resistance indicator (`rait0/rait5/rait10`)**[a] – Stands with fire resistance below a predefined threshold need a wider road to ensure access for firefighting.

[a] The variable to be used is the one corresponding to the shrub cleaning periodicity prescribed.

2.3.2 Integrating Prescription Data into the Solutions

Each of the three solutions must be integrated with the prescription data to extract essential attributes: `Vremovido`, `rait0`, `rait5`, `rait10`, and the corresponding harvesting `period`. This was accomplished by merging the relevant prescription data with each solution.

For the MaxRes and MaxWood solutions, the integration process began by filtering the prescription dataset to retain only those rows where the columns `Na sol MaxRES` or `Na sol MaxWood`, respectively, indicated inclusion in the solution. Then, `pandas.merge` was used with an outer join to combine the filtered prescription data with the corresponding solution, matching rows based on the `UG` and `Presc` columns. This outer join ensured that all rows from both tables were retained. The results of these merges are exemplarily illustrated in Table 2.8 and Table 2.9, with only the relevant columns shown.

Unlike the other solutions, the stakeholder solution lacks an explicit indicator in the prescription data identifying which prescriptions are included. As a result, no filtering was performed prior to merging. Instead, an outer join was applied directly, and any rows containing *NaN* values in the merged result were removed in a subsequent step. The resulting merged stakeholder solution is exemplarily illustrated in Table 2.10.

Table 2.8: MaxRes solution merged with prescription data

UG	Presc	species	period	Vremovido (ton)	rait0	rait5	rait10	ShPer
1	1601	Ec	1	2999.2720	110.7646	115.0248	85.2036	5
...	...	...	...	...	...	...	...	...

Length: 8212 rows

Table 2.9: MaxWood solution merged with prescription data

UG	Presc	species	period	Vremovido (ton)	rait0	rait5	rait10	ShPer
1	3502	Ec	1	3968.3560	136.3257	149.1062	119.285	0
...	...	...	...	...	...	...	...	...

Length: 7670 rows

Table 2.10: Stakeholder solution merged with prescription data

UG	Presc	species	period	Vremovido (ton)	rait0	rait5	rait10	ShPer
69	1	Ec	1	2999.2719	110.7646	115.0248	85.2036	5
...	...	...	...	...	...	...	...	...

Length: 8403 rows

The reader might wonder why, starting from 1,512 selected prescriptions in each solution (1,514 for stakeholder), the integration with the prescription data results in tables of completely different lengths — none of which equal the product of the number of prescriptions times the number of periods:

$$\underbrace{1512}_{\text{\# prescriptions}} \times \underbrace{5}_{\text{\# periods}} = \underbrace{7560}_{\text{expected length of integrated solution/prescription data}}$$

This discrepancy arises because the harvesting schedule data includes special cases where, after a total harvest, a different species is planted. In these cases, two rows of data are generated for the period of change instead of one, which explains the increased number of rows. A detailed explanation of the impact of such special cases on the data is provided in section 2.3.4.

With these nuances accounted for, the integration process ensures that all necessary information is now in place, setting the stage for the next analytical step and allowing us to confidently assign the appropriate fire resistance values based on the harvesting schedules data.

2.3.3 Assigning the Appropriate Fire Resistance Value

Each prescription specifies a shrub cleaning periodicity (`ShPer`) — either 0, 5, or 10 years — which directly affects the stand's fire resistance. Correspondingly, the dataset includes three fire resistance indicators: `rait0`, `rait5`, and `rait10`. Since decisions on road access requirements depend on a single fire resistance value per stand, only one of these indicators can be retained and used for each record.

To address this, a new column, `Rait`, was created. For each record, the appropriate fire resistance value was assigned by selecting the fire resistance indicator that corresponds to its specified `ShPer`, as illustrated in Table 2.11.

Table 2.11: Visualization of assigned fire resistance values (`Rait`) based on shrub cleaning periodicity (`ShPer`), selected values highlighted in yellow

ShPer	rait0	rait5	rait10	Rait
...	...	...	...	...
5	6.5650	7.8357	6.7768	7.8357
0	136.3257	149.1062	119.2850	136.3257
...	...	...	...	...

With all relevant attributes now available in the dataset for each stand and period, the data is nearly ready for use in the model. However, before moving on to formatting and storage considerations (see section 2.3.5), it is important to address a few special cases present in the data — namely, species conversions and mixed stands.

2.3.4 Notable Special Cases: Species Conversions and Mixed Stands

In preparing the dataset for model input, two noteworthy structural patterns emerged that require to be addressed: **species conversions** and **mixed stands**. Although they may appear similar at first glance — both involving more than one species in a stand — their data structure and implications are fundamentally different.

Each case is described in detail below, followed by an assessment of its impact on the size of the merged datasets.

Species Conversions

A species conversion occurs when, within a single prescription and stand, different species are listed for the same period.[12] This reflects a harvest-replant sequence within the 10-year period: the original species (e.g., *Ec*) is harvested and then replaced by a new species (e.g., *Ct*). These rows represent consecutive steps within the same prescription.

Table 2.12 illustrates an example: During period 1, prescription no. 1312 in stand no. 1 prescribes a change from *Ec* to *Ct*. Characteristically for such species conversions is that the "old" species shows a high `Vremovido` value, while the newly planted species has a `Vremovido` value of zero. Additionally, the space previously occupied by the "old" species is fully allocated to the "new" species (see `Area`).

Table 2.12: Example of a species conversion

UG	Presc	Species	Area	Period	Vremovido (ton)
1	1312	Ct	42.602	1	0.0
1	1312	Ec	42.602	1	2999.27
1	1312	Ct	42.602	2	0.0
1	1312	Ct	42.602	3	50.84

Mixed Stands

A mixed stand refers to a case where, in the same stand and period, *two different prescriptions* with distinct species are applied simultaneously. In contrast to species conversions, both

[12] It happens the first time a stand is harvested. Conversion can only happen once in the planning horizon, and there can be no conversion back into Eucalypt.

species are present together — each covering a different portion of the stand's area.

As illustrated in Table 2.13, during the first two periods, stand no. 1015 contains the species *Ec* along with the second species *Pb*, each applied via a different prescription (no. 3 and no. 503) on only a fraction of the total area.

Table 2.13: Example of a mixed stand

UG	Presc	Species	Area	Period
1015	3	Ec	2.934	1
1015	503	Pb	8.036	1
1015	3	Ec	2.934	2
1015	503	Pb	8.036	2

Impact on Row Counts

These structural variations explain the peculiarities mentioned previously: Each solution's prescriptions apply only to a total of $1,406$ stands (see Table 2.3, Table 2.4 and the corresponding remarks below). Still, the solutions consist of $1,512$ rows (stakeholder solution: $1,514$) - which is caused by the presence of mixed stands: 106 (stakeholder solution: 108) stands in the solution have two prescription instead of one.

Secondly, the species conversions affect the number of rows in the merged solution datasets (Table 2.8, Table 2.9, Table 2.10). Each species conversion results in two entries per `[UG, Period, Presc]` combination, thus inflating row counts beyond the baseline of $1,512 \times 5 = 7,560$ (stakeholder solution: $1,514 \times 5 = 7570$) expected for five periods per prescription. The numbers of species conversions occurrences in each solution dataset are as follows: 652 in the MaxRes solution, 110 in the MaxWood solution, and 833 in the Stakeholder solution - which explains the table lengths:

MaxRes solution: $7,560 + 652 = 8,212$ rows,

MaxWood solution: $7,560 + 110 = 7,670$ rows,

Stakeholder solution: $7,570 + 833 = 8,403$ rows.

Remark. We observe that the Stakeholder and MaxRes solutions — both prioritizing fire resistance — feature significantly more species conversions than the MaxWood solution, which instead optimizes for timber yield. This difference arises from variations in growth rates among species. In particular, species conversions away from eucalyptus, which is currently dominant and grows rapidly, are rare in the MaxWood solution due to its emphasis on maximizing productivity. Notably, eucalyptus replanting is restricted under Portuguese regulations.[13]

2.3.5 Determining and Storing Access Requirements by Stand and Period

At this stage, preprocessing of the harvesting schedule data is essentially complete. Data pertaining to stands that fall outside the scope of the study can therefore be removed.

Filter for Relevant Data Only (Inaccessible Stands in Paiva) As outlined in section 2.2.1, the study area is only Castelo de Paiva. However, no separate filtering step was needed for that: As stated in section 2.2.4, of the 687 stands located in Castelo de Paiva, only the 315 that are classified as inaccessible are relevant for the road access analysis, while all accessible stands shall be excluded from the problem. Consequently, the preprocessed harvesting schedule data were filtered to retain only data on the 315 inaccessible stands in Paiva (see illustration in fig. D.8).

General Access Determination Rules Road access requirements for each stand and period were determined based on the following two key variables:

- `Vremovido`: Indicates the volume of timber removed during a given period. If the value is greater than zero, at least timber road access (i.e., the construction of at least a 3-meter road) is required.
- `Rait`: Indicates the level of fire resistance in a given period. If the value falls within the lowest 5% quantile[14], wide firefighting road access (i.e., the construction of a 6-meter road) is required.

Remark. The `Rait` values used in this step are static and reflect the current infrastructure. Dynamically updating `Rait` in response to changes in the network over time could better capture fire resistance development of the stands and could be considered in future work.

[13] Following the devastating wildfires in 2017, the Portuguese government enacted legislation (Decreto-Lei n.º 11/2019) restricting eucalyptus planting, particularly its reintroduction in areas previously occupied by other species, with the aim to protect native forests and reduce fire risk.

[14] This threshold is illustrative and adjustable. Final selection should reflect expert and stakeholder judgment.

Handling Stands with Multiple Prescriptions In some cases, a stand may have more than one prescription per period due to special conditions such as species conversions or mixed stands (as discussed in section 2.3.4), leading to multiple `Vremovido` and `Rait` values. These cases were handled as follows:

- `Vremovido`: If multiple values exist for a given stand and period and any prescription of them has a value greater than zero, access is considered required.
- `Rait`: If multiple values exist for a stand and period, the minimum `Rait` value is used.[15]

For each harvesting schedule, a road access needs matrix was constructed. Each matrix identifies, for every stand and period, whether road access is required and of which type. The first column lists the stands, the next five columns indicate timber road access for periods $t = 1, \ldots, 5$ via binary values (1 = access needed, 0 = not needed), and the last five columns indicate firefighting road access for the same periods.

2.3.6 Summary of Preprocessed Harvesting Schedules: Access Needs

The completed preprocessing of the harvesting schedule data resulted eventually in the wanted road access requirements by stand and period, stored in form of matrices (315x11).

Here presented exemplarily for the MaxRes solution:

$$\begin{array}{c}
\begin{array}{ccccccccccc}
\text{UG} & t_1^{\text{timber}} & t_2^{\text{timber}} & t_3^{\text{timber}} & t_4^{\text{timber}} & t_5^{\text{timber}} & t_1^{\text{wide}} & t_2^{\text{wide}} & t_3^{\text{wide}} & t_4^{\text{wide}} & t_5^{\text{wide}}
\end{array} \\
\begin{pmatrix}
840 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\
841 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
1620 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0
\end{pmatrix}
\end{array}$$

[15] The minimum was preferred, rather than the average or weighted average. This conservative, worst-case approach ensures that fire resistance is not overestimated, particularly in periods with species conversions or pending harvests, which could otherwise result in high economic losses in case of fire. In future work, alternative approaches could be explored.

2.4 Costs for Road Construction and Maintenance

The final pillar of information required to address the problem at hand concerns the costs associated with constructing and maintaining forest roads. As outlined before, the model is designed with the objective is to minimize these costs. Accordingly, a preparatory step involves quantifying the cost per kilometer for each of the following actions:

- Constructing a timber road,
- Constructing a wider fire road road,
- Maintaining an existing timber road,
- Maintaining an existing wider fire road,
- Upgrading (Widening) a timber harvesting path to a wider fire road.

This section begins by outlining the available cost data (section 2.4.1), followed by the assumptions used to fill data gaps (section 2.4.2), and concludes with the derived cost estimates applied in the optimization model (section 2.4.3).

2.4.1 Available Cost data

The Portuguese Monitoring Commission for Forest Operations (Comissão de Acompanhamento para as Operações Florestais (CAOF)) provides a reference matrix detailing minimum and maximum costs for key forestry activities, including manual, mechanical, and mixed afforestation operations; reforestation; infrastructure development; and maintenance within forest territories. This matriz de (re)arborização (CAOF 2022) includes cost estimates for the construction and the maintenance of one kilometer of timber road, categorized by three slope classes.

The CAOF values refer to forest paths of standard width suitable for mechanized timber harvesting operations. However, this study also accounts for wider fire roads, assumed to be twice the width of standard timber roads. As not all actions considered in this work are explicitly covered in the cost listings of CAOF's matriz de (re)arborização, several assumptions[16] were necessary to estimate the remaining costs for the additional configurations.

[16]These assumptions were developed in consultation with the thesis supervisors who are considered subject matter experts.

2.4.2 Cost Estimation Assumptions

(i) The cost of constructing a wider fire road is assumed to be 150% of the cost of constructing a standard timber road:

$$\text{CostC}_{\text{wide}} = 1.5 \cdot \text{CostC}_{\text{timber}} \tag{2.1}$$

(ii) The cost of maintaining a wider fire road is assumed to be the same as maintaining a standard timber road:

$$\text{CostM}_{\text{wide}} = \text{CostM}_{\text{timber}} \tag{2.2}$$

(iii) The cost of upgrading a standard timber road to a wider fire road is assumed to be 75% of the cost of constructing a standard timber road:

$$\text{CostsU} = 0.75 \times \text{CostC}_{\text{timber}} \tag{2.3}$$

(iv) The cost of operating with a cross slope between 5% and 25% is approximated by averaging the minimum and maximum slope category costs:

$$\text{Cost}_{(5<\text{slope}<25\%)} = \frac{\text{Cost}_{(\text{slope}\leq 5\%)} + \text{Cost}_{(\text{slope}\geq 25\%)}}{2} \tag{2.4}$$

2.4.3 Result: Estimated Costs per Action per Road Kilometer

Table 2.14 presents the estimated costs in EUR per kilometer, based on the values provided by CAOF and the assumptions outlined above. The color coding in the table corresponds to the colors used in the assumptions, indicating which assumption was applied to derive each value.

Using this data, we can determine the construction, maintenance, and upgrade costs for all potential road segments considered in this study, as detailed in section 2.6. Specifically, for each segment, once the appropriate slope category has been identified, its length is multiplied by the corresponding cost per kilometer to obtain the total cost. The obtained values are incorporated into the IP model as input parameters. Given that the problem is multi-period, the fixed values shown in the Table 2.14 were discounted at a rate of 3.5% during implementation.

Table 2.14: Estimated Costs in EUR per action per kilometer; with highlighted entries color-coded to their respective assumptions.

Slope [%]	$\text{CostC}_{\text{timber}}$	$\text{CostM}_{\text{timber}}$	$\text{CostC}_{\text{wide}}$	$\text{CostM}_{\text{wide}}$	CostU
≤ 5	2147	1073.5	3220.5	1073.5	1610.25
$5 < \cdots < 25$	4830.75	1878.625	7246.125	1878.625	3623.0625
≥ 25	7514.5	2683.75	11271.75	2683.75	5635.875

2.5 Segmentation of the Problem into Independent Subproblems

The problem was decomposed into smaller, more manageable subproblems through implementation of a graph-based spatial segmentation, as mentioned conceptually in section 3.3. Each inaccessible forest stand can be represented as a node $v \in V$ in an undirected graph Graph $G = (V, E)$,[17], with spatial adjacency relationships between the stands forming the "edges" $e \in E$ of the graph. Connected component analysis was used to partition this graph into geographically cohesive components (section 2.5.1). Each component corresponds to a spatial cluster of adjacent inaccessible stands that can be treated as an independent optimization subproblem. Around each of these inaccessible clusters, the ring of neighboring stands along which the inaccessible ones will be connected to the road segment were identified (section 2.5.2). To maintain consistency in the modeling, spatially adjacent components with potentially shared access roads were combined (section 2.5.3). This allows that road infrastructure will be shared, where geographically possible.

Remark. Note that natural geographical barriers, if existing, (e.g., rivers, streams, or steep terrain) should be additionally incorporated in a previous segmentation step. Such barriers may prevent connectivity between stands even if they are spatially adjacent, and therefore must be accounted for to ensure a realistic decomposition.

2.5.1 Identification of Clusters of Inaccessible Stands

To systematically identify clusters of inaccessible stands, a graph-based method using the Python library `NetworkX` was used. Each inaccessible stand in the dataset was represented

[17]This graph is not to be confused with the graph representing potential future roads, they are entirely different, just the same mathematical concept is applied.

as a node in an undirected graph, uniquely identified by its `ID_UG` attribute. An edge was created between two nodes if the stands they represented were spatially adjacent. Adjacency was determined by applying a 1-meter buffer to each stand polygon and checking for overlaps between these buffered geometries. Stands with overlapping buffers were considered neighbors and connected by an edge in the graph. This procedure resulted in a graph capturing the spatial relationships among inaccessible stands (see fig. 2.3). Connected components of the graph – subsets of nodes connected to each other but disconnected from other such subsets – were then computed. Each connected component corresponds to a contiguous cluster of inaccessible stands. Using this method, **51 clusters of inaccessible stands** were identified.

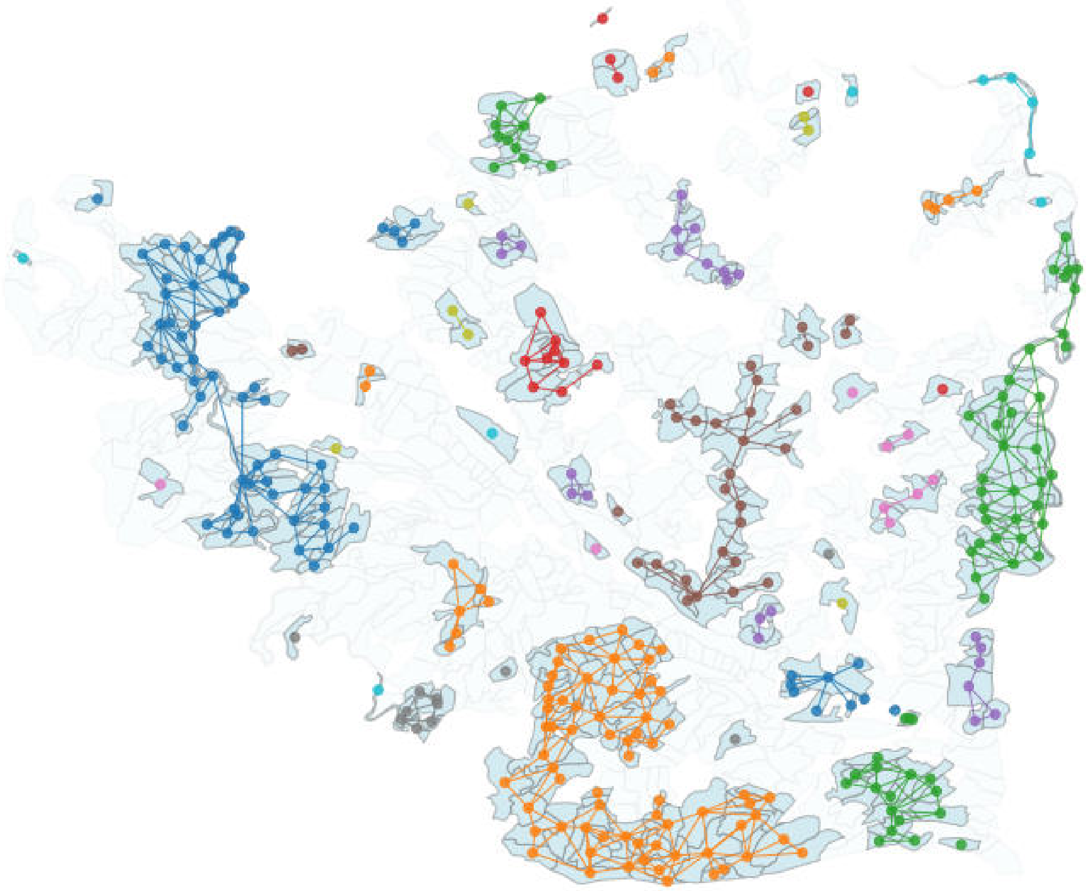

Figure 2.3: Inaccessible stands with their graph representation, forming 51 clusters

2.5.2 Merging Inaccessible Clusters with Surrounding Accessible Neighbors

The inaccessible patches had to be combined with their neighboring stands already connected to the road network, as new access roads would inevitably follow the boundaries of these accessible stands. To achieve this, each inaccessible cluster was dissolved into a single geometry, and a 1-meter buffer was applied to identify neighboring stands (their "ring of neighbors," see fig. 2.4 in green). These neighbors form the immediate interface through which future access roads can connect to the existing infrastructure. After combining each inaccessible component with its neighboring stands, the boundaries of this newly created components were stored for further analysis.

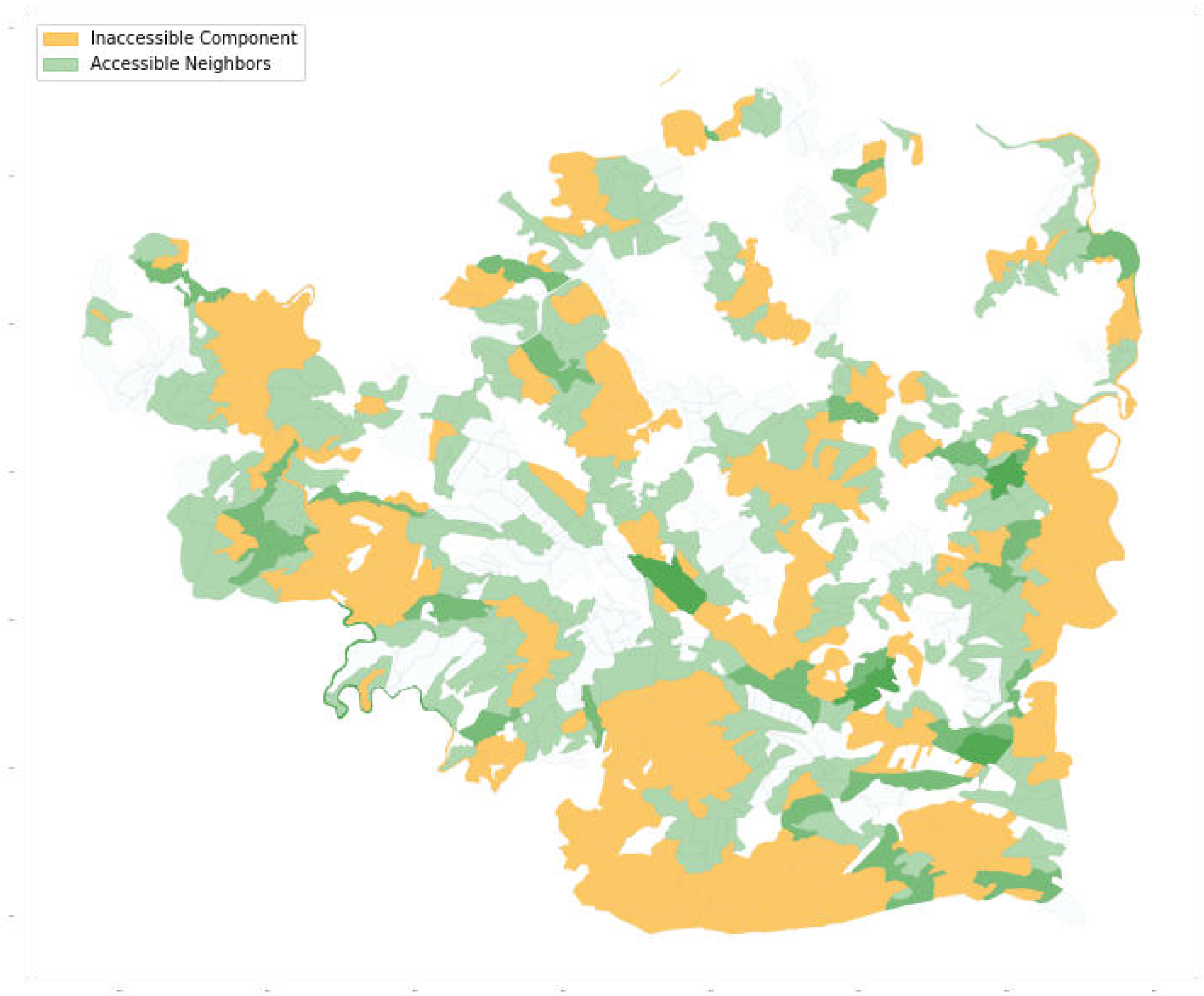

Figure 2.4: Inaccessible areas (yellow) surrounded by their neighboring already accessible stands (green). Darker shades of green reflect that an accessible stand is neighboring several inaccessible clusters.

As indicated by the darker green tones in fig. 2.4, several accessible stands are adjacent to more than one inaccessible component, which requires particular attention.

2.5.3 Handling Overlaps Between Rings of Accessible Neighboring Stands

Overlaps occur when an accessible stand is adjacent to more than one cluster of inaccessible stands, thereby belonging to multiple of the components created in section 2.5.2. To avoid redundant infrastructure planning, such overlapping components must be merged whenever access roads to one cluster could also serve another. In contrast, if two clusters share an accessible stand but lie on opposite sides of the existing road network – so that their future access roads cannot realistically overlap – their planning can remain independent, since access to one does not affect access to the other. Similarly, if two inaccessible clusters touch along their boundaries but connecting them would require a significantly longer detour than linking each directly to the network, they are treated as independent “opposite-side” cases.

Applying this rule, overlapping components were merged into joint units for access planning. This procedure yielded a decomposition into **35 independent areas**, which are presented in appendix D.

2.5.4 Result: Independent Subproblems for Road Network Planning

Figures D.10 to D.43 illustrate the segmentation of the study area into 35 independent components. Each segment consists of at least one of the components created in section 2.5.2 – inaccessible stands combined with their accessible neighboring stands – along whose boundaries the inaccessible areas are to be connected to the pre-existing road network. The resulting segments provide the basis for the subsequent optimization in CPLEX, with each segment treated as a separate decision problem whose solutions will later be integrated.

Remark. One component (comp 17, see fig. D.21) can not be processed within the current study using the chosen approach, as it is not adjacent to any existing roads and has no neighboring stands whose boundaries could guide the modeling of future access. For road construction decisions in this area, stakeholders need to coordinate with the owners of surrounding lands through which future connections to the road network could be established.

2.6 Creation of the Network of Potential Road Segments

Next, for each area segment (”component”) determined in section 2.5.4, the graph structure representing the network of potential forest road segments (mathematically introduced in chapter 3, definition 3.6) was created. What are the key requirements on the design of that road network?

- Each road segment must align with the borders between forest stands to ensure that the impact of road construction is distributed among different forest stands and their respective owners.
- The network must guarantee that every forest stand has at least one viable connection to the existing public road infrastructure.[18]

To construct the graph for this network, each component determined in section 2.5.1 was processed to extract stand boundaries and first generate an initial graph representing the road network via nodes and edges (section 2.6.1), assign respective costs (section 2.6.3), and then simplify the graph by retaining only the necessary nodes to improve computational efficiency.

In the following sections the key steps from the workflow are outlined. Additionally, data storage, visualization, and verification steps were performed which will not be explicitly detailed here, but can be reviewed in the accompanying notebooks. Throughout the construction and modification of the graph, special attention was given to the integrity of the spatial and cost-related attributes. At each major step — such as the addition of exit nodes, the splitting of edges, and the introduction of centroid-to-boundary connections — cumulative metrics (total edge length and aggregated costs) were tracked. This continuous monitoring served both as a validation mechanism to detect unintended topological distortions and as a verification tool to ensure that cost manipulations (e.g., assigning zero-cost to imaginary edges) were correctly applied.

2.6.1 Extracting Nodes and Edges from Stand Boundaries to Create a Road Network Graph

To construct the road network graph, nodes and edges were extracted directly from the boundaries of forest stands. The graph nodes were identified, accessed and stored in form of the

[18]Without this, the model would not be able to generate feasible solutions, as certain stands would remain inaccessible regardless of the selected road segments.

coordinate pairs they represent. The process starts by parsing the polygonal geometries within the `GeoDataFrame` of stands of the component. Each polygon's exterior boundary is analyzed to derive graph vertices (nodes) and the linear segments between them (edges), along with associated attributes such as edge length and slope. It iterates through each feature's geometry and checks whether it is a `Polygon`.[19] Each exterior coordinate is snapped to a fixed spatial grid to improve precision and ensure consistency across adjacent stands. These snapped coordinates form the graph's nodes. Edges are constructed by connecting each consecutive pair of exterior points. For each edge, two main attributes are calculated:

- **Edge length**, computed as the Euclidean distance between the two vertices.
- **Slope**, inherited from the "`Declive`" attribute of the associated stand polygon.[20]

This returns five key elements: A list of all unique boundary vertices (nodes), a list of edges (pairs of nodes), a list of dictionaries storing edge attributes (length, slope, etc.), a list of exit points, where stand boundaries intersect the existing road network, and a mapping of nodes to the stands they belong to. These are then used to construct an undirected graph using the `networkx` library, where nodes represent snapped boundary coordinates, and edges represent the physical path segments along stand boundaries. An illustration of one resulting graph component is shown in fig. D.44.

Remark. It is critical for this step that the polygon geometries in the shapefile are complete and not corrupted. This requirement enforced the extensive preprocessing steps described in section 2.2.2, whose completion ensured valid, clean geometries such that the boundary extraction process was eventually working without weird things happening.

2.6.2 Creating Exit Nodes

To integrate potential exit points into the network model, a series of spatial procedures were applied to ensure correct mapping to the graph topology. Each candidate exit point is associated with the closest graph node based on Euclidean distance. If the nearest node lies within

[19] In this study's dataset, all geometries were simple `Polygons`. In case the data includes `MultiPolygons`, it must be ensured that only the outer boundaries are used. Interior "holes" are enclosed within a single stand and do not connect to neighboring stands or the existing road network, making them irrelevant for graph construction.

[20] Although this per-polygon slope approximation may seem coarse, the data provider indicated that the stands were designed to be relatively homogeneous with respect to slope. To avoid excessively increasing the scope of this master's dissertation, which is already quite extensive, improving the approximation using additional data sources was left for future research (see also the discussion in chapter 4).

a 10-meter threshold, the exit point is merged with that node; otherwise, it is added as a new node. If the exit point does not coincide with an existing node, the graph is modified to introduce a new vertex at the closest edge by splitting it into two segments connected via the new exit node. The new edges retain key attributes like slope and are flagged with a `has_exit` attribute for downstream use. See fig. D.45 as an example.

2.6.3 Assigning Cost Attributes to Network Edges

The costs for each road segment in each period were calculated using the base costs per action performed on road kilometer as estimated in section 2.4 and their individual length and slope. The respective costs were assigned as attribute to the edges. Based on the distinction made by CAOF, three slope categories were distinguished:

- **Flat Terrain** ($\leq 5°$)
- **Moderate Slope** (between 5 and $25°$)
- **Steep Terrain** ($\geq 25°$)

The function created to assign construction, maintenance, and upgrade costs to each edge in the road network graph G based on its slope and length iterates over all edges. Each edge is classified into one of three slope categories, and the to build and maintain for both widths, and upgrade of the road are computed multiplying the predefined base cost for each action performed on a kilometer of road (see Table 2.14) with the length of each edge. The resulting costs are stored as edge's attributes.

Remark. To account for the time value of money, all cost parameters were discounted at a rate of 3.5%. The discounting was applied during implementation of the model, see section 3.5.

2.6.4 Eliminating Irrelevant Nodes and Merging Edges

As shown in fig. D.44, some edges in the graph created are quite short, resulting in unnecessarily fragmented road segments. There is no justification to keep this over-fragmentation, it occurs purely due to the shape of the geometries. To simplify the network and improve efficiency (model strenghtening), all nodes that are not exit nodes or road junctions (degree $>$ 2) were removed and their edges merged accordingly, which reduces the number of decision variables.[21] The resulting graph with all the created additional information was then stored to be used in the next step.

Remark. Please note that the order of the processing steps matters: the exit nodes and the length of each road segment must be determined using the geographically accurate data before the graph can be abstracted like that.

The function `merge_edges()` iteratively simplifies the graph G by merging adjacent edges while preserving essential structural nodes: The algorithm examines each edge and attempts to remove intermediate nodes that are neither exit points nor intersections. To ensure topological consistency, nodes with a degree greater than two (representing road crossings) and nodes marked as exits are maintained. When an irrelevant node is identified, the function connects its neighboring nodes directly, effectively replacing two consecutive edges with a single, aggregated edge.

For each successful merge, the function:

- Determines the nearest neighboring edge to the candidate node and combines the two corresponding edge segments.
- Computes the new edge attributes by summing relevant parameters such as `edgelength`, `Build5m`, `Maintain5m`, `Build10m`, `Maintain10m`, and `Upgrade`, while setting auxiliary attributes (e.g., `slope`, `has_source`, and `has_exit`) accordingly.
- Records removed intermediate nodes in a dedicated attribute (`removed_nodes`) to maintain merge traceability.
- Deletes the original edges and the merged intermediate node, then inserts the newly created edge connecting the remaining endpoints.

This process repeats iteratively until no additional merges occur. The result is a topologically consistent but simplified representation of the original network, where redundant short segments are collapsed into longer, continuous links. While preserving the connectivity and key structural features of the road system, this reduction in graph complexity supports more efficient network analysis.

After merging edges, the integrity of the network was verified by visually comparing pre- and post-merged graphs, and especially the relevant information (like, road intersections and exit

[21] This is actually part of the model strengthening and could be explained in section 3.3, but to better understand the whole process it is explained here.

nodes are maintained). Furthermore, key attributes, such as network length, and total costs, were checked to ensure that no inconsistencies were introduced.

Remark. Please note that after this abstraction, the edges in the graph no longer represent the stands' borders with geographical accuracy. However, this was the exact goal and is not a concern, as the essential information (edge length for verification purposes and cost attributes for optimization) remain preserved. This aligns with the fundamental principle of modeling: simplifying the representation while retaining the necessary elements.

2.6.5 Adding Stand-internal Edges

The centroid was calculated for each stand polygon and added as a node in the graph. From this centroid, edges were created to connect it with the polygon's boundary vertices. Although these edges do not represent actual physical paths, they were intentionally added because the network model later requires these internal connections to correctly formulate all necessary constraints. Since these stand-internal edges are not meant to be constructed in reality and therefore do not contribute to the total cost of building or maintaining the road network, each of them was assigned a cost value of zero.

Remark. Assigning zero cost to certain edges obviously affects the optimization model and can lead to illogical selections – for instance, the repeated construction of zero-cost road segments (instead of maintenance in the following period), or choosing to construct the same segment in the same period for both possible road widths, or simultaneously maintaining, upgrading, and constructing it. Such outcomes are unrealistic, as only one of these actions can occur for a given segment. While this kind of behavior does not generally pose a problem – since the stand-internal edges are not intended to be constructed anyway – it makes it considerably more difficult to verify the solution against the expected logic or behavior. Additional constraints were introduced (see section 3.2.4.F) to prevent such undesired behavior.

2.6.6 Summary: Network Evolution Across Processing Steps & Characteristics

The evolution of the network in terms of nodes, edges, and associated attributes at each processing step is summarized in Table 2.15, providing an overview of the network characteristics after each processing step. From here on, all analyses were performed using the fully preprocessed network (row 5 of Table 2.15).

Table 2.15: Overview of network graph characteristics after each processing step.

Step	Description	#Nodes	#Edges	#Sources	#Exits	Length [km]	Costs [€]
1	Initial network	25,939	26,895	0	0	1,172.88	0
2	+ exit nodes	27,078	28,034	0	1,960	1,172.88	0
3	+ costs	27,078	28,034	0	1,960	1,172.88	4,973,712
4	simplified network	3,681	4,637	0	1,960	1,172.88	4,973,712
5	+ source nodes	4,605	11,915	924	1,960	1,172.88	4,973,712

The difference between the results of step 3 and step 4 show clearly how the network simplification procedure of only retaining structural relevant nodes reduces network complexity while preserving the key attributes edge length and cost. Eliminating structurally irrelevant nodes reduced the number of edges by about 83%. The contribution of each component to the totals is shown in table 2.16. The table highlights that some components are considerably larger than others, implying that solving the optimization problem for these components is substantially more computationally demanding than for the smaller ones.

Table 2.16: Component network graph statistics sorted by number of edges (descending).

Component	Nodes	Edges	Sources	Exits	Length [km]	Costs [€]
comp_2_44_merged	651	1,760	123	294	151.41	628,353
comp_5_39_merged	527	1,445	130	157	137.61	602,838
comp_20_25_28_33_merged	439	1,153	89	175	109.98	458,994
comp_14_15_48_merged	376	1,019	89	111	105.74	472,577
comp_4_11_12_13_51_merged	334	876	66	146	81.25	354,727
comp_9	267	691	44	141	54.99	221,252
comp_31_34_merged	184	475	36	79	44.99	182,163
comp_23	122	300	27	43	30.96	132,878
comp_10	114	287	25	46	28.99	131,755
comp_16	110	283	23	46	31.29	124,241
comp_22	112	277	19	56	27.60	123,274
comp_45	117	288	18	70	21.62	96,991
comp_21_40_merged	94	244	19	43	24.63	101,761
comp_49	91	236	20	33	24.87	117,184
comp_7	83	196	11	52	16.06	86,045
comp_36_50_merged	77	193	14	36	35.33	134,635
comp_6	66	163	13	28	19.33	80,883
comp_24	58	141	14	19	15.96	79,789
comp_30	47	110	9	24	14.00	50,732
comp_27	44	106	10	16	12.25	51,695
comp_1	43	98	7	23	12.55	45,458

Continued on next page

Table 2.16 – continued from previous page

Component	Nodes	Edges	Sources	Exits	Length [km]	Costs [€]
comp_8	41	94	8	18	9.47	31,831
comp_19_41_merged	38	87	8	17	9.92	35,939
comp_46	37	84	6	20	10.76	36,822
comp_35	24	53	4	10	6.85	31,538
comp_29	16	37	4	5	4.78	17,320
comp_18	15	32	4	4	4.00	18,077
comp_47	7	14	2	2	2.57	14,503
Totals	4,605	11,915	924	1,960	1,172.88	4,973,712

Remark. The 924 source nodes (see Tables 2.15 and 2.16) representing forest stands exceed the 687 unique stands because some accessible stands neighbor multiple inaccessible stands (section 2.5.3). When inaccessible stands lie on different sides of the preexisting road that made the accessible stand reachable, their network designs do not influence each other, so their components need not be merged – modeling them together cannot improve the solution. Only overlaps on the same side of a road lead to merged components. Consequently, some accessible stands appear in multiple components. An example for this is the overlap between component 3_43 and component 10 (see fig. 2.5 and fig. 2.6). Many similar cases exist, which explains why duplicates inflate total source nodes and account for the difference between 924 source nodes and 687 stands.

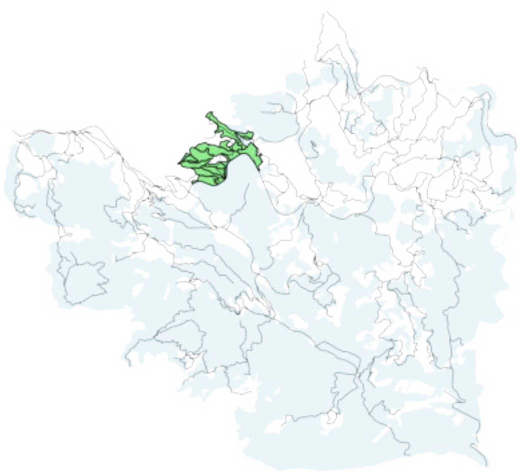

Figure 2.5: Component 3_43 merged

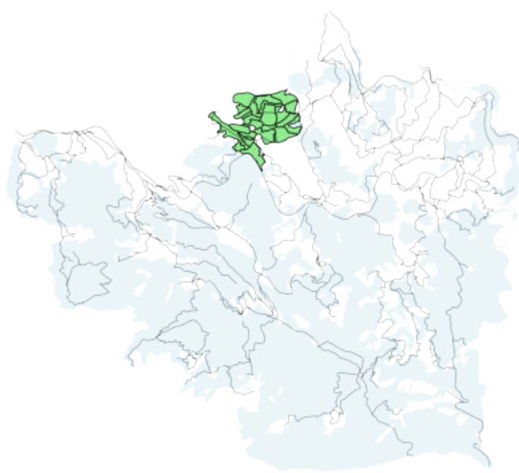

Figure 2.6: Component 10

3

Optimization Model and Solution Approach

Contents

THIS chapter presents and explains the model developed to address the problem outlined in the introduction and the necessary fundamentals. A single-objective IP approach is adopted, allowing the usage of an efficient solver to find the cost-minimal solution to the problem defined. In section 3.1, fundamental definitions and basic terminology are introduced. In section 3.2, the model is formulated, to cover all aspects of the real life problem which are relevant to the optimization. Section 3.3 briefly outlines the model-strengthening methods used to enhance convergence toward a viable solution, while section 3.4 provides a concise overview of the formulated model.

The model was then implemented and solved using Python and the CPLEX Python API (section 3.5): Each of the subproblems was solved independently, and their solutions were subsequently aggregated to form a complete road network for the entire study area, with the total cost obtained by summing the results across all components. This was performed for each of the three different harvesting schedules, but due to page limitations in this Master's dissertation, only the results of the stakeholder-based solution are presented in this document. The results for all three schedules are available in the accompanying GitHub repository.

3.1 Basic Definitions & Terminology

The network of road segments that can be selected for construction, maintenance and upgrade will be modeled as a Graph $G = (V, E)$, where a junction is represented by a vertex $v \in V$, and road segments are represented by so called "edges" (pairs of vertices $\{u, v\} \in E$). Throughout this chapter, we will work with the following sets and indices:

Sets

T	set of harvesting periods t,
S	set of forest stands,
$B(s)$	the *Boundary* of a stand $s \in S$,
V_{source}	set of *Source Nodes*, representing the forest stands,
V_{exit}	set of *Exit Nodes*, representing connection points to the existing road network,
V_{transit}	set of *Transit Nodes*, representing all other nodes (neither exit nor source),
V	set of **all** *Vertices/Nodes*, $V := V_{\text{source}} \dot{\cup} V_{\text{exit}} \dot{\cup} V_{\text{transit}}$,
E	set of *Edges* $\{u, v\}$, representing road segments,
A_{fwd}	set of *Forward arcs* (u, v) starting at u and ending at v,
A_{bwd}	set of *Backward arcs* (v, u) starting at v and ending at u,
A	set of **all** *Arcs* (u, v) and (v, u), $A := A_{\text{fwd}} \dot{\cup} A_{\text{bwd}}$,
$N(v)$	Neighborhood of vertex $v \in V$ (nodes adjacent to v),
$B(s)$	*Boundary* of stand s (nodes on the stand's geographical border).

Indices

w	the *road width* (either *timber* or *wide*)
t	a *harvesting period* (from 1 to 5)
s	a forest stand (identified via its associated source node)
v	a *node* (junction in the road network)
u	a *node* (junction in the road network)
$e := \{u, v\}$	an *edge*, denoting the road segment connecting the nodes u and v
$a := (u, v)$	an *arc* (directed edge), denoting the road segment from start u to end v

Before defining the decision variables, objective function and constraints, we must establish the framework for accurately representing the network of forest road segments that can be constructed. In this context, we differentiate between various types of vertices (nodes):

Definition 3.1 (Source Node)**.** A *source node (root)* is a point located at the center of a forest stand. Each forest stand has exactly one source node, and each source node uniquely identifies the forest stand surrounding it.

Definition 3.2 (Exit Node)**.** An *exit node (sink)* is a point that connects a forest road segment to the public road infrastructure.

Definition 3.3 (Transit Node)**.** A *transit node* is a point that is neither a source node nor an exit node.

By definition, each node can be only one option of the three above. With all node types defined, we can represent the network of potential roads segments formally.

Definition 3.4 (Vertices/Nodes)**.** Let the set of vertices

$$V := V_{\text{source}} \dot\cup V_{\text{exit}} \dot\cup V_{\text{transit}} = \Big\{(x, y) \text{ coordinate pairs}\Big\},$$

be the disjoint union of all source nodes (V_{source}), exit nodes (V_{exit}), and transit nodes (V_{transit}).

Definition 3.5 (Edges)**.** Let

$$E := \Big\{\{u, v\} \mid u \in V, v \in V \text{ and } u \neq v\Big\}$$

be the set of *edges*, linking the vertices, that represent undirected road segments.

Definition 3.6 (Network of potential road segments)**.** With the sets V and E as defined above, the graph

$$G = (V, E)$$

represents the *network of potential road segments*.

To formulate certain constraints, it is necessary to incorporate information about the direction of *flow* along each edge. For this purpose, we define a set of directed edges, referred to as *arcs*. While an edge simply denotes a connection between two nodes without implying direction, an arc explicitly specifies the direction by indicating a *start* node and an *end* node. This distinction allows us to represent the movement of entities along a segment in a directed manner.

Definition 3.7 (Arcs)**.** Given the set of vertices V, as defined in definition 3.6, we define the set of **arcs** (directed edges) as the set of ordered pairs

$$A := \{(u, v) \in V \times V \mid u \neq v\},$$

where each pair (u, v) represents a directed connection in a road segment (arc) from node u to node v.

We additionally define a digraph $D = (V, A)$ that captures the directionality of movement. For each undirected edge representing a potential road segment, the digraph contains two directed arcs (one in each direction), rather than a single undirected edge. Moreover, the reader may have noticed that the defined road network includes segments connecting the interior of a stand s to points on its boundary $B(s)$. However, we do not intend to construct such segments within the stands. Why, then, are they included in the model?

These segments serve a conceptual purpose: they are *virtual* road segments, introduced to facilitate the formulation of certain constraints that ensure the existence of required access paths. These virtual segments are assigned zero cost and are never intended to be physically constructed, maintained, or upgraded.

While the solution to the linear model may formally select some of these segments for construction, maintenance, or upgrade, this selection is purely auxiliary and does not correspond to real-world implementation. These virtual segments primarily support the representation of real-world requirements.

Definition 3.8 (Directed network of potential road segments)**.** The digraph

$$D = (V, A)$$

represents the *directed network of potential road segments*.

The set of arcs A can be partitioned into two equally sized subsets representing the two possible directions along each undirected edge. Specifically, if $\{u, v\} \in E$, then $(u, v) \in A_{\text{fwd}}$ and $(v, u) \in A_{\text{bwd}}$ represent the forward- and backward-directed arcs, respectively.

Definition 3.9 (Forward- and backward-directed arcs)**.** Let A be the set of directed arcs corresponding to an undirected edge set E. We define two subsets of A as follows:

$$A_{\text{fwd}} = \{(u, v) \in A \mid \{u, v\} \in E\},$$
$$A_{\text{bwd}} = \{(v, u) \in A \mid \{u, v\} \in E\},$$

where A_{fwd} contains the forward-directed arcs and A_{bwd} contains the corresponding backward-directed arcs. These subsets satisfy:

$$A_{\text{fwd}} \cap A_{\text{bwd}} = \emptyset, \quad A_{\text{fwd}} \cup A_{\text{bwd}} = A.$$

Let us now introduce the concept of a *neighborhood*:

Definition 3.10 (Neighborhood)**.** Let $G = (V, E)$ be a graph and let $v \in V$ be a vertex. The **neighborhood** of v, denoted $N(v)$, is the set of all vertices adjacent to v:

$$N(v) = \{u \in V \mid (v, u) \in E\}.$$

We refer to the neighborhood $N(s)$ of a source node $s \in V_{\text{source}}$ as the *boundary* $B(s)$. It contains all the nodes situated along the geographical border of stand s:

Definition 3.11 (Boundary)**.** The **boundary** $B(s)$ is the neighborhood of a source node $s \in V_{\text{source}}$, representing the nodes located along the geographical border of the forest stand s.

Let us further agree on some basic terminology:

Definition 3.12 (Road Existence)**.** A road segment is considered to *exist* in period t at width w if it undergoes one of the following actions during that period:

- construction at width w,
- maintenance at width w,
- upgrade to width w.

3.2 Model Formulation

In this section, we define the decision variables representing the choices to be made (see section 3.2.1) and the necessary parameters (see section 3.2.2), before formulating the objective function (see section 3.2.3). The objective function is subject to several constraints, which are explained in section 3.2.4.

While developing the cost-minimization objective function is relatively straightforward, the more complex challenge lies in formulating appropriate constraints that ensure the model accurately reflects real-world requirements. For the scenario under study, these constraints must, on the one hand, guarantee the existence of roads that facilitate timber harvesting and transport during each period where necessary, and on the other hand, ensure fire truck accessibility to stands with low wildfire resistance.

3.2.1 Decision Variables

A variety of decision variables is necessary to accurately represent the problem. Each additional distinction from the ”real world” increases the number of variables in the model, as will be demonstrated below. We begin by defining the binary decision variables.

3.2.1.A Binary Decision Variables

For each road segment, binary decisions must be made regarding five possible actions: construction at timber or wide width[1], maintenance at timber or wide width[2], and upgrade[3]:

[1] Construct the road? (Y/N)

[2] Maintain the existing road? (Y/N)

[3] Upgrade the existing timber road to a wide road? (Y/N)

Definition 3.13 (Decision variables, binary)**.** For $e \in E$, $w \in \{\text{timber}, \text{wide}\}$, and $t \in \{1, 2, 3, 4, 5\}$, define:

$$C^w_{e,t} := \begin{cases} 1 & \text{if edge } e \text{ is } \textbf{selected for construction} \text{ at width } w \text{ in period } t, \\ 0 & \text{otherwise;} \end{cases}$$

$$M^w_{e,t} := \begin{cases} 1 & \text{if edge } e \text{ is } \textbf{selected for maintenance} \text{ at width } w \text{ in period } t, \\ 0 & \text{otherwise;} \end{cases}$$

$$U_{e,t} := \begin{cases} 1 & \text{if edge } e \text{ is } \textbf{selected for upgrade} \text{ to wide width in period } t, \\ 0 & \text{otherwise.} \end{cases}$$

Considering a harvesting schedule spanning multiple time periods, these decisions must be made iteratively for each period – leading to five decisions per possible action, per road segment. Additionally, the construction and maintenance decisions must be made for both possible road widths, which increases the number of corresponding decision variables. In summary, we have

$$\underbrace{5 \cdot 2}_{\text{construction}} + \underbrace{5 \cdot 2}_{\text{maintenance}} + \underbrace{5}_{\text{upgrade}} = 25 \tag{3.1}$$

decisions to take per road segment e, leading to a total of $25 \cdot |E|$ binary decision variables, represented by $C^w_{e,t}, M^w_{e,t}$, and $U_{e,t}$ – see Table C.2.

3.2.1.B Integer Decision Variables

To model access feasibility, we introduce auxiliary integer decision variables, commonly referred to as *flow variables*. While such variables in network optimization typically represent the movement of goods or traffic along arcs, here they only indicate whether an arc will be used for traffic and for which type of traffic (see constraints in section 3.2.4.B). Specifically, the *timber flow* and *wide flow variables* on each directed arc a, listed in Table C.1, count the number of forest stands whose access path uses that arc during a given time period.

Definition 3.14 (Auxiliary variables, integer)**.** For $a \in A$, $w \in \{\text{timber}, \text{wide}\}$ and $t \in \{1, 2, 3, 4, 5\}$:

$$\mathit{flow}_{a,t}^{\text{timber}} := \text{number of stands whose timber traffic traverses arc } a \text{ in period } t;$$
$$\mathit{flow}_{a,t}^{\text{firefighting}} := \text{number of stands whose fire vehicle traffic traverses arc } a \text{ in period } t.$$

These variables allow us to ensure connectivity: Via the flow variables, we can create constraints (as described in section 3.2.4.C) that encode stand-level access requirements by limiting the outgoing flow from a Source Node s to either 0 (stand does not need access) or 1 (needs access) via the access needs parameters (see definition in definition 3.16), and then ensure that each stand requiring access gets linked to an exit point via a valid sequence of road segments selected for construction (see constraints in section 3.2.4.E).

Since flow may occur in both directions — forward (fwd) and backward (bwd) — and since each edge yields two directed arcs, we have 20 flow variables per undirected edge e: two types of flow (timber and wide), two directions per flow type, across five time periods. This results in a total of:

$$\underbrace{5 \cdot 2}_{\text{timber (fwd + bwd)}} + \underbrace{5 \cdot 2}_{\text{wide (fwd + bwd)}} = 20 \cdot |E| = 10 \cdot |A| \tag{3.2}$$

integer variables, as illustrated in Table C.1.

Remark. While the flow variables refer to directed arcs a, with $A_{\text{fwd}} \ni (u, v) \neq (v, u) \in A_{\text{bwd}}$, they are by definition direction-sensitive, so it is $flow_{(u,v),t} \neq flow_{(v,u),t}$. In contrast, the binary decision variables refer to undirected edges e, where $\{u, v\} = \{v, u\}$, so it is $C^w_{\{u,v\},t} = C^w_{\{v,u\},t}$; $M^w_{\{u,v\},t} = M^w_{\{v,u\},t}$; and $U_{\{u,v\},t} = U_{\{v,u\},t}$.

3.2.2 Parameters

To represent the costs associated with the construction, maintenance, and upgrade of each road segment, we use the parameters $CostC^w_{e,t}, CostM^w_{e,t}, CostU^w_{e,t}$.

Definition 3.15 (Parameters to represent costs)**.**

$$CostC^w_{e,t} := \text{Cost of constructing road segment } e \text{ at width } w \text{ in period } t,$$

$$CostM^w_{e,t} := \text{Cost of maintaining road segment } e \text{ at width } w \text{ in period } t,$$

$$CostU_{e,t} := \text{Cost of upgrading road segment } e \text{ to wider width in period } t.$$

Secondly, we introduce the parameters $need\_road^{\text{timber}}_{s,t}$ and $need\_road^{\text{wide}}_{s,t}$ to represent the varying access requirements of forest stands across time periods. If a stand s has timber to harvest in period t — that is, it requires an access path — we set $need\_road^{\text{timber}}_{s,t} = 1$. Similarly, if stand s has low fire resistance and requires a wide access path in period t, we set $need\_road^{\text{wide}}_{s,t} = 1$.

Definition 3.16 (Parameters to represent stand access needs)**.**

$$need\_road^{\text{timber}}_{s,t} := \begin{cases} 1 & \text{if stand } s \text{ has timber to harvest during period } t, \\ 0 & \text{otherwise.} \end{cases}$$

$$need\_road^{\text{wide}}_{s,t} := \begin{cases} 1 & \text{if the fire resistance in } s \text{ is below the defined limit} \\ 0 & \text{otherwise.} \end{cases}$$

3.2.3 Objective Function

To identify the minimum cost configuration among all feasible road networks, we define the objective function Z to represent the total costs of road construction, maintenance, and upgrades across all road segments, time periods, and road widths. The goal is to determine values for the decision variables such that the total cost Z is minimized.

Definition 3.17 (Objective Function)**.** Minimize

$$Z(C^w_{e,t}, M^w_{e,t}, U_{e,t}) = \sum_{t \in T} \left(\sum_{w \in W} \left(\sum_{e \in E} (C^w_{e,t} \cdot CostC^w_{e,t} + M^w_{e,t} \cdot CostM^w_{e,t} + U_{e,t} \cdot CostU_{e,t}) \right) \right)$$

3.2.4 Constraints

The objective function is subject to a set of constraints that reflect the real-world requirements of the problem. These include logical relationships between road construction, maintenance, and upgrades, as well as the necessity of ensuring access to all forest stands. Since we distinguish between timber and wide roads, each requirement is formulated through two corresponding constraints. As previously explained, fire vehicle access requires wide roads, while timber transport can be handled by narrower ones. Because fire access roads can also accommodate timber transport, the two constraints derived from each real-world requirement are not necessarily identical in form.

3.2.4.A Maintenance or Upgrade Only When Road Already Exists

Any maintenance or upgrade action must build on an existing road. Specifically, constraint (3.3) ensures that a road segment can only be maintained at timber width or upgraded to wide width in period t if, in the previous period $t-1$, it was either constructed or maintained at timber width:

$$\forall t \in T,\ \forall e \in E: \quad M_{e,t}^{\text{timber}} + U_{e,t} \leq C_{e,t-1}^{\text{timber}} + M_{e,t-1}^{\text{timber}} \tag{3.3}$$

Similarly, constraint (3.4) ensures that a road segment can only be maintained at wide width in period t if, in the previous period $t-1$, it was either constructed or maintained at wide width, or upgraded from timber to wide width:

$$\forall t \in T,\ \forall e \in E: \quad M_{e,t}^{\text{wide}} \leq C_{e,t-1}^{\text{wide}} + M_{e,t-1}^{\text{wide}} + U_{e,t-1} \tag{3.4}$$

The defined set T contains the harvesting periods 1 through 5. In constraints (3.3) and (3.4), the time index t appears together with $t-1$. To ensure that these constraints are well-defined in the first period ($t = 1$), we need to **initialize all variables indexed with $t = 0$ to zero**, representing the non-existence of the road segments previous to the first period.

Road Maintenance and Upgrade Constraints

For all $t \in T$ and all $e \in E$:

$$M_{e,t}^{\text{timber}} + U_{e,t} \leq C_{e,t-1}^{\text{timber}} + M_{e,t-1}^{\text{timber}} \tag{3.3}$$

$$M_{e,t}^{\text{wide}} \leq C_{e,t-1}^{\text{wide}} + M_{e,t-1}^{\text{wide}} + U_{e,t-1} \tag{3.4}$$

with Initialization for $t = 0$:

$$C_{e,0}^{w} = 0, \quad M_{e,0}^{w} = 0, \quad U_{e,0} = 0 \qquad \forall e \in E,\ \forall w \in W$$

3.2.4.B Flow only on Existing Roads

The flow of fire suppression forces or timber through a road segment is only permitted if two conditions are met:

1. The road segment exists.
2. The road segment meets the minimum required width.

Accordingly, we must establish dependencies between the relevant decision variables. This model focuses on whether *any* flow occurs from a given source. This introduces a binary decision framework: either timber flow (firetruck flow, respectively) from a specific stand s occurs (flow = 1), or it does not occur (flow = 0).

Consequently, for each road segment $e = \{u, v\} \in E$ and each period $t \in T$, the total timber flow (fire truck flow, respectively) in both directions, $flow_{(u,v),t} + flow_{(v,u),t}$, is bounded by the number of stands requiring road access (wide-road access, respectively) during that period. If, in period t, the road segment e is to be used for timber harvest/transport, then it must exist (be constructed, maintained or upgraded) during that period. Conversely, if no such timber road exists between u and v in period t, then the timber flow along that segment must be zero, see (3.5):

$\forall t \in T \quad \forall e \in E$

$$\underbrace{flow_{(u,v),t}^{\text{timber}} + flow_{(v,u),t}^{\text{timber}}}_{\in \mathbb{N}_0} \leq \underbrace{\overbrace{\left(\sum_{s \in S} need\_road_{s,t}^{\text{timber}}\right)}^{\text{\# of stands needing timber road}}}_{\in \mathbb{N}} \cdot \underbrace{\overbrace{\left(C_{e,t}^{\text{timber}} + M_{e,t}^{\text{timber}} + C_{e,t}^{\text{wide}} + M_{e,t}^{\text{wide}} + U_{e,t}\right)}^{\text{Existence of timber or wide road segment } e}}_{\in \{0,1\}} \tag{3.5}$$

where e denotes the edge $\{u, v\}$, while $(u, v) \in A_{\text{fwd}}$ and $(v, u) \in A_{\text{bwd}}$ denote the corresponding directed arcs. This constraint (3.5) is formulated as an inequality because not every stand requiring firetruck access must use the same road segment. The number of stands with low fire resistance may exceed the actual firetruck flow over any given segment, as access routes can differ across the network.

Similarly, if, in period t, the road segment between u and v shall enable firetruck flow, the road segment e must exist at wide width during that period. And if in period t no road segment between u and v exists, the firetruck flow between u and v must be zero during that period, enforced by (3.5).

$\forall t \in T, \quad \forall e \in E:$

$$flow^{\text{firefighting}}_{(u,v),t} + flow^{\text{firefighting}}_{(v,u),t} \leq \overbrace{\left(\sum_{s \in S} need\_road^{\text{wide}}_{s,t}\right)}^{\text{\# of stands requiring wide-road access}} \cdot \overbrace{\left(C^{\text{wide}}_{e,t} + M^{\text{wide}}_{e,t} + U_{e,t}\right)}^{\text{existence of wide road segment } e} \tag{3.6}$$

Let us summarize:

Maximum Capacity Constraints (Flow only on existing roads)

$\forall t \in T \quad \forall e \in E$

$$flow^{\text{timber}}_{(u,v),t} + flow^{\text{timber}}_{(v,u),t} \leq \left(\sum_{s \in S} need\_road^{\text{timber}}_{s,t}\right)\left(C^{\text{timber}}_{e,t} + M^{\text{timber}}_{e,t} + C^{\text{wide}}_{e,t} + M^{\text{wide}}_{e,t} + U_{e,t}\right) \tag{3.5}$$

$$flow^{\text{firefighting}}_{(u,v),t} + flow^{\text{firefighting}}_{(v,u),t} \leq \left(\sum_{s \in S} need\_road^{\text{wide}}_{s,t}\right)\left(C^{\text{wide}}_{e,t} + M^{\text{wide}}_{e,t} + U_{e,t}\right) \tag{3.6}$$

3.2.4.C Access Points for Stands That Need Road Access

Now we benefit from the imaginary road segments (explained at the end of **??**). Each stand's access road (and its timber/firetruck flow) must begin somewhere —- specifically, at a point along the boundary of that stand. Thus, we must enforce the existence of a "stand access point": During period t, for each stand s requiring an access road, at least one imaginary outgoing arc from inside the stand to a node in its boundary $B(s)$ must have flow. This boundary node then becomes the entrance to stand s, because non-zero flow already enforces the existence of the corresponding road segment in that period (see 3.2.4.B; thanks to (3.5) and (3.6), it suffices to enforce flow from stands when road existence is needed).

If stand s has timber to harvest in period t ($need\_road^{\text{timber}}_{s,t} = 1$), then one of the outgoing arcs from s must carry a flow of 1 during that period. If s has no timber to harvest, then all the outgoing timber flow from s (on all segments) must be zero., as represented via equation (3.7):

$$\forall t \in T \quad \forall s \in V_{\text{source}} : \quad \overbrace{\sum_{v \in B(s)} flow^{\text{timber}}_{(s,v),t}}^{\text{outgoing timber flow from } s} = need\_road^{\text{timber}}_{s,t} \tag{3.7}$$

Similarly, if stand s has low fire resistance in period t ($need\_road^{\text{wide}}_{s,t} = 1$), we ensure that some ”firetruck flow” exits s through one of its boundary arcs. If the stand’s fire resistance is not particularly low, then no such flow is permitted; represented by equation (3.8).

$$\forall t \in T \quad \forall s \in V_{\text{source}} : \quad \overbrace{\sum_{v \in B(s)} flow^{\text{firefighting}}_{(s,v),t}}^{\text{outgoing firetruck flow from } s} = need\_road^{\text{wide}}_{s,t} \tag{3.8}$$

Remark. In contrast to the general flow variables, which allow integer values, the outgoing flow on the stand-internal auxiliary road segments is binary due to constraints (3.7) and (3.8): The flow from a source node to a boundary node can only take values of either 0 or 1.

Constraints ensuring that stands needing access have an access point

$\forall t \in T, \quad \forall s \in V_{\text{source}}$:

$$\sum_{v \in B(s)} flow^{\text{timber}}_{(s,v),t} = need\_road^{\text{timber}}_{s,t} \tag{3.7}$$

$$\sum_{v \in B(s)} flow^{\text{firefighting}}_{(s,v),t} = need\_road^{\text{wide}}_{s,t} \tag{3.8}$$

3.2.4.D Avoiding In-flow into Stands

Would any forest owner be pleased if another stand’s access road cut right through their own stand? Clearly not. However, to build access paths for other stands, the model would “prefer” to select the imaginary road segments located inside the other stands, since they have zero cost assigned – ”minimum cost”. We need to forbid the selection any of the imaginary road segments for construction, except when the stand containing that segment has outgoing flow —- in which case the access point is genuinely needed.[4]

[4]As mentioned in **??**, the imaginary segment from the access point in the boundary into the stand will never actually be built, which is why the assigned costs are zero. They are purely theoretical and very useful to represent our real world constraints in mathematical terms (as seen in section 3.2.4.C).

What we must ensure is that the model cannot incorporate such segments into the access path of another stand. To prevent these stand-crossing access roads, we require that the total incoming flow into each Source Node s be zero (3.9):

Constraints forbidding in-flow into stands

$$\forall t \in T \quad \forall s \in V_{\text{source}} \qquad \overbrace{\sum_{u \in B(s)} \left(flow^{\text{firefighting}}_{(u,s),t} + flow^{\text{timber}}_{(u,s),t} \right)}^{\text{total incoming flow to stand} s} = 0 \tag{3.9}$$

3.2.4.E Flow Conservation

To ensure that each access path forms a complete and uninterrupted connection to the public road network, the model must preserve flow continuity at all internal points. In other words, the total *incoming flow* to each Transit Node must equal the total *outgoing flow*.

Specifically, for every transit node $v \in V_{\text{transit}}$, the total incoming firetruck flow must match the total outgoing firetruck flow. This guarantees that a complete route from a stand access point to an exit point in the public road system is possible, see equation (3.10). Similarly, the total incoming timber flow to each transit node must equal the total outgoing timber flow (3.11):

$$\forall t \in T \quad \forall v \in V_{\text{transit}} \qquad \overbrace{\sum_{u \in N(v)} flow^{\text{firefighting}}_{(u,v),t}}^{\text{incoming "firetruck flow" to } v} = \overbrace{\sum_{u \in N(v)} flow^{\text{firefighting}}_{(v,u),t}}^{\text{outgoing "firetruck flow" from } v} \tag{3.10}$$

$$\forall t \in T \quad \forall v \in V_{\text{transit}} \qquad \overbrace{\sum_{u \in N(v)} flow^{\text{timber}}_{(u,v),t}}^{\text{incoming "timber flow" to } v} = \overbrace{\sum_{u \in N(v)} flow^{\text{timber}}_{(v,u),t}}^{\text{outgoing "timber flow" from } v} \tag{3.11}$$

These constraints ensure that any road built — initiated by an outgoing flow from a stand access point via (3.6) and (3.5) — will ultimately connect to the public road network. This prevents access roads from ending in dead-ends and guarantees that each constructed path forms a valid and continuous connection.

Remark (Transit nodes versus Exit nodes)**.** The effectiveness of the above constraints relies on the clear separation between transit nodes and exit nodes. Since flow conservation applies exclusively to transit nodes, the model is naturally incentivized to route flows to exit nodes in the most cost-effective manner. At exit nodes, flow conservation is not enforced, allowing paths to terminate there without incurring additional costs, thereby ensuring efficient and realistic network solutions.

Flow conservation

$\forall t \in T \quad \forall v \in V_{\text{transit}}$

$$\sum_{u \in N(v)} flow^{\text{firefighting}}_{(u,v),t} = \sum_{u \in N(v)} flow^{\text{firefighting}}_{(v,u),t} \tag{3.10}$$

$$\sum_{u \in N(v)} flow^{\text{timber}}_{(u,v),t} = \sum_{u \in N(v)} flow^{\text{timber}}_{(v,u),t} \tag{3.11}$$

3.2.4.F Additional Constraints introduced due to Zero-Cost Imaginary Edges

While implementing the model, two additional constraints were introduced due to the presence of imaginary edges with costs assigned as zero. In models where all edges have positive costs assigned, the objective function of cost minimization naturally prevents invalid selections, such as double construction or overlapping usage, without the need for additional constraints. However, in our problem, there are imaginary edges with costs set to zero. For these edges, the usual cost-minimization logic does not automatically prevent undesirable configurations. To ensure the solution remains consistent, verifiable, and reasonable, we therefore introduce two additional constraints.

The first additional constraint enforces that at most one type of action variable can be active at a given time step:

$$\forall t \in T\ \forall e \in E: \quad C^{timber}_{e,t} + M^{timber}_{e,t} + C^{wide}_{e,t} + M^{wide}_{e,t} + U_{e,t} \leq 1 \tag{3.12}$$

The second constraint extends this logic to consecutive time steps, preventing the ”double” construction of a road segment that already exists:

$$\forall t \in T\ \forall e \in E: \quad C^{timber}_{e,t} + C^{wide}_{e,t} + C^{timber}_{e,t-1} + M^{timber}_{e,t-1} + C^{wide}_{e,t-1} + M^{wide}_{e,t-1} + U_{e,t-1} \leq 1 \tag{3.13}$$

Although these constraints may seem redundant in a purely positive-cost model, they ensure that the solution remains consistent, reasonable, and verifiable in the presence of zero-cost imaginary edges. Additionally, explicitly formulating such constraints helps reduce the feasible solution space, guiding the solver more efficiently through the branch-and-bound process and enhancing computational reliability.

3.3 Model Strengthening

The high number of decision variables (see section 3.2.1.A) underscores the need for model strengthening to reduce computational effort. With the implementation already described in chapter 2, the problem was decomposed into smaller, logically distinct subproblems by geographically separating areas whose decisions are unlikely to influence each other. Each subproblem corresponds to a geographically cohesive cluster of inaccessible forest stands, separated by already-accessible areas. Since road network decisions in one cluster do not affect others, the subproblems can be solved independently without compromising global optimality. This segmentation reduces the number of decision variables and constraints per instance, leading to faster solution times, while ensuring realistic network designs that respect practical road-building constraints and enabling parallel or sequential optimization.

In addition to geographical decomposition, further model strengthening was achieved by simplifying the underlying graph representation. Only the minimal set of nodes – necessary to represent the road segments along the boundaries of each inaccessible component and its neighboring accessible stands until reaching an "exit point" (connection to pre-existing road network) – was retained, reducing computational complexity without affecting the logical structure of the optimization problem. This approach further enhances solving efficiency by shrinking the number of decision variables, and thus, possible solution combinations.

In addition, the explicit constraints from section 3.2.4.F can also be seen as a form of model strengthening. While they were primarily required to maintain solution consistency and verifiability, they also reduce and prune the feasible solution space, guiding the solver more efficiently.

Remark. While the current model relies on the problem-specific strategies described above to reduce the feasible solution space and improve solver performance, additional general mixed-integer programming strengthening techniques could be explored in future work. For instance, symmetry-breaking constraints, variable fixing, or cutting planes derived from LP relaxations were not applied here. Incorporating such techniques could potentially further enhance computational efficiency, particularly for larger instances or more complex network structures, and may provide additional guidance to the solver in navigating the solution space.

3.4 Model Summary

Sets

S	Inaccessible forest stands
$T := \{1, 2, 3, 4, 5\}$	Harvesting periods
$W := \{\text{timber, wide}\}$	Road width options
V_{source}	set of *Source Nodes*, representing the forest stands,
V_{exit}	set of *Exit Nodes*, representing connection points to the existing road network,
V_{transit}	set of *Transit Nodes*, representing all other nodes (neither exit nor source),
V	set of all *Vertices/Nodes*, $V := V_{\text{source}} \dot\cup V_{\text{exit}} \dot\cup V_{\text{transit}}$,
E	set of undirected *Edges* $\{\{u, v\} \mid u \in V, v \in V\}$ (road segments)
A	set of all directed *Arcs* (u, v) and (v, u)
$N(v)$	*Neighborhood* of vertex $v \in V$ (nodes adjacent to v)
$B(s)$	nodes that are part of the *Boundary* of stand s

Decision Variables

$C^w_{e,t}, M^w_{e,t}, U_{e,t}$	$\in \{0, 1\}$	Decision variables (road construction, maintenance, upgrade)
$flow^{\text{timber}}_{a,t}, flow^{\text{firefighting}}_{a,t}$	$\in \mathbb{N}_0$	Auxiliary variables ("timber flow", "firefighting flow")

Parameters

$CostC^w_{e,t}, CostM^w_{e,t}, CostU_{e,t}$	$\in \mathbb{R}_{\geq 0}$	Cost parameters for road construction, maintenance, upgrade
$need\_road^{\text{timber}}_{s,t}, need\_road^{\text{wide}}_{s,t}$	$\in \{0, 1\}$	Parameters for access requirements per stand s and period t

Objective function

Minimize

$$Z(C^w_{e,t}, M^w_{e,t}, U^t_{e,t}) = \sum_{t \in T} \left(\sum_{w \in W} \left(\sum_{e \in V} (C^w_{e,t} \cdot CostC^w_{e,t} + M^w_{e,t} \cdot CostM^w_{e,t} + U^t_{e,t} \cdot CostU_{e,t}) \right) \right)$$

subject to

Constraints

- Maintenance or Upgrade Only When Road Already Exists

$\forall t \in T,\ \forall e \in E:$

$$M^{\text{timber}}_{e,t} + U_{e,t} \leq C^{\text{timber}}_{e,t-1} + M^{\text{timber}}_{e,t-1} \tag{3.3}$$

$$M^{\text{wide}}_{e,t} \leq C^{\text{wide}}_{e,t-1} + M^{\text{wide}}_{e,t-1} + U_{e,t-1} \tag{3.4}$$

- Flow only on Existing Roads

$\forall t \in T \quad \forall e \in E:$

$$flow^{\text{timber}}_{(u,v),t} + flow^{\text{timber}}_{(v,u),t} \leq \left(\sum_{s \in S} need\_road^{\text{timber}}_{s,t}\right)\left(C^{\text{timber}}_{e,t} + M^{\text{timber}}_{e,t} + C^{\text{wide}}_{e,t} + M^{\text{wide}}_{e,t} + U_{e,t}\right) \tag{3.5}$$

$$flow^{\text{firefighting}}_{(u,v),t} + flow^{\text{firefighting}}_{(v,u),t} \leq \left(\sum_{s \in S} need\_road^{\text{wide}}_{s,t}\right)\left(C^{\text{wide}}_{e,t} + M^{\text{wide}}_{e,t} + U_{e,t}\right) \tag{3.6}$$

- Access Points for Stands That Need Road Access

$\forall t \in T,\ \forall w \in W,\ \forall s \in V_{\text{source}}:$

$$\sum_{v \in B(s)} flow^{\text{timber}}_{(s,v),t} = need\_road^{\text{timber}}_{s,t} \tag{3.7}$$

$$\sum_{v \in B(s)} flow^{\text{firefighting}}_{(s,v),t} = need\_road^{\text{wide}}_{s,t} \tag{3.8}$$

- Avoiding In-flow into Stands

$$\forall t \in T \quad \forall s \in V_{source}: \quad \sum_{u \in B(s)} \left(flow^{\text{firefighting}}_{(u,s),t} + flow^{\text{timber}}_{(u,s),t}\right) = 0 \tag{3.9}$$

- Flow Conservation

$\forall t \in T \quad \forall v \in V_{\text{transit}}:$

$$\sum_{u \in N(v)} flow^{\text{firefighting}}_{(u,v),t} = \sum_{u \in N(v)} flow^{\text{firefighting}}_{(v,u),t} \tag{3.10}$$

$$\sum_{u \in N(v)} flow^{\text{timber}}_{(u,v),t} = \sum_{u \in N(v)} flow^{\text{timber}}_{(v,u),t} \tag{3.11}$$

- Additional Constraints introduced due to Zero-Cost Imaginary Edges

$\forall t \in \ \forall e \in E:$

$$C^{timber}_{e,t} + M^{timber}_{e,t} + C^{wide}_{e,t} + M^{wide}_{e,t} + U_{e,t} \leq 1 \tag{3.12}$$

$$C^{timber}_{e,t} + C^{wide}_{e,t} + C^{timber}_{e,t-1} + M^{timber}_{e,t-1} + C^{wide}_{e,t-1} + M^{wide}_{e,t-1} + U_{e,t-1} \leq 1 \tag{3.13}$$

Initialization

$$C^{w}_{e,0} = 0, \quad M^{w}_{e,0} = 0, \quad U_{e,0} = 0 \qquad \forall e \in E,\ \forall w \in W$$

3.5 Model Implementation and Solution using CPLEX Python API

The model was implemented and solved using Python and the Python API of CPLEX, provided through the `CPLEX` package[5]. This interface enables direct interaction with the IBM® ILOG® CPLEX® Optimization Studio (CPLEX) solver engine, which must be installed locally. The theoretical formulation, summarized in Section 3.4, was translated into a computational representation compatible with CPLEX.

The implementation follows a modular workflow that iterates over each harvesting schedule and all spatial components identified during preprocessing (see section 2.5.4 and figures in appendix D). For each combination of harvesting schedule and component, the script loads the corresponding graph data (nodes, edges, and attributes), classifies node types (source, transit, exit), and prepares edge attributes reflecting operational and economic parameters. Prior to model construction, automated verification steps ensure data consistency – such as matching node identifiers, validating bidirectional arc pairs, and confirming that all required constraints can be instantiated.

Decision variables were defined for road construction, maintenance, and upgrading actions across all planning periods, as well as for the flow of timber and wide-access needs between sources and exits. Constraints were programmatically added to enforce flow conservation, logical dependencies between activities (e.g., a road must exist before it can be maintained), and non-overlapping construction periods. Cost parameters were discounted at a rate of 3.5% to account for the time value of money, applied to the midpoint of each 10-year period (i.e., years 5, 15, 25, 35, and 45) according to

$$\text{Cost}(t) = \frac{\text{Cost}}{(1+0.035)^{(10t-5)}}.$$

Solver configurations were applied uniformly across all components. When optimality was not reached within the given thresholds, the solver was rerun using a warm start from the previous solution with an extended time limit. Solver logs, model files (LP, MPS, SAV), and summaries of the solution status were exported for each component to ensure reproducibility and transparency. The modular structure allows independent solution of components, which could be executed sequentially or in parallel to enhance computational efficiency.

Finally, component-level solutions were aggregated per harvesting schedule and combined to obtain the full model results for each of the three optimization scenarios, forming the basis for the analysis presented in Chapter 4.

[5] https://pypi.org/project/CPLEX/

4

Results and Discussion

Contents

This chapter presents and interprets the main results of the five-period, dual-purpose forest road network optimization model under the three harvesting schedule scenarios for the study area Paiva. It integrates a discussion of model outputs with an analysis of their practical relevance, underlying assumptions, and methodological limitations.

Fundamental characteristics of the full road network for the case study area, from which the model selects the cost-optimal combination of segments for construction, maintenance, or upgrade, were presented earlier in chapter 2, Table 2.15. The network comprises 4,605 nodes (including 924 source nodes and 1,960 exit nodes) and 11,915 edges, with a total length of 1,172.88 km and total costs of €4,973,712. As reported previously in table 2.16 (Chapter 2), the contributions of individual components to these totals are evident.

4.1 Core Metrics and Computational Performance

All computational results were obtained with a maximum time limit of four hours per component on a laptop computer[1] equipped with an Intel® Core™ i5-7300U processor (2.6 GHz, 2 cores / 4 logical processors) and 8 GB of RAM. The solver was initially configured for an optimality gap (MIP gap) of 0.0001 (0.01%). All model components were first solved with a time limit of 10 min per component. For components where the solver did not reach optimality within that limit, a warm start strategy from that previous solution was employed to attempt improvement, while extending the runtime first to one hour and then to four hours.

Table 4.1 presents the global performance metrics under each of the three harvesting schedule scenarios for the solver run with a time limit of ten minutes. The Total Upper Bound (UB) column reports the sum of the objective values of the best feasible solutions across all components, while Total Lower Bound (LB) shows the sum of the theoretical lower bounds found by the solver (both represent costs in EUR, rounded to the nearest integer). Aggregated MIP Gap[2]. gives the relative difference between Total UB and Total LB, and Mean Runtime [s] indicates the average solver runtime per component. Given the variability in component and model sizes, the results in Table 4.1 indicate that the solutions obtained with a 10-minute runtime per component are not yet of sufficient quality, as indicated by the high aggregated MIP gaps. Consequently, a second solver run was performed using a warm start from the previous solution and a time limit of 1 hour per component. Finally, for the remaining two components per harvesting schedule that remained without optimal solution, a four-hour solver run was conducted.

[1]The laptop used was a Lenovo ThinkPad X270, released in 2017 and designed for business professionals who require a compact, reliable, and robust device.

Table 4.1: Global solver performance metrics for all three harvesting schedules, runtime limit = 10min.

Harvesting Schedule	Total UB [€]	Total LB [€]	Agg. MIP Gap	Avg. Runtime [s]
MaxRes	384,462	299,888	22.0%	97.0
MaxWood	395,866	321,604	18.76%	96.83
Stakeholder	408,429	318,890	21.92%	95.26

Tables 4.2, 4.4, and 4.3 summarize the solver results per component given each of the *Stakeholder*, *MaxWood*, and *MaxRes* harvesting schedules, using a 10-minute per-component time limit. Components that reached an optimality gap below 0.01% within the time limit are denoted with a "-" in the MIP gap column. Several patterns emerge from the tables: Larger components or highly connected networks (those with many edges) generally required longer runtimes and occasionally failed to reach optimality within the time limit. Conversely, smaller, less complex components were typically solved within seconds with a zero MIP gap. The tables indicate that the four largest components are primarily responsible for the observed optimality gaps. These components were re-solved using an extended time limit, resulting in the improved solutions presented in Tables 4.5, 4.6, 4.7 (1h time limit); and 4.8, 4.9 and 4.10 (4h time limit).

Notably, Component 7 exhibits a trivial solution, i.e. all variables are zero, not only in Table 4.2, but in all three harvesting schedule scenarios (see also Table 4.4 and Table 4.3), which at first glance appears implausible. A review of the preprocessing steps, in particular the accessibility analysis and the access requirements of the included stands reveals that this component contains two stands categorized as inaccessible (IDs 1508 and 1318), and each is requiring road access in certain periods. So why do we end up with a zero solution? The zero solution results from the graph construction process: the snapping-to-grid procedure (see section 2.6.1) applied to stand boundaries slightly altered the polygons, causing the previously inaccessible stands to be effectively treated as accessible with exit points. Given that the underlying spatial data required manual adjustments to align layers accurately, this discrepancy reflects the inherent imprecision of the dataset rather than a substantive modeling error. Therefore, redoing the entire analysis to correct this minor artifact is unnecessary, and the solution is considered acceptable despite this anomaly. Importantly, this does not indicate a flaw in the model itself; it performs as intended, and the irregularity arises solely from preprocessing during graph creation.

Table 4.2: Solver performance for *Stakeholder* harvesting schedule, time limit 10min.

Component Name	UB [€]	LB [€]	MIP Gap	Runtime [s]
comp_5_39_merged	140,566	62,149	55.79%	600.17
comp_14_15_48_merged	58,345	51,254	12.15%	600.17
comp_20_25_28_33_merged	25,402	22,772	10.35%	600.11
comp_2_44_merged	30,894	29,496	4.53%	600.20
comp_16	16,309	16,308	0.01%	309.68
comp_10	19,598	19,596	0.01%	437.48
comp_23	12,603	12,603	-	17.39
comp_24	9,206	9,206	-	16.17
comp_31_34_merged	11,224	11,224	-	37.15
comp_4_11_12_13_51_merged	13,399	13,399	-	4.73
comp_49	12,946	12,946	-	2.26
comp_19_41_merged	7,631	7,631	-	2.35
comp_21_40_merged	3,793	3,793	-	1.93
comp_32	2,791	2,791	-	1.53
comp_6	4,587	4,587	-	3.30
comp_9	3,114	3,114	-	0.48
comp_38	4,579	4,579	-	0.68
comp_3_43_merged	3,295	3,295	-	0.15
comp_1	1,552	1,552	-	0.13
comp_18	2,280	2,280	-	0.22
comp_26	825	825	-	0.08
comp_27	2,559	2,559	-	0.33
comp_29	2,993	2,993	-	0.50
comp_30	574	574	-	0.08
comp_35	2,004	2,004	-	0.09
comp_36_50_merged	1,249	1,249	-	0.28
comp_37	1,915	1,915	-	0.23
comp_42	462	462	-	0.08
comp_45	555	555	-	0.15
comp_46	321	321	-	0.06
comp_47	2,473	2,473	-	0.03
comp_7	0	0	-	0.05
comp_8	478	478	-	0.06

Table 4.3: Solver performance for the *MaxRes* harvesting schedule, time limit 10 min

Component Name	UB [€]	LB [€]	MIP Gap	Runtime [s]
comp_5_39_merged	134,087	61,185	54.37%	600.19
comp_14_15_48_merged	57,430	49,092	14.52%	600.26
comp_20_25_28_33_merged	24,389	22,466	7.88%	600.07
comp_16	16,362	15,250	6.80%	600.04
comp_2_44_merged	29,140	28,844	1.02%	600.14
comp_19_41_merged	7,114	7,113	0.01%	0.92
comp_24	8,980	8,979	0.01%	7.47
comp_35	1,707	1,707	-	0.08
comp_1	1,552	1,552	-	0.29
comp_10	15,480	15,480	-	239.59
comp_18	2,280	2,280	-	0.12
comp_21_40_merged	3,694	3,694	-	1.32
comp_22	7,073	7,073	-	0.92
comp_23	11,455	11,455	-	16.06
comp_26	825	825	-	0.08
comp_27	2,559	2,559	-	0.11
comp_29	2,769	2,769	-	0.38
comp_30	490	490	-	0.42
comp_31_34_merged	9,191	9,191	-	16.52
comp_32	2,377	2,377	-	0.40
comp_36_50_merged	1,135	1,135	-	0.17
comp_37	1,631	1,631	-	0.46
comp_38	4,579	4,579	-	0.53
comp_3_43_merged	2,905	2,905	-	0.17
comp_42	462	462	-	0.29
comp_45	555	555	-	0.25
comp_46	208	208	-	0.06
comp_47	2,473	2,473	-	0.03
comp_49	11,553	11,553	-	1.02
comp_4_11_12_13_51_merged	12,755	12,755	-	3.59
comp_6	4,177	4,177	-	5.25
comp_7	0	0	-	0.04
comp_8	419	419	-	0.11
comp_9	2,655	2,655	-	0.72

Table 4.4: Solver performance for the *MaxWood* harvesting schedule, runtime limit 10 min.

Component Name	UB [€]	LB [€]	MIP Gap	Runtime [s]
comp_5_39_merged	127,346	66,932	47.44%	600.17
comp_14_15_48_merged	61,300	50,984	16.83%	600.12
comp_20_25_28_33_merged	24,568	22,726	7.50%	600.46
comp_2_44_merged	30,894	29,207	5.46%	600.20
comp_16	16,309	16,308	0.01%	319.44
comp_10	19,598	19,596	0.01%	487.96
comp_1	1,552	1,552	-	0.11
comp_18	2,280	2,280	-	0.14
comp_19_41_merged	7,114	7,114	-	2.82
comp_21_40_merged	3,793	3,793	-	1.87
comp_22	7,073	7,073	-	0.48
comp_23	12,603	12,603	-	15.74
comp_24	9,206	9,206	-	13.39
comp_26	825	825	-	0.10
comp_27	2,559	2,559	-	0.12
comp_29	2,993	2,993	-	0.20
comp_30	574	574	-	0.11
comp_31_34_merged	11,110	11,110	-	37.69
comp_32	2,791	2,791	-	0.65
comp_35	2,004	2,004	-	0.08
comp_36_50_merged	1,249	1,249	-	0.18
comp_37	1,915	1,915	-	0.24
comp_38	4,579	4,579	-	0.80
comp_3_43_merged	3,295	3,295	-	0.18
comp_42	462	462	-	0.09
comp_45	555	555	-	0.16
comp_46	321	321	-	0.06
comp_47	2,473	2,473	-	0.07
comp_49	12,946	12,946	-	2.03
comp_4_11_12_13_51_merged	13,399	13,399	-	4.45
comp_6	4,587	4,587	-	1.36
comp_7	0	0	-	0.19
comp_8	478	478	-	0.07
comp_9	3,114	3,114	-	0.61

Table 4.5: Solver performance for extended run (1h, stakeholder harvesting schedule).

Component Name	UB [€]	LB [€]	MIP Gap	Runtime [s]	#Variables	#Nonzero
comp_5_39_merged	136,948	82,862	39.49%	3600.63	72,250	3,031
comp_14_15_48_merged	58,345	51,792	11.54%	3600.28	50,950	1,786
comp_2_44_merged	30,893	30,584	1,56%	3600.82	88,000	2,427
comp_20_25_28_33_merged	25,402	25,402	-	3551.94	57,650	1,327

Table 4.6: Solver performance for extended run (4h, stakeholder harvesting schedule).

Component Name	UB [€]	LB [€]	MIP Gap	Runtime [s]	#Variables	#Nonzero
comp_5_39_merged	127,520	84,646	33.62%	14,404.2	72,250	3,038
comp_14_15_48_merged	58,345	53,113	8.97%	14,401.44	50,950	1,786

Table 4.7: Solver performance for extended run (1h, MaxWood harvesting schedule).

Component Name	UB [€]	LB [€]	MIP Gap	Runtime [s]	# Variables	# Nonzero
comp_5_39_merged	123,597	79,638	35.57%	3,600.42	72,250	3,025
comp_14_15_48_merged	58,345	52,994	10.88%	3,600.84	50,950	1,749
comp_2_44_merged	30,881	30,542	1.09%	3,600.79	88,000	2,468
comp_20_25_28_33_merged	24,568	24,566	0.01%	946.43	57,650	1,315

Table 4.8: Solver performance for extended run (4h, MaxWood harvesting schedule).

Component Name	UB [€]	LB [€]	MIP Gap	Runtime [s]	# Variables	# Nonzero
comp_5_39_merged	123,393	79,827	35.3%	14,402.66	72,250	3,038
comp_14_15_48_merged	58,345	53,844	7.71%	14,407.04	50,950	1,749

Table 4.9: Solver performance for extended run (1h, MaxRes harvesting schedule).

Component Name	UB [€]	LB [€]	MIP Gap	Runtime [s]	# Variables	# Nonzero
comp_5_39_merged	130,240	75,942	41.69%	3,600.49	72,250	2,759
comp_14_15_48_merged	56,372	51,479	8.68%	3,601.42	50,950	1,524
comp_20_25_28_33_merged	24,385	24,382	0.01%	1,402.29	57,650	1,211
comp_16	16,040	16,038	0.01%	2,488.43	14,150	372

Table 4.10: Solver performance for extended run (4h, MaxRes harvesting schedule).

Component Name	UB [€]	LB [€]	MIP Gap	Runtime [s]	# Variables	# Nonzero
comp_5_39_merged	121,666	77,629	36.19%	14,408.14	72,250	2,726
comp_14_15_48_merged	56,372	53,321	5.41%	14,402.77	50,950	1,524

Table 4.11: Global solver performance metrics for all three harvesting schedules.

Harvesting Schedule	Total UB [€]	Total LB [€]	Aggregated MIP Gap	Mean Runtime [s]
MaxRes			%	
MaxWood			%	
Stakeholder			%	

fill the table after final solver run when decided how to deal with the suboptimal components

placeholder for final phrase on computational results

4.2 Practical Relevance

The proposed model enables integrated planning of forest road networks by balancing economic efficiency with fire resilience. It provides a quantitative basis for prioritizing road construction and maintenance, helping planners identify routes that serve both functions – timber harvesting access where extraction is planned and ground-based firefighting access in vulnerable stands – while selecting appropriate road widths. Practically, these outputs can guide investment decisions and inform policy development for infrastructure planning in fire-prone landscapes, supporting transparent and evidence-based decision-making among stakeholders. While the model shows clear potential for practical application, adoption may be limited by unfamiliarity with optimization tools and concerns about data quality or reliability. Maintaining accurate, well-documented data and close communication with subject-matter experts is essential to ensure the model's effective use.

4.3 Model Scope, Assumptions & Limitations

This work was developed for Vale do Sousa, an area in Northern Portugal where forests are primarily managed for timber harvesting. The model is therefore **limited to timber-oriented forests** and may not directly apply to national parks or forests with other objectives, particularly in areas with no pre-existing road networks or where no roads are required for timber extraction.

Within this model, **only currently inaccessible stands are targeted** for new roads due to practical and methodological reasons. Limiting the model scope to inaccessible stands improves tractability and simplifies the optimization problem, but additional motivations exist: first, there is significant data uncertainty, as the width (and thus usability) of existing small roads is unclear (see section 2.1.3); second, forest owners prioritize creating new access over upgrading existing roads; third, inaccessible stands represent the most urgent connectivity gaps for both timber extraction and firefighting operations; fourth, constructing new roads in isolated areas is generally more cost-efficient than upgrading small existing roads, because more stands benefit from gaining road access via newly built roads than by extending small existing ones; and fifth, many existing roads are publicly owned, complicating coordination for upgrades, whereas new roads on privately managed land allow more straightforward implementation. Nevertheless, a future expansion of this work could include "timber-accessible" stands that are already road-accessible but do not meet minimum width requirements for ground-based firefighting.

While the Douro and Paiva rivers define the study area boundaries, **minor streams are assumed to be seasonal and pose no obstacle to forestry operations or road crossings**. A hydrographic dataset from the Copernicus EU-Hydro River Network Database was identified; however, it lacked key attributes such as depth, permanence, and seasonal variation. Moreover, some mapped streams may not exist in reality or carry water only intermittently. For this reason, no water lines were considered as natural boundaries in this work. Consequently, the segmentation algorithm does not (yet) account for potentially disconnected components caused by unbridgeable terrain. With more complete data on waterlines, including the relevant attributes, this limitation could be addressed in future work.

In addition to assumptions regarding natural boundaries, the model relied on **estimated road widths for segments where data was missing or implausible.** As described in section 2.1.3, zero values in the `LARGURA` column were deemed erroneous and corrected based on the `REDE_DFCI` classification system. It was assumed that, segments with `REDE_DFCI` = 1 have a road width of 6 m; segments with `REDE_DFCI` = 2 have a width of 4 m; and segments with `REDE_DFCI` = 3 have a road width of 3 m.

4.4 Limitations and Data Considerations

A key limitation of this work is the presence of **data gaps** in the stands dataset due to **differences in land ownership**. The forest stands considered in this work are not fully contiguous, as certain areas are owned by other stakeholders. This creates challenges for modeling, particularly when inaccessible stands under the target ownership are separated from the existing road network by stands belonging to different owners. In such cases, the model cannot propose new road segments crossing these areas, since road construction decisions cannot be made for land outside the modeled ownership domain. **As a result, potential access routes in these regions cannot be generated, and their road network problem remains unresolved.** One such case occurred in this project (see Component 17 in fig. D.21). Addressing this limitation in future work would require closer collaboration with the respective landowners to enable data sharing and coordinated access planning across ownership boundaries, or alternatively, enlarging the ZIF Paiva[3] to include those areas currently under different ownership.

The dataset used was originally collected for purposes other than this study.[4] Although supplementary sources were reviewed, certain granularity limitations remain. As in any data-driven analysis, **model outputs are inherently limited by the accuracy of the underlying assumptions**, and results depend heavily on data quality. **Inaccuracies in input data can compromise the real-world applicability of the findings**:

- Inaccuracies in the geospatial representation of the stands or road data may compromise accessibility and neighborhood analyses. Such distortions can propagate through subsequent modeling steps, leading to incorrect clustering and improper problem segmentation – where areas that should be treated separately may be grouped together, or vice versa – and ultimately resulting in biased or invalid optimization outcomes.
- Incorrect road width values of pre-existing roads may affect connectivity assumptions and the usability of newly built roads. Underestimating the width of existing roads could lead to impractical situations, such as a newly planned wide road connecting to an existing road that was assumed to be wide enough, while in reality being too narrow for fire vehicles.
- The accuracy of cost estimates directly affects the optimality of the model's solutions. Imprecise cost data/estimates may result in a cost-minimal solution that does not reflect the real-world optimum.

[3] https://www.afvs.ws/zif-paiva.html

[4] For any additional information beyond what's described in this work, please contact the data provider smarques@isa.ulisboa.pt

- It was assumed that the study area contains no waterlines that hinder road construction, except the Douro River (northern boundary) and the Paiva River (eastern boundary). Minor streams were considered seasonal and non-obstructive, and no water lines were treated as natural boundaries due to incomplete or uncertain hydrographic data. If this assumption is incorrect – i.e., some streams are impassable – the segmentation into subproblems would be incomplete, and affected components would need to be split along the waterlines with their road networks optimized separately in future work.
- The slope data used were relatively coarse, potentially introducing inaccuracies in the assigned road construction costs. Although the stand boundaries were, according to the data provider, designed to represent relatively homogeneous slopes, using stand-averaged slope values may not capture local variations that are relevant for selecting the appropriate cost category for each road segment.

4.5 Methodological Reflections

The multi-period, dual-purpose forest road network optimization problem in this study is inherently deterministic, with clearly defined decision variables, parameters, and linear constraints. For such problems, IP provides a robust, transparent, and mathematically rigorous framework capable of guaranteeing global optimality within a known gap, making it particularly suitable for infrastructure and resource allocation tasks where feasibility and interpretability are critical. The CPLEX solver was chosen for its performance, numerical stability, and advanced presolve and branching techniques for Mixed Integer Linear Programming (MILP), efficiently handling large-scale linear and mixed-integer formulations while providing optimality certificates and bounds that allow explicit control over solution quality. Access via its Python API enabled seamless integration into the computational workflow, enhancing automation, transparency, and reproducibility.

Although AI-based and heuristic optimization methods are increasingly applied across various fields, MILP was preferred here due to the deterministic, structured nature of the problem. MILP is well-suited for constraint-driven problems with linear or piecewise-linear relationships, offering guaranteed optimal solutions (or bounded approximations) and full constraint satisfaction. In contrast, AI and heuristic approaches – such as genetic algorithms, simulated annealing, or reinforcement learning – are typically favored for highly nonlinear, uncertain, or dynamic problems. These methods efficiently explore large search spaces without requiring linearity, but rely on heuristics, need parameter tuning, and do not guarantee optimality. In this work, MILP with CPLEX was chosen for its interpretability, verifiable optimality, and computational feasibility, even on modest personal hardware, with explicit control over solution precision. Artificial Intelligence (AI) or metaheuristic approaches could be explored for larger-scale or stochastic variants, e.g., accounting for fire spread, climate variability, or other dynamic processes.

4.6 Potential Future Works

Future analyses could enhance model accuracy and applicability by systematically addressing key data gaps. Incorporating high-resolution digital elevation models (DEMs) would better capture local slope variations, while improving road width and condition data would strengthen accessibility and connectivity analyses, increasing the real-world relevance of predictions. Additionally, enriching geospatial datasets with accurate hydrographic information would ensure correct segmentation of subproblems with respect to waterlines, further improving the model's utility for forest infrastructure planning and wildfire management.

Building on the core methodology, a two-stage decision process could extend the scope of this work to target not only currently inaccessible stands, but also stands that are already accessible via existing timber roads. In the first stage, new roads would be constructed in inaccessible areas that need access, as implemented in this study. In a potential second stage, existing small timber roads could be evaluated for widening to improve fire accessibility. Any decisions regarding road widening would need to be coordinated with the respective road or landowners. Another potential extension is to assign costs to the virtual edges introduced in this work, which currently connect the centroid of each stand to the boundary nodes. These costs could represent intra-stand timber extraction effort, environmental impact, or operational constraints such as difficult terrain or fuel load. Incorporating these costs would allow the model to more realistically capture trade-offs within each stand, potentially influencing both road layout decisions and harvesting strategies.

Future research could explore hybrid approaches combining the strengths of MILP and AI-based methods. Given the strict constraints of the forest road network problem, MILP could first generate feasible, high-quality candidate solutions, which heuristic or AI algorithms – such as genetic algorithms or particle swarm optimization – could then refine, adapt, or explore alternative network configurations under uncertain or dynamic conditions. MILP outputs could also serve as benchmarks or training data for machine learning models approximating optimal decisions in stochastic or time-varying scenarios. This two-stage framework would merge the precision and verifiability of MILP with the scalability and flexibility of AI methods.

5

Conclusion

This master's dissertation presented a multi-period, dual-purpose forest road network optimization model addressing both timber harvesting and ground-based wildfire accessibility in fire-prone Mediterranean forests in Northern Portugal. The model was applied to the Castelo de Paiva case study, part of the ZIF Paiva, which is representative of northern Portuguese forests with small, scattered stands and a high incidence of wildfires. Three different harvesting schedule scenarios were evaluated. Jupyter notebooks were developed to process and format the obtained data, run the model, and analyze the resulting solutions.

Key Findings and Contributions The main contribution of this work is the development of a single-objective integer programming model to optimize a dual-use forest road network over multiple harvesting periods. Unlike typical approaches that focus primarily on timber harvesting efficiency, this model explicitly incorporates accessibility requirements for ground-based firefighting forces, ensuring that less fire-resilient areas can be reached quickly in the event of a wildfire. The model integrates construction, maintenance, and upgrade decisions for road segments, distinguishes between timber and wide roads, and accounts for varying access needs of forest stands, including both timber transport and fire vehicle mobility. Auxiliary flow variables were introduced to guarantee connectivity from each stand to the public road network while preserving realistic flow patterns and preventing traversal through other stands, and virtual, zero-cost road segments were incorporated to facilitate constraint formulation without affecting physical implementation. Implemented in Python using the CPLEX API, the model leverages geographical decomposition and graph simplification to reduce computational complexity and enhance solver performance.

By integrating multi-period economic planning with dual-use road network design, the work provides a tool that simultaneously supports timber harvesting and wildfire mitigation, identifies infrastructure investments that satisfy multiple objectives, and incorporates realistic constraints related to stand accessibility and terrain limitations to generate actionable recommendations for forest managers. Essentially, the framework bridges the gap between optimization results and practical management, enabling road network planning to support actionable decisions in forestry and wildfire management, embedding model-driven decisions within real-world management processes rather than treating them as a purely academic exercise. The use of MILP with the CPLEX solver guarantees globally optimal solutions with a known optimality gap under deterministic, linear constraints, ensuring interpretability, full constraint satisfaction, and explicit control over solution quality. Together, these contributions constitute a rigorously formulated, implementable optimization framework capable of generating cost-minimal, feasible road networks that address practical needs for both forestry operations and wildfire management, with the novelty of explicitly supporting ground-based firefighting access to vulnerable areas.

Practical Implications The study provides actionable guidance for forest management:

- Forest managers can prioritize road construction and maintenance to serve multiple purposes, balancing economic efficiency with safety objectives.
- The model informs the strategic placement of roads over multiple periods, enhancing wildfire response to the least resilient areas while supporting timber extraction where needed.
- Stakeholders can use model outputs to support transparent, quantitative decision-making and facilitate collaborative scenario evaluation.

Limitations Several limitations of this study should be acknowledged. Some forest stands remain inaccessible due to land ownership constraints, which prevents the model from proposing new road segments in these areas and leaves their road network problem unresolved. The analysis assumed that waterlines do not hinder road construction, except for the Douro and Paiva rivers defining the study boundaries, while minor streams were considered seasonal and non-obstructive. If some streams are in fact impassable, road network segmentation may be incomplete, requiring future splitting and optimization along waterlines. Inaccuracies in the geospatial representation of stands or roads could compromise accessibility analyses, clustering, and segmentation, potentially biasing optimization results. Similarly, estimated road widths for pre-existing segments may be incorrect, affecting connectivity assumptions and the usability of newly built roads, particularly for fire vehicles. Coarse slope data can introduce errors in construction cost estimation, as stand-averaged slopes do not capture local variability relevant for assigning cost categories. Finally, uncertainties in cost estimates mean that solutions appearing cost-minimal may, in practice, be suboptimal.

Future Work Potential directions to expand and further develop this work include:

- Address data and model limitations identified in this study, such as improving geospatial accuracy, validating road width and slope data, to produce more robust and implementable road network plans.

- Implement a staged road development strategy: first constructing new roads in currently inaccessible areas, then evaluating existing roads for widening based on fire risk and accessibility.
- Assign costs to virtual edges within stands to represent intra-stand extraction effort, environmental impacts, or operational constraints, increasing the realism of road network and harvesting trade-offs.
- Explore hybrid optimization approaches combining MILP with heuristic or AI-based methods to improve scalability, handle stochastic or dynamic scenarios, and address uncertainty in fire behavior.
- Incorporate natural barriers and more detailed terrain constraints to improve road feasibility and construction planning.
- Integrate firebreaks and fire behavior models to enhance the effectiveness of dual-use roads.
- Consider broader ecological and social factors, including sensitive habitats and community access.

Synthesis In summary, this dissertation advances forest road network planning by presenting a multi-period, dual-use optimization framework that balances economic efficiency with wildfire mitigation needs. By explicitly integrating access for ground-based firefighting into road network design, the work extends beyond traditional timber-focused approaches and offers a practical, implementable tool for forest managers. While data limitations and simplifications impose some constraints, the framework provides clear guidance for infrastructure investment, operational planning, and strategic decision-making. The proposed model, together with the outlined directions for future work, lays a foundation for increasingly resilient, cost-effective, and ecologically aware forest road networks. Ultimately, the study demonstrates how model-driven road network planning can support both forestry operations and enhanced wildfire preparedness, bridging the gap between academic optimization research and real-world forest management practice.

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A

The FIRE-RES project

FIRE-RES is a four-year European project (2021–2025) dedicated to developing integrated forest management strategies to address the challenges posed by Extreme Wildfire Events (EWEs). It aims to harmonize wildfire emergency response with landscape-level planning and evaluate the economic feasibility of resilience-enhancing measures. The project's overarching goal is to develop tools, processes, and methodologies that promote integrated approaches to managing EWEs. It further advocates for proactive governance to improve societal resilience, raise risk awareness, and support effective communication—ultimately fostering fire-resilient landscapes and communities.

FIRE-RES is implemented through eleven Living Labs, which serve as open innovation ecosystems located in fire-prone regions across Europe and beyond. These include: Bulgaria, the Canary Islands, Catalonia and Galicia (Spain), Chile, Germany/The Netherlands, Greece, Norway/Sweden, Nouvelle-Aquitaine (France), Portugal, and Sardinia (Italy). Measures and approaches developed within these contexts are intended for broader replication and scaling.

For more information, see: https://fire-res.eu.

B

Forest Growth Projections

To provide context for the forest growth projections underlying the harvesting schedules the present work used and was built upon, the following description, coming from Marques, Rodrigues, et al. (2024), is reproduced from that source:

> ”Forest growth was assessed via specific empirical growth and yield models over a planning horizon of 100 years, allowing us to estimate the evolution of wood production volumes originated by thinnings, and accounting with different assortments of biomass and carbons. Each prescription is a schedule of silvicultural activities (e.g., planting, thinning, regeneration, fertilization, or harvesting), which when implemented on a stand is expected to achieve the desired outcomes. Usually, a myriad of different prescriptions is technically or biologically possible for each stand and were then simulated using species-specific empirical growth and yield models or yield tables. The *Globulus 3.0* and the *Pinaster* models, both implemented in the `standsSIM-MD` module included in the `SIMFLOR` platform (Lisbon, Portugal), were

used for *E. globulus* and *P. pinaster*, respectively. The *SUBER v5.0*, implemented in the same platform, was used for the growth simulation of *Q. suber*. The *Sim-Galiza* simulator, developed for *Q. robur* stands in Galicia (Spain), was also used, considering the similar soil and climate conditions from that region and our case study area. For *C. sativa*, yield tables were used. We obtained, from riparian stand national databases, structural parameters over the planning horizon."

(Adapted from Marques, Rodrigues, et al. (2024))

C

Large Tables

Flow type	Flow direction	Period	Decision Variable
timber	backwards	1	$flow^{\text{timber}}_{(v,u),1}$
timber	backwards	2	$flow^{\text{timber}}_{(v,u),2}$
timber	backwards	3	$flow^{\text{timber}}_{(v,u),3}$
timber	backwards	4	$flow^{\text{timber}}_{(v,u),4}$
timber	backwards	5	$flow^{\text{timber}}_{(v,u),5}$
timber	forwards	1	$flow^{\text{timber}}_{(u,v),1}$
timber	forwards	2	$flow^{\text{timber}}_{(u,v),2}$
timber	forwards	3	$flow^{\text{timber}}_{(u,v),3}$
timber	forwards	4	$flow^{\text{timber}}_{(u,v),4}$
timber	forwards	5	$flow^{\text{timber}}_{(u,v),5}$
wide	backwards	1	$flow^{\text{wide}}_{(v,u),1}$
wide	backwards	2	$flow^{\text{wide}}_{(v,u),2}$
wide	backwards	3	$flow^{\text{wide}}_{(v,u),3}$
wide	backwards	4	$flow^{\text{wide}}_{(v,u),4}$
wide	backwards	5	$flow^{\text{wide}}_{(v,u),5}$
wide	forwards	1	$flow^{\text{wide}}_{(u,v),1}$
wide	forwards	2	$flow^{\text{wide}}_{(u,v),2}$
wide	forwards	3	$flow^{\text{wide}}_{(u,v),3}$
wide	forwards	4	$flow^{\text{wide}}_{(u,v),4}$
wide	forwards	5	$flow^{\text{wide}}_{(u,v),5}$

Table C.1: Flow variables

Action	Road Width	Period t	Decision Variable
Construction	timber	1	$C^{\text{timber}}_{e,1}$
Construction	timber	2	$C^{\text{timber}}_{e,2}$
Construction	timber	3	$C^{\text{timber}}_{e,3}$
Construction	timber	4	$C^{\text{timber}}_{e,4}$
Construction	timber	5	$C^{\text{timber}}_{e,5}$
Construction	wide	1	$C^{\text{wide}}_{e,1}$
Construction	wide	2	$C^{\text{wide}}_{e,2}$
Construction	wide	3	$C^{\text{wide}}_{e,3}$
Construction	wide	4	$C^{\text{wide}}_{e,4}$
Construction	wide	5	$C^{\text{wide}}_{e,5}$
Maintenance	timber	1	$M^{\text{timber}}_{e,1}$
Maintenance	timber	2	$M^{\text{timber}}_{e,2}$
Maintenance	timber	3	$M^{\text{timber}}_{e,3}$
Maintenance	timber	4	$M^{\text{timber}}_{e,4}$
Maintenance	timber	5	$M^{\text{timber}}_{e,5}$
Maintenance	wide	1	$M^{\text{wide}}_{e,1}$
Maintenance	wide	2	$M^{\text{wide}}_{e,2}$
Maintenance	wide	3	$M^{\text{wide}}_{e,3}$
Maintenance	wide	4	$M^{\text{wide}}_{e,4}$
Maintenance	wide	5	$M^{\text{wide}}_{e,5}$
Upgrade	wide	1	$U_{e,1}$
Upgrade	wide	2	$U_{e,2}$
Upgrade	wide	3	$U_{e,3}$
Upgrade	wide	4	$U_{e,4}$
Upgrade	wide	5	$U_{e,5}$

Table C.2: Binary Decision Variables

D

Visualizations

D.1 Road Data Preprocessing

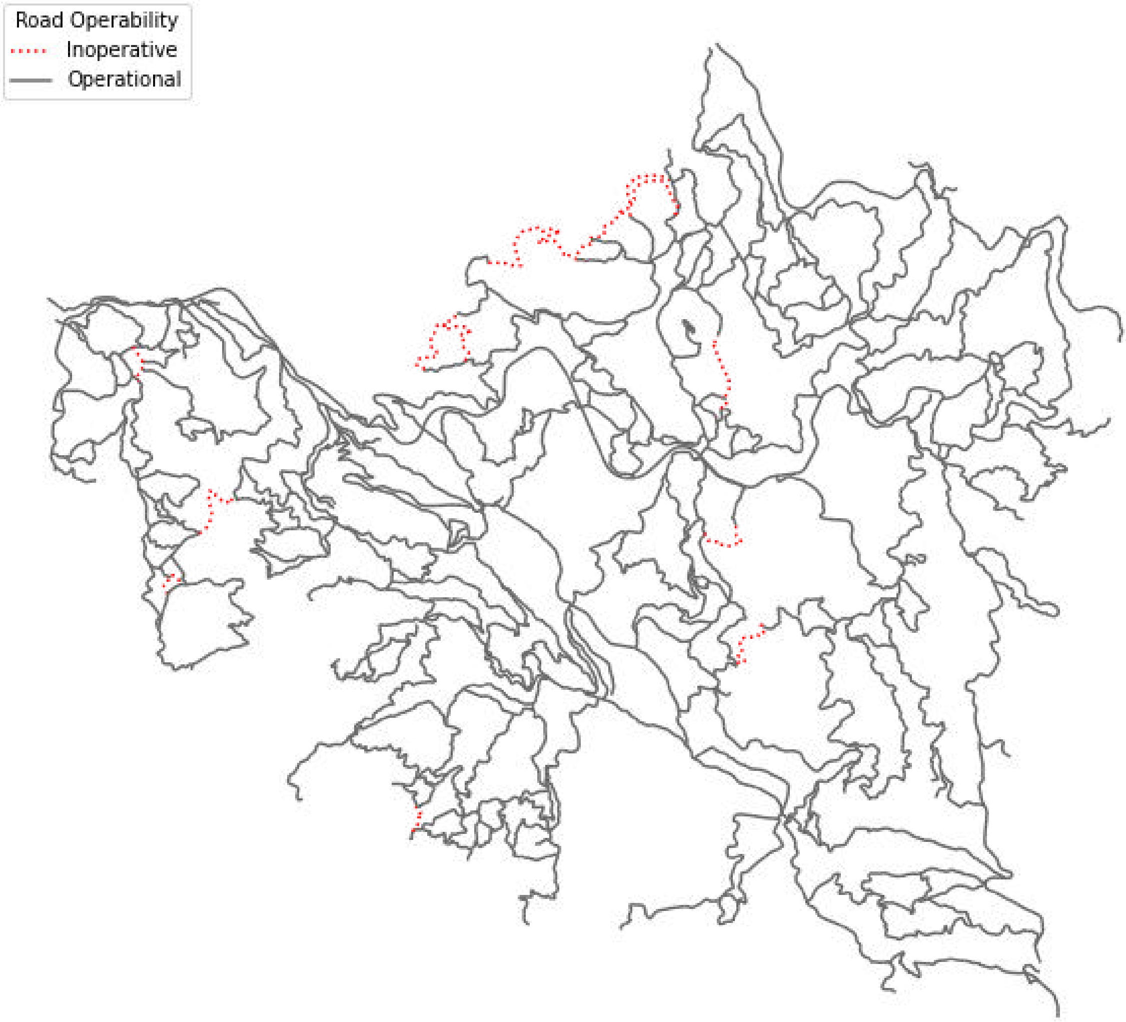

Figure D.1: Operability status of the road network in Paiva; inoperative roads shown as dotted red lines.

Figure D.2: Visualization of implausible road width information on operable road segments, segments with road width of zero shown as dashed lines in orange

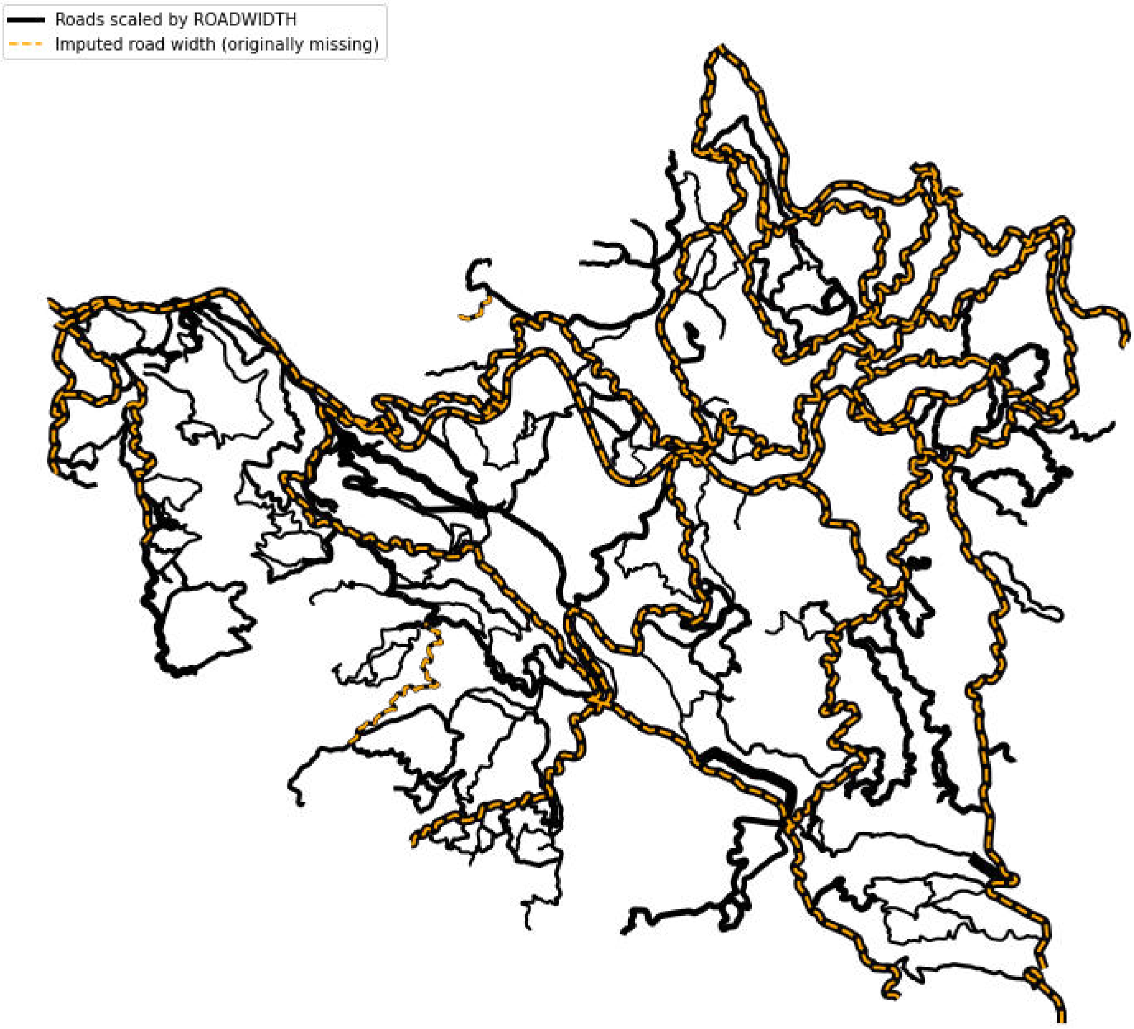

Figure D.3: Operable roads displayed with line widths proportional to road widths; imputed road width values highlighted as dashed lines in orange

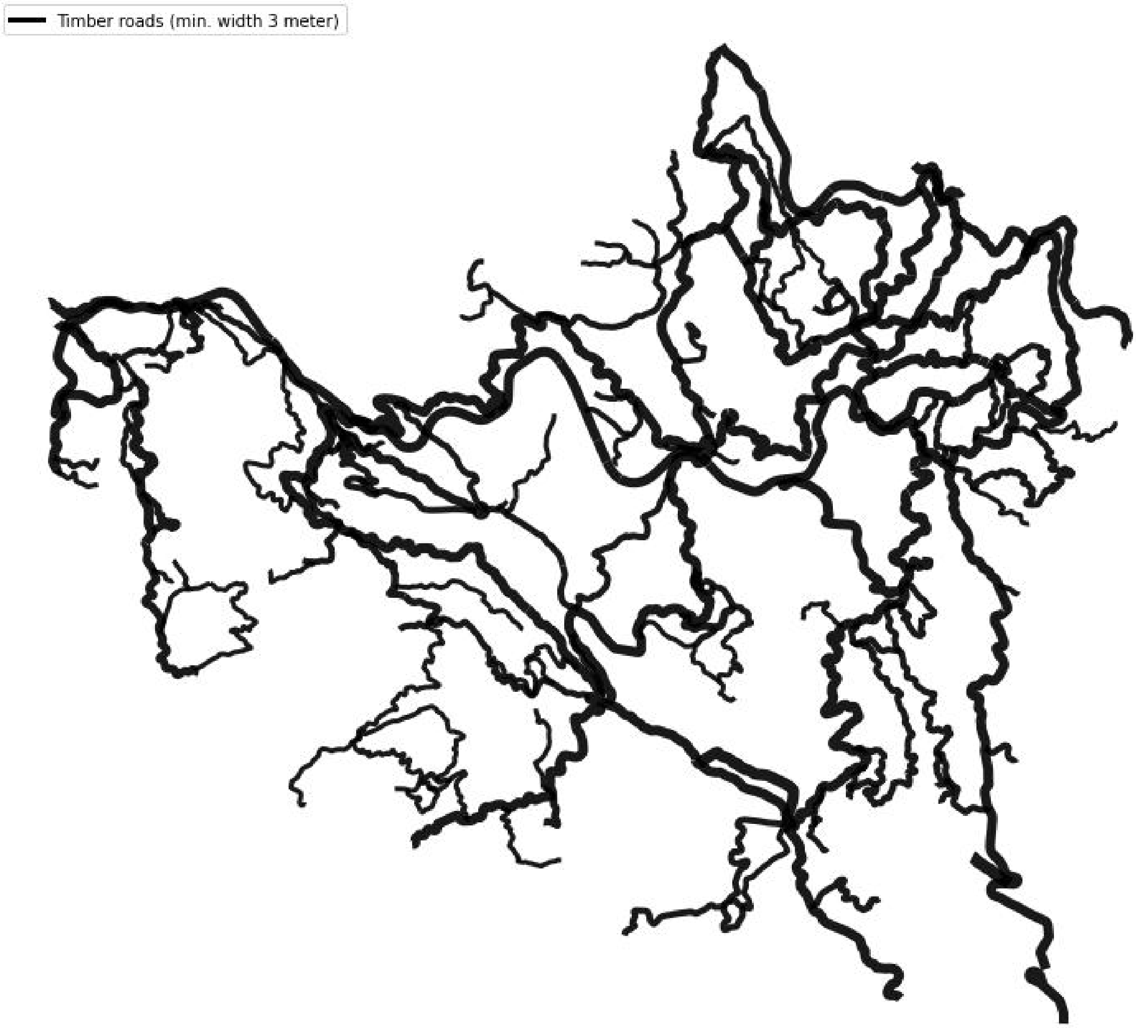

Figure D.4: Network of operable roads with widths greater than 3 m ("timber roads"), visualized with line width proportional to assigned road width values and colored by road type.

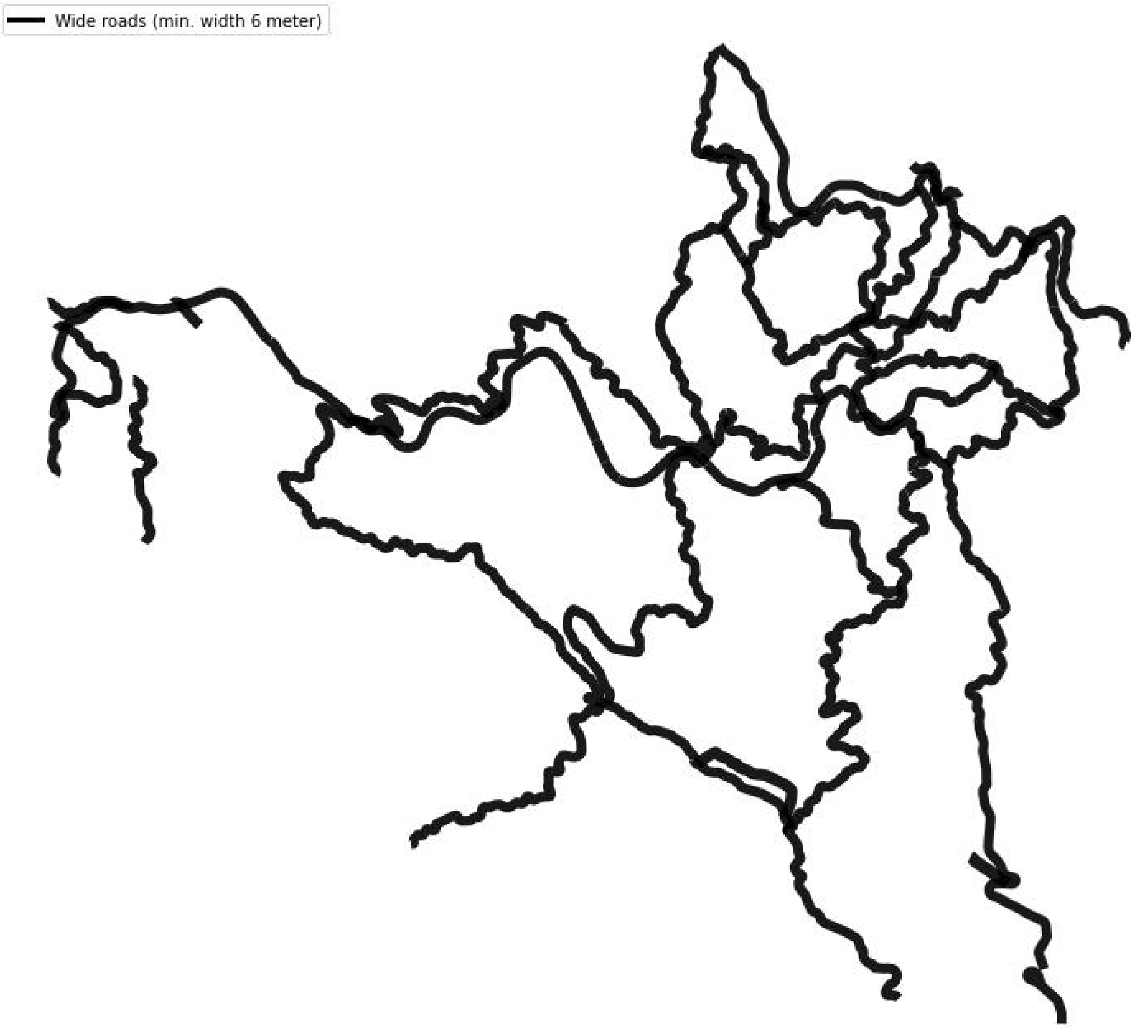

Figure D.5: Network of operable roads with widths greater than 6 m (”fire roads”), visualized with line width proportional to road width values.

D.2 Forest Stand Road Accessibility Analysis

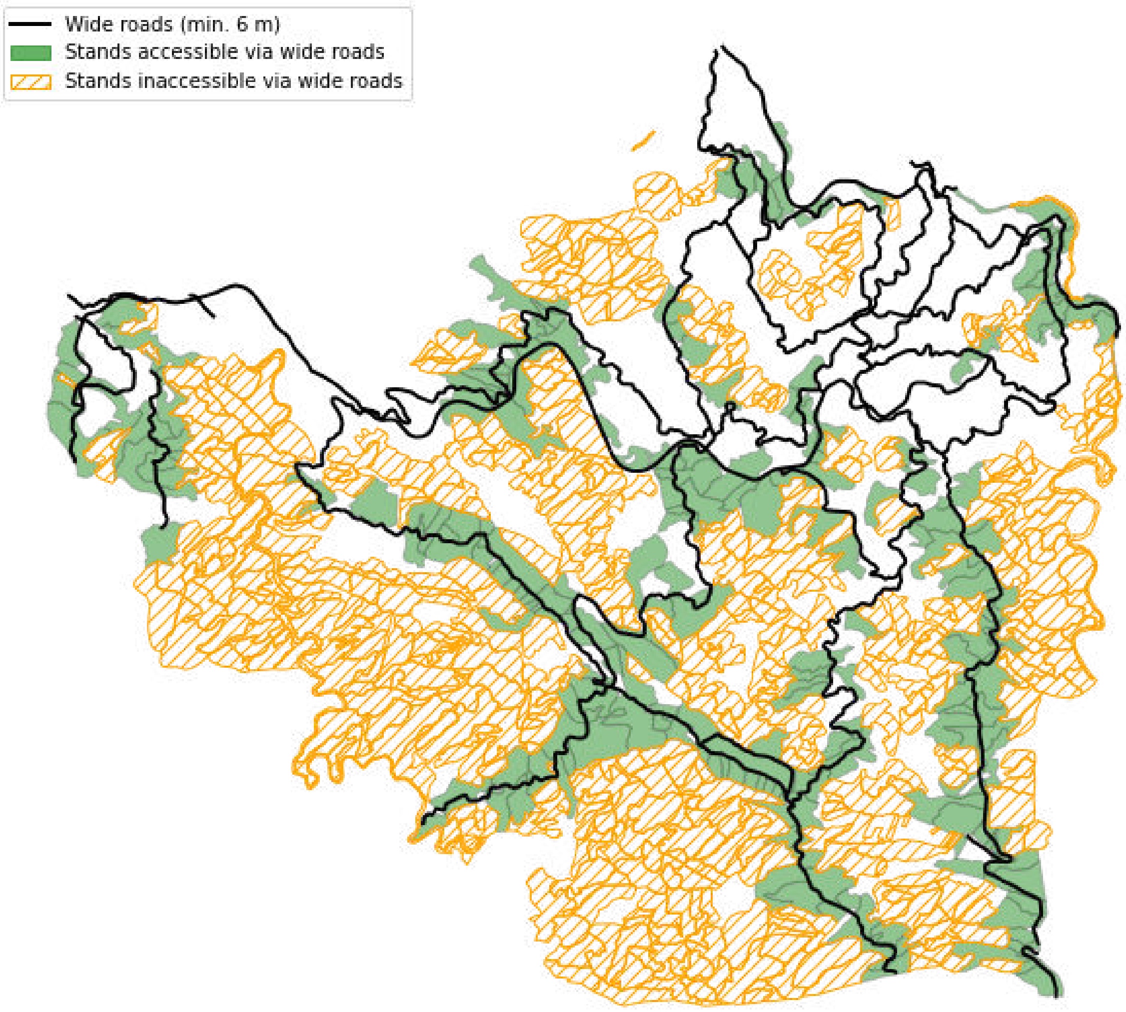

Figure D.6: Stand accessibility via the existing network of wide roads

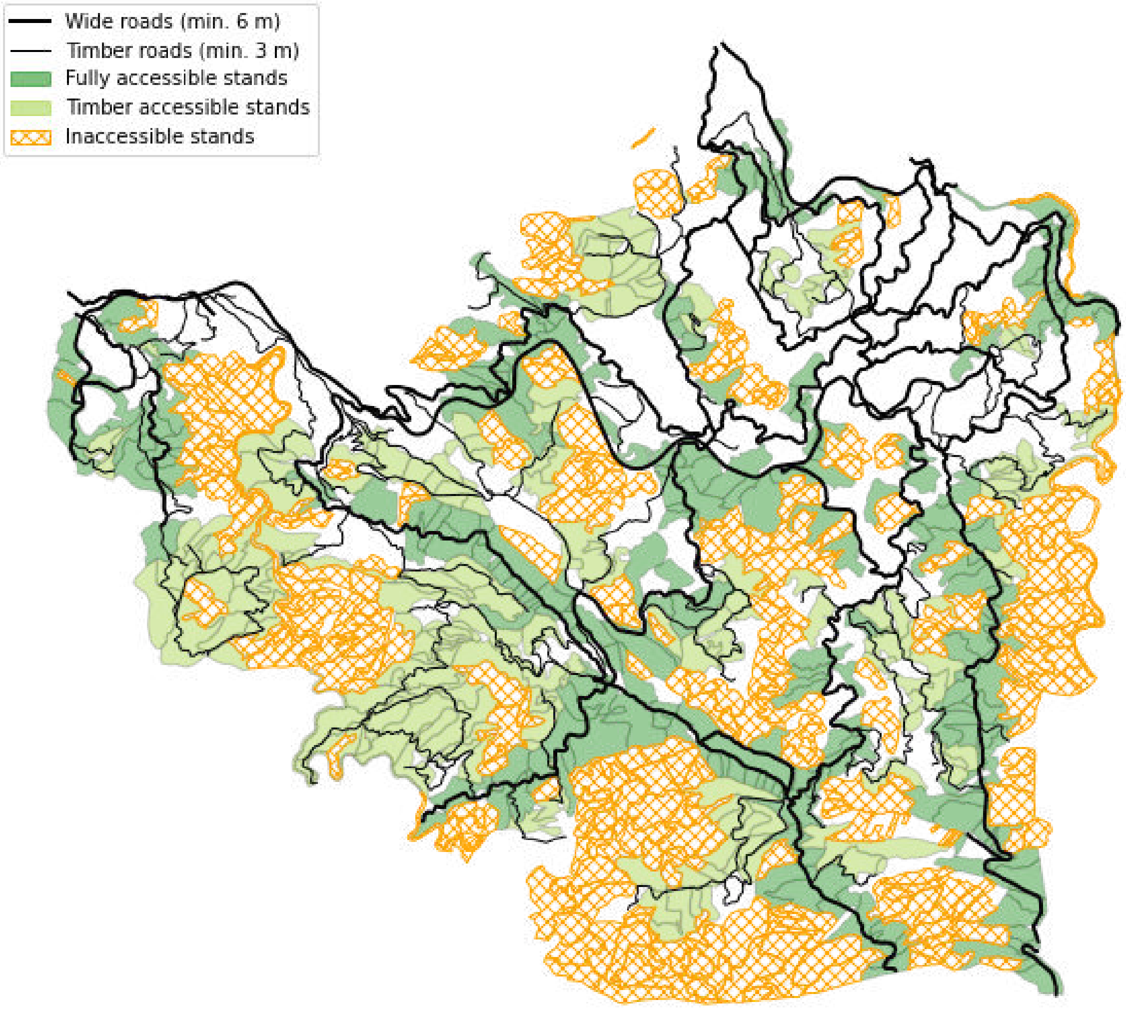

Figure D.7: Stand accessibility via the existing timber road network

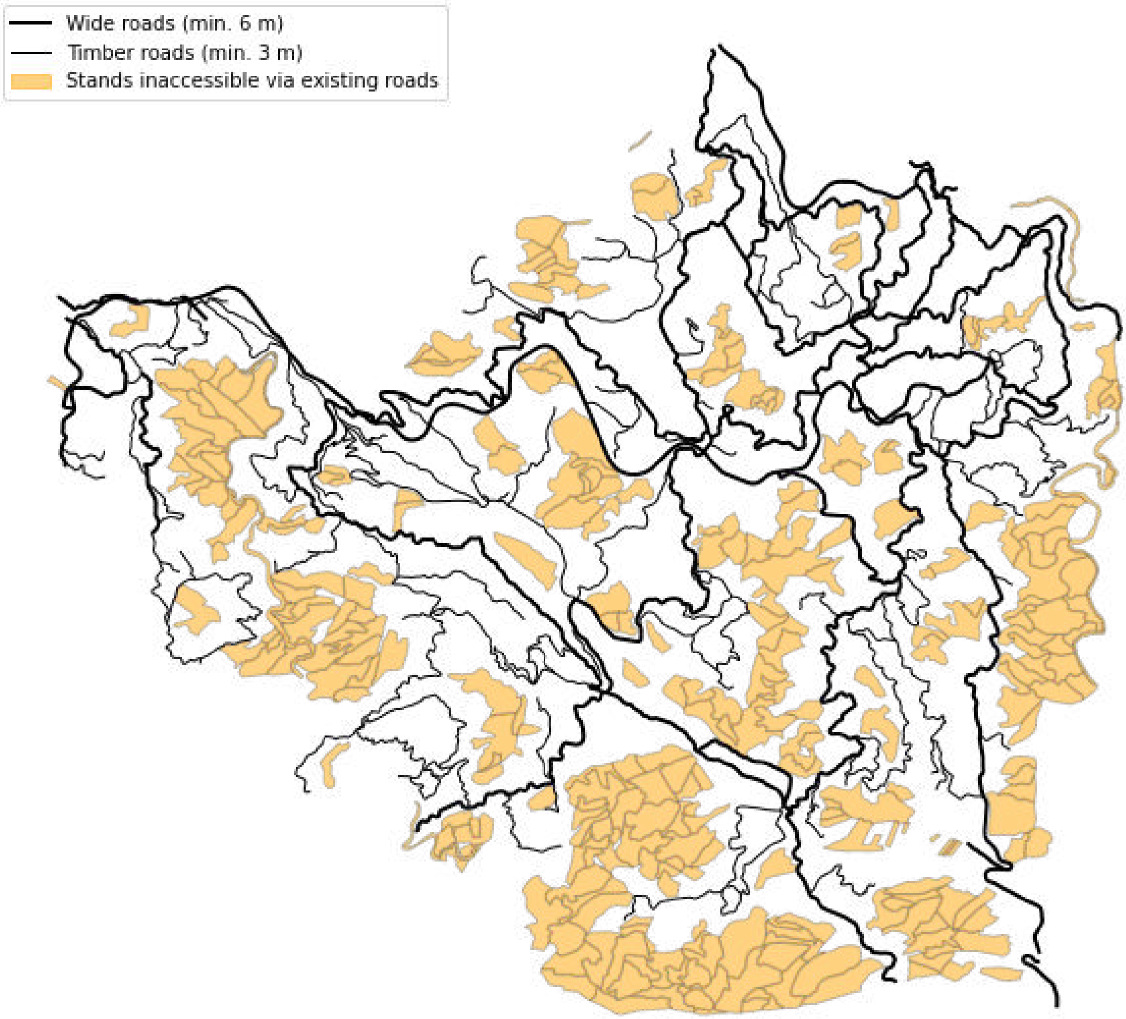

Figure D.8: Sets of stands whose access needs are considered in the decision problem

D.3 Segmentation Results

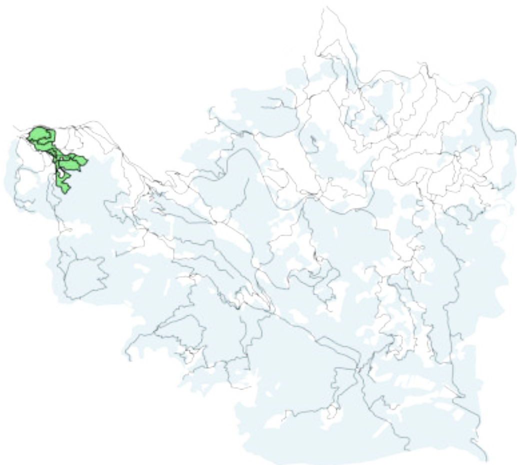

Figure D.9: Component 1

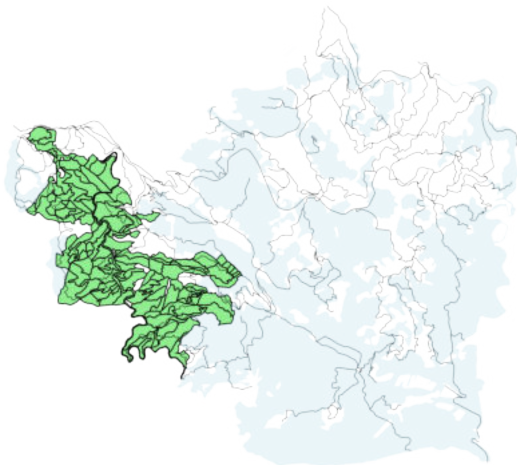

Figure D.10: Component 2_44 merged

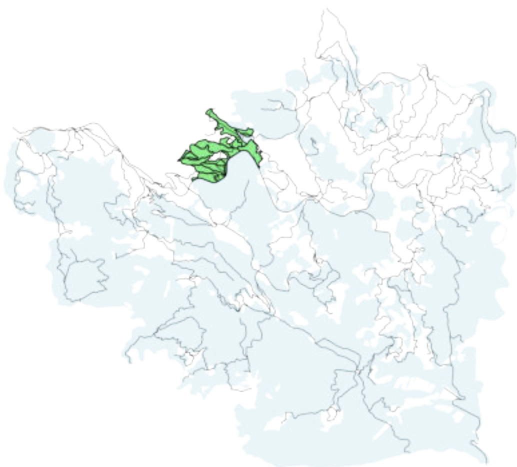

Figure D.11: Component 3_43 merged

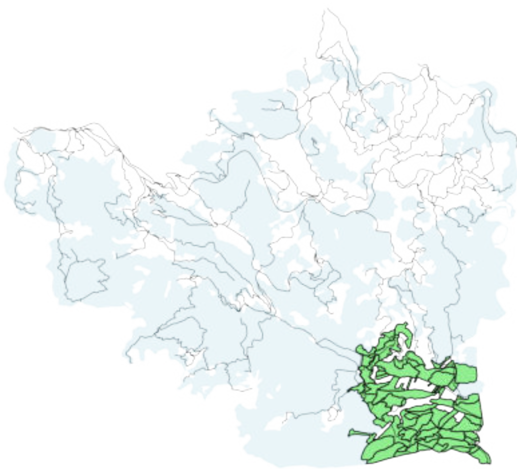

Figure D.12: Component 4_11_12_13_51 merged

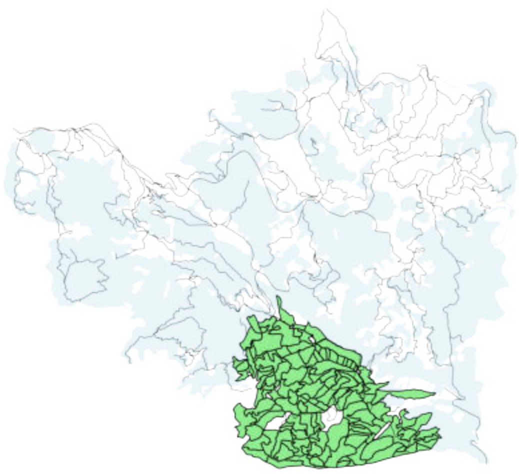

Figure D.13: Component 5_39 merged

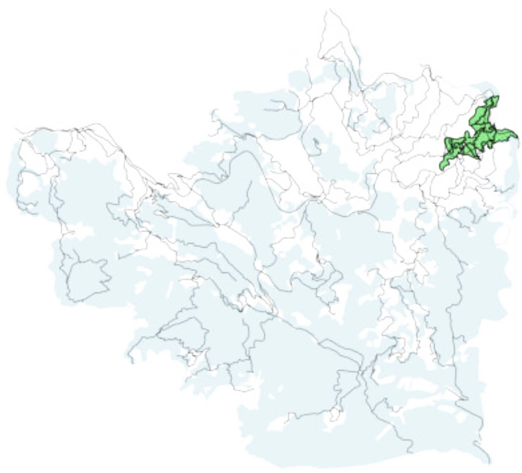

Figure D.14: Component 6

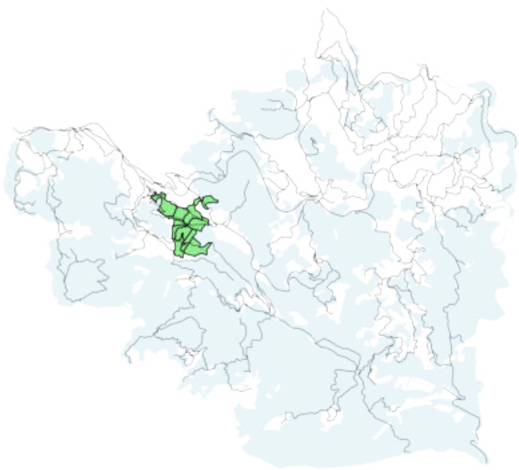

Figure D.15: Component 7

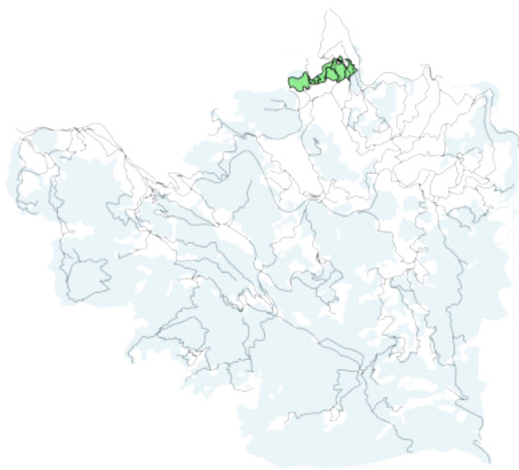

Figure D.16: Component 8

Figure D.17: Component 9

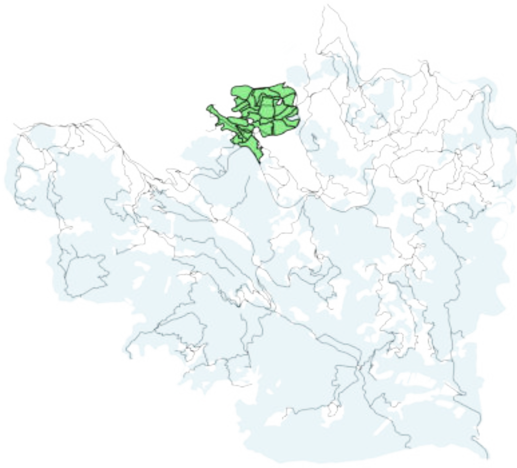

Figure D.18: Component 10

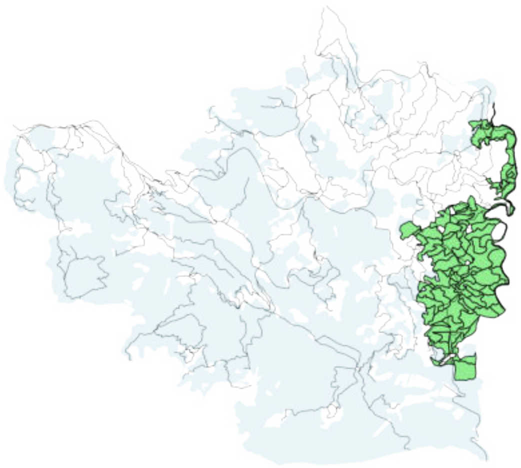

Figure D.19: Component 14_15_48 merged

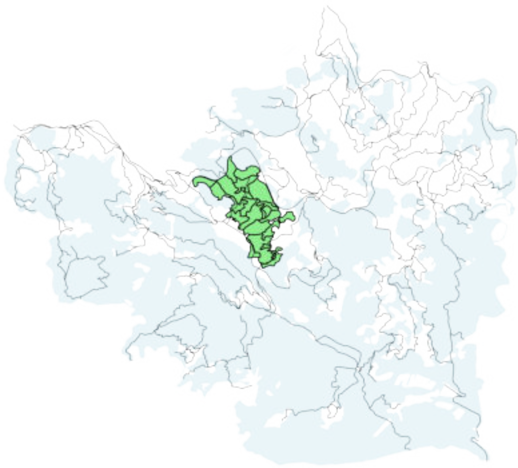

Figure D.20: Component 16

Remark. Component 17, the tiny green spot on the very top of fig. D.21, could not be addressed in the current project, as the area between this inaccessible component and the existing road network is under different ownership.

Figure D.21: Component 17

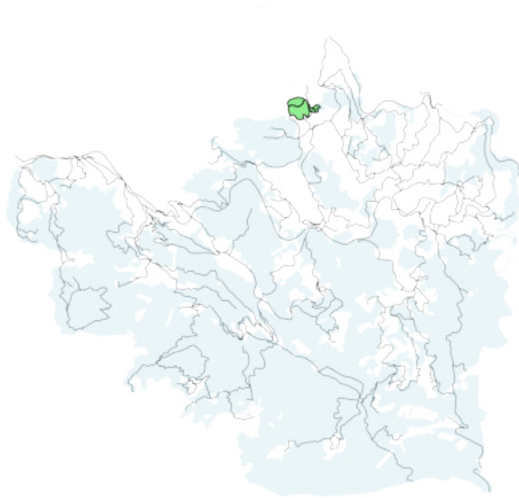

Figure D.22: Component 18

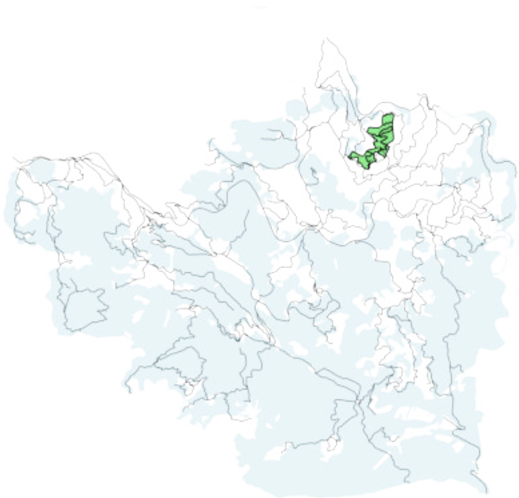

Figure D.23: Component 19_41 merged

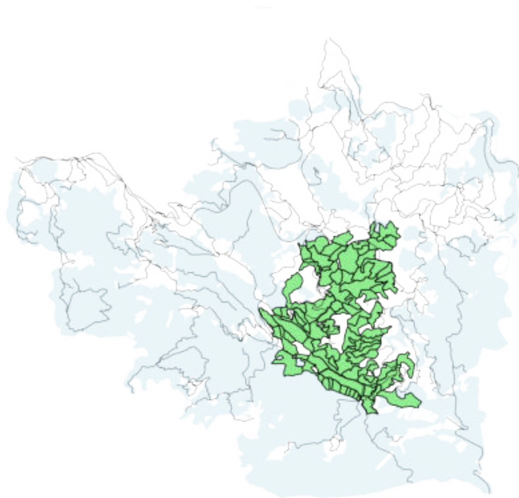

Figure D.24: Component 20_25_28_33 merged

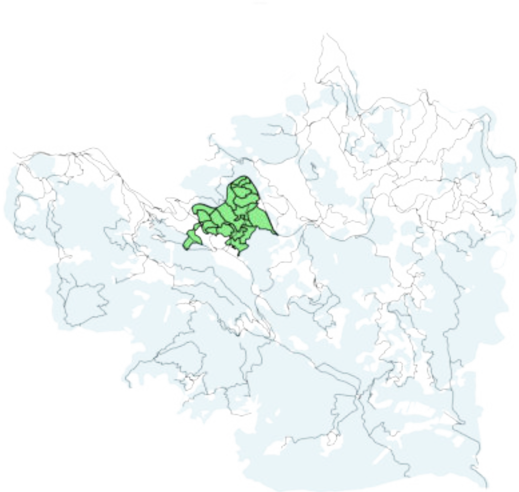

Figure D.25: Component 21_40 merged

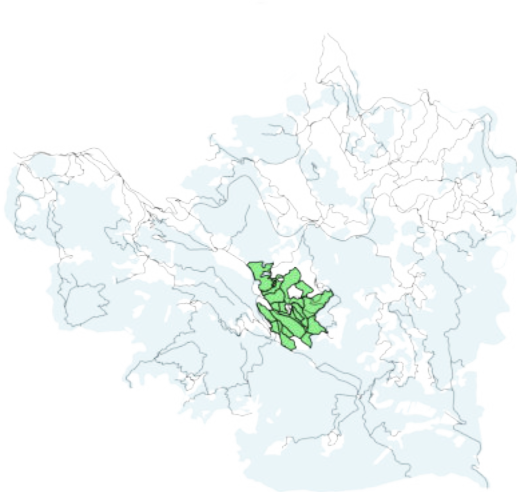

Figure D.26: Component 22

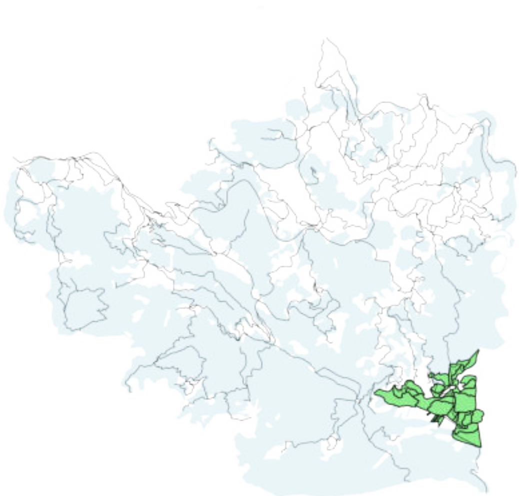

Figure D.27: Component 23

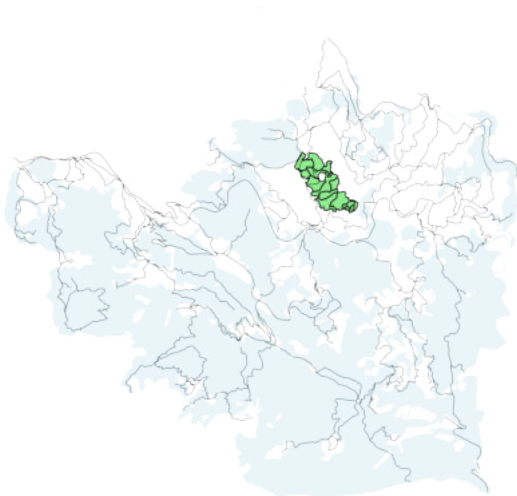

Figure D.28: Component 24

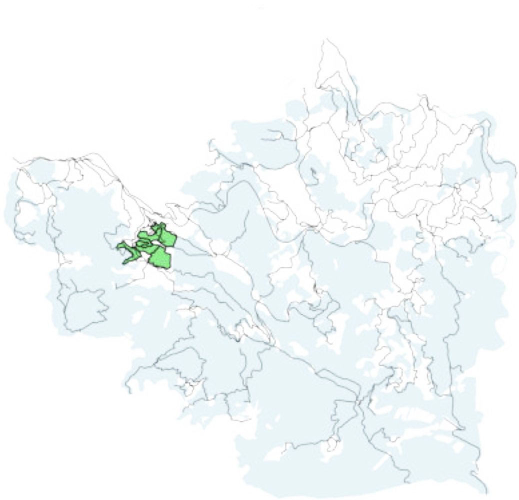

Figure D.29: Component 26

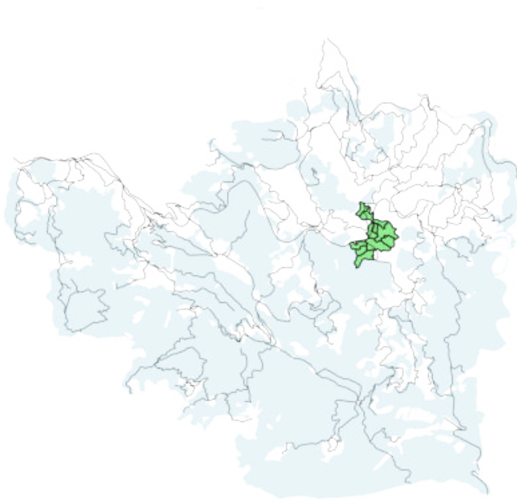

Figure D.30: Component 27

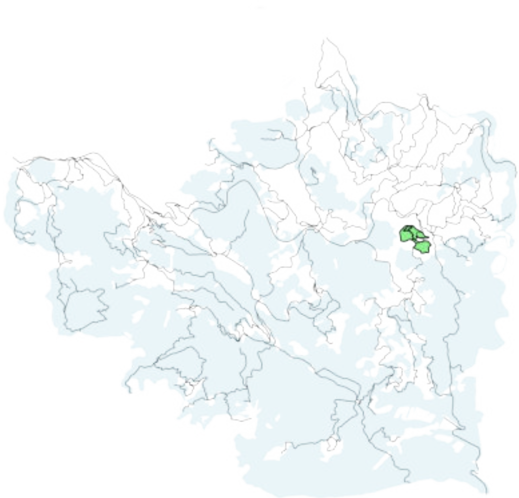

Figure D.31: Component 29

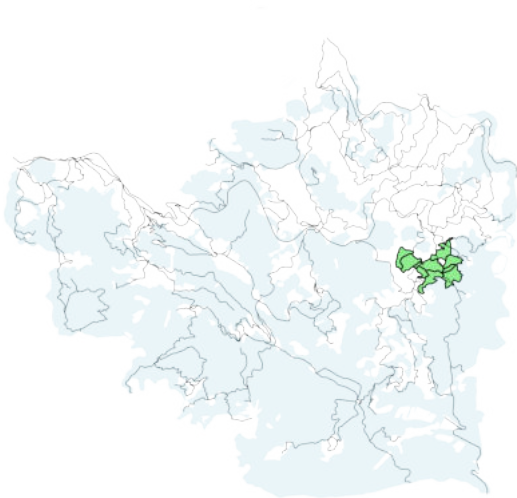

Figure D.32: Component 30

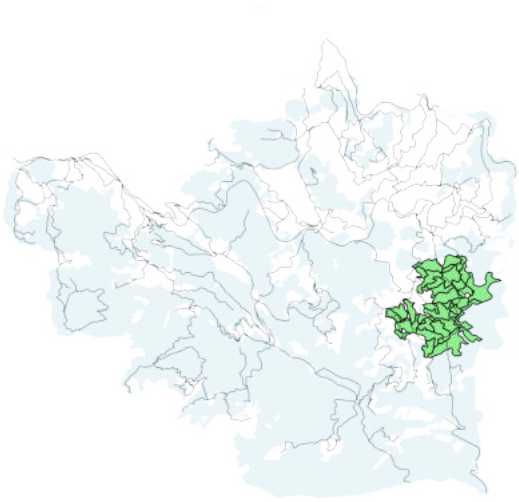

Figure D.33: Component 31_34 merged

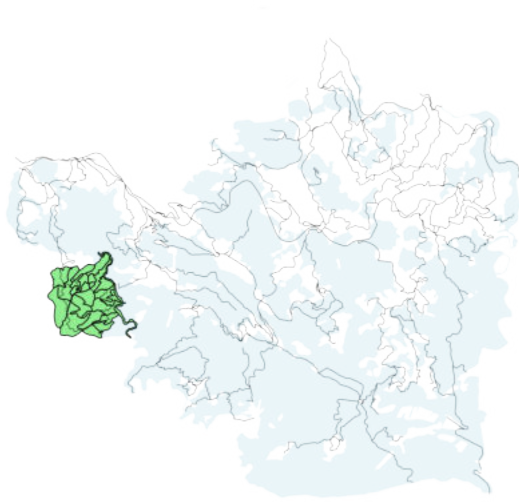

Figure D.34: Component 32

Figure D.35: Component 35

Figure D.36: Component 36_50 merged

Figure D.37: Component 37

Figure D.38: Component 38

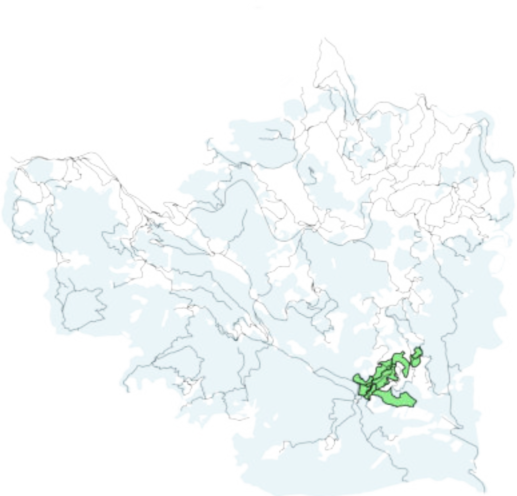

Figure D.39: Component 42

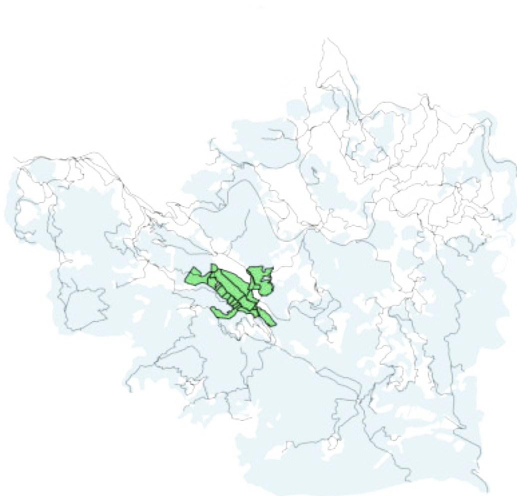

Figure D.40: Component 45

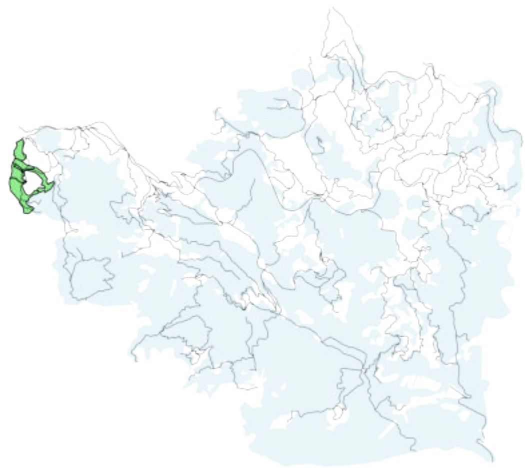

Figure D.41: Component 46

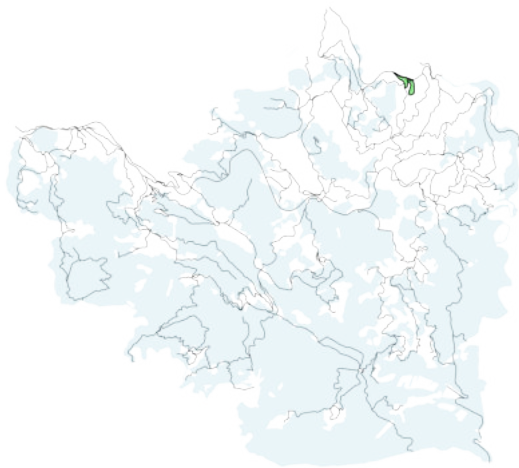

Figure D.42: Component 47

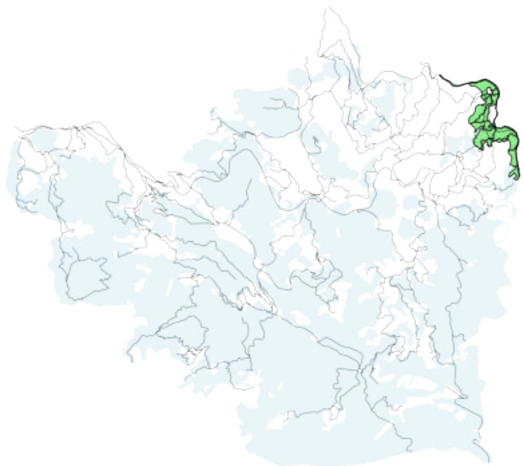

Figure D.43: Component 49

D.4 Road Network Graph Development - Example: Component 1

Figure D.44: Road network graph for component 1

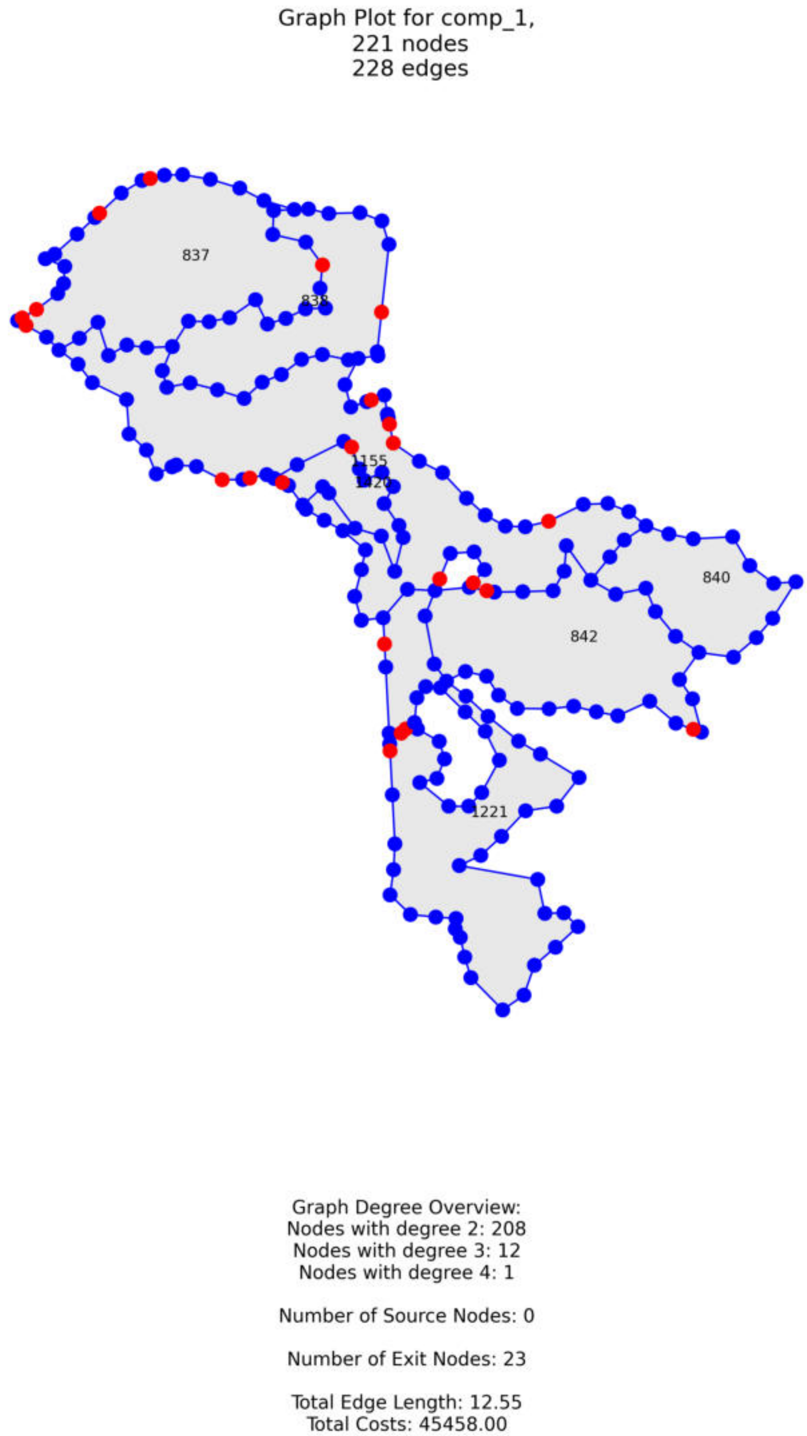

Figure D.45: Road network graph for component 1 with exit nodes in red

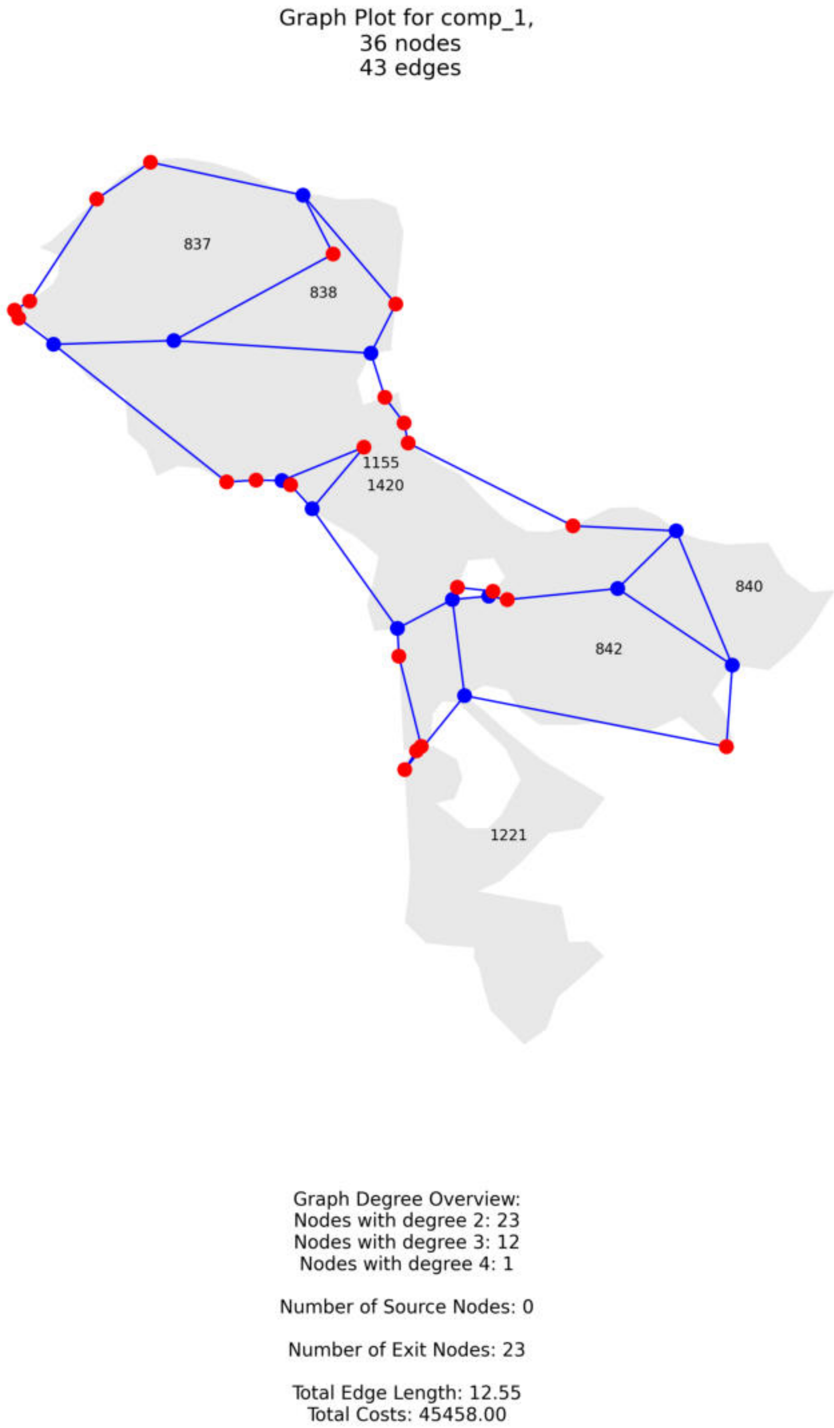

Figure D.46: Abstracted road network graph of component 1

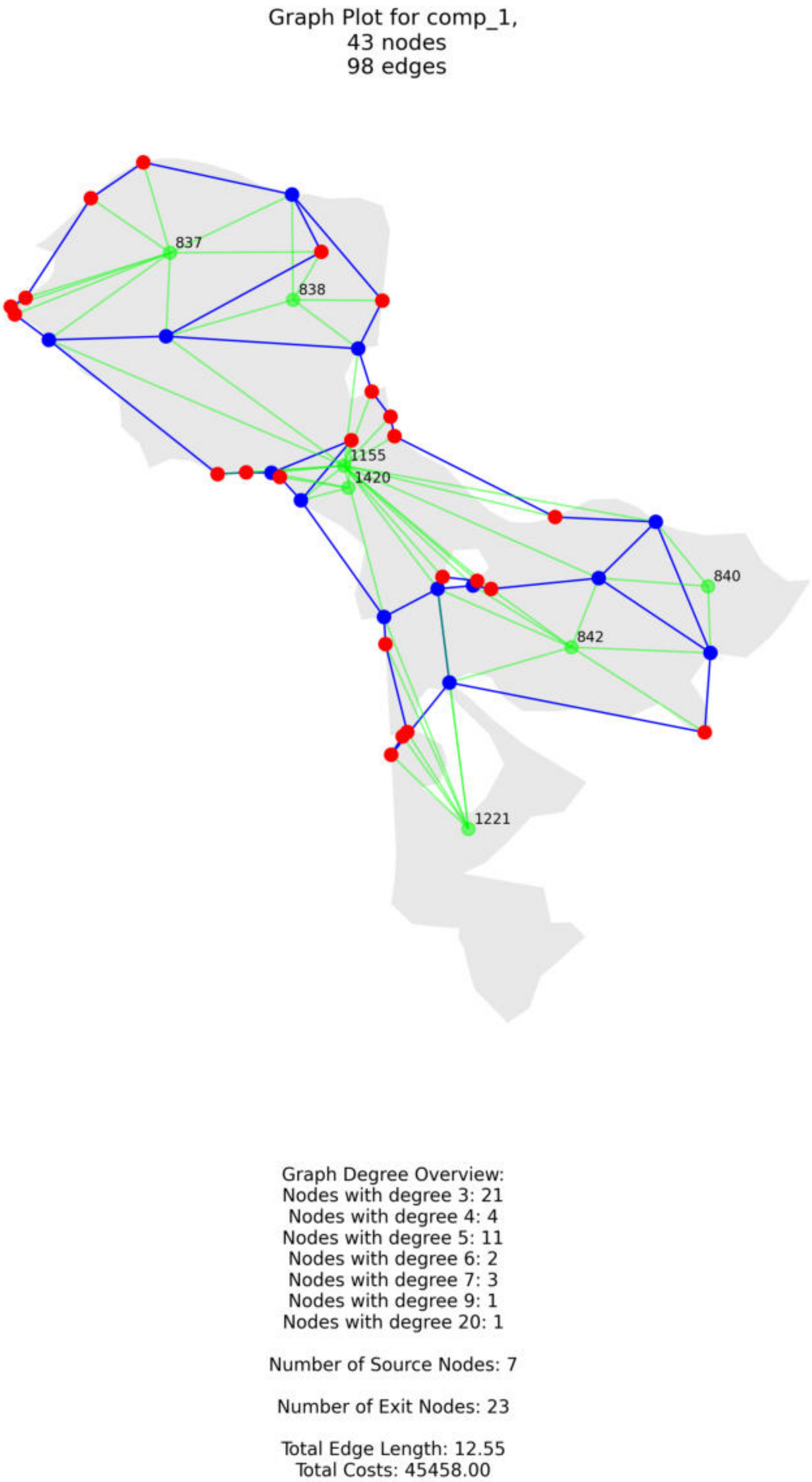

Figure D.47: Abstracted road network graph of component 1 with source nodes and edges in green